AWS Lab 02: Theory

- Data Centre
- Auto Scaling
  - Scale-up
  - Scale-down
- Cloud -> Make things scalable according to the usage in real-time and robust
- Horizontal scaling and vertical scaling
- Load Balancer: Maximum concurrent users or asking for something/lag in DB “Slowing machine because of any reason” so we balance load
- Docker -> Containerization
- Use case of Docker: Making the environment portable.
- Lab Today: Setting up load balancer and how it works

Elastic Load Balancing (ELB)

Dividing load to different machine

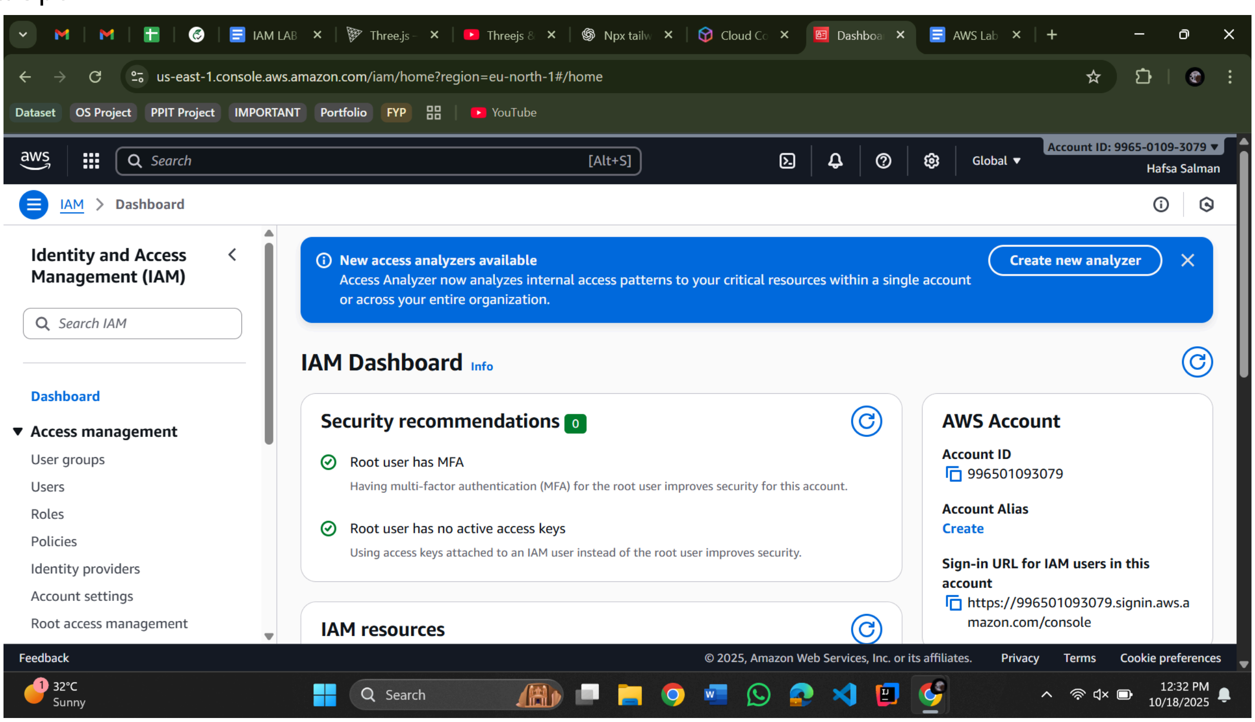
Security group -> Define inbounds and outbounds

Target group -> Set load balancer to define which machine to give/take services

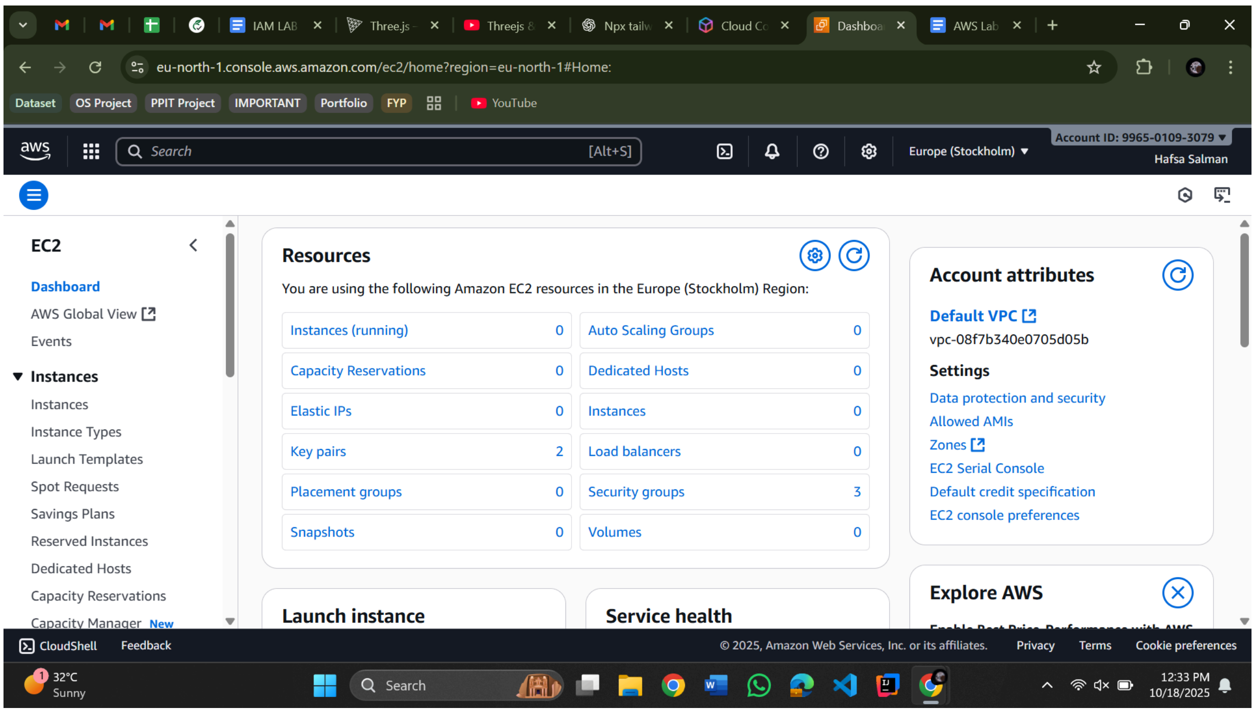
//User Account

- Login using root account
- Open incognito
- Login through IAM by entering Account ID (copy it from root account), user name (the one you made in root account) and user password
- Follow the following steps

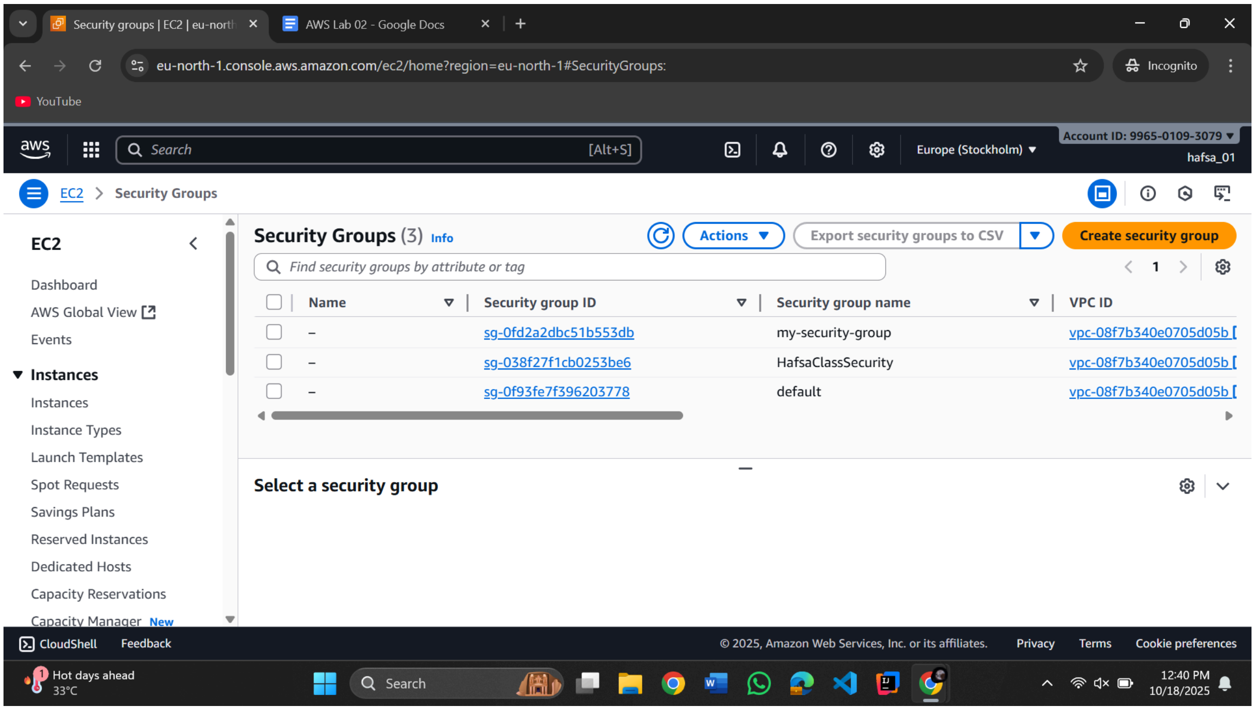
//Open IAM

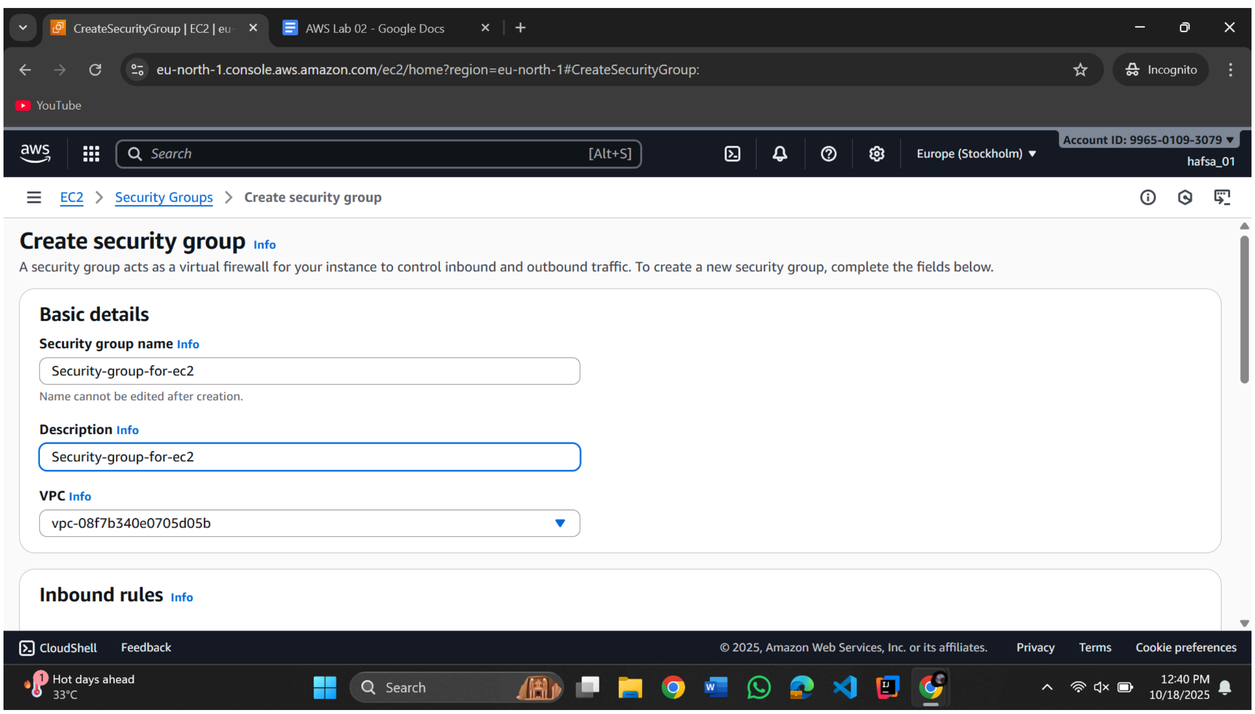


//Open EC2

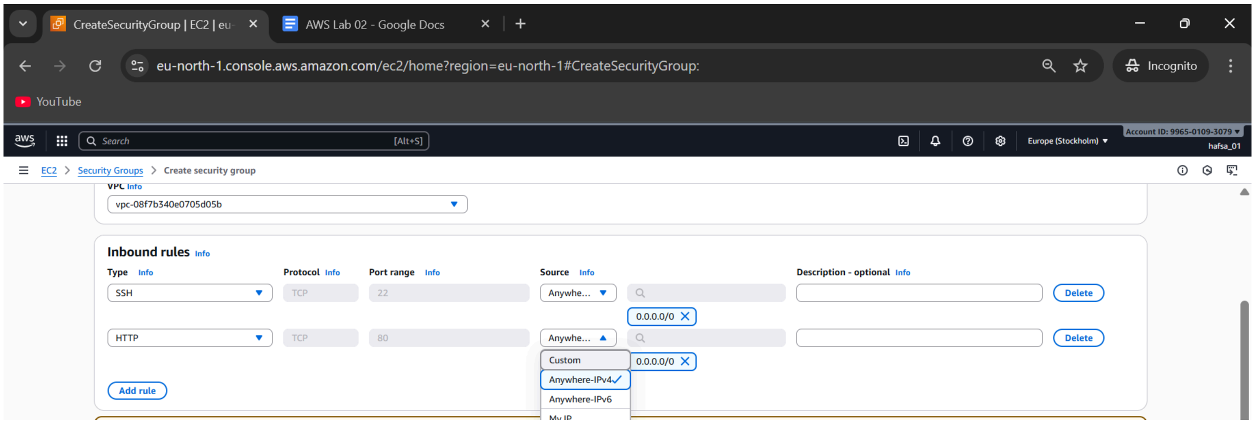


//create Security groups (top right)

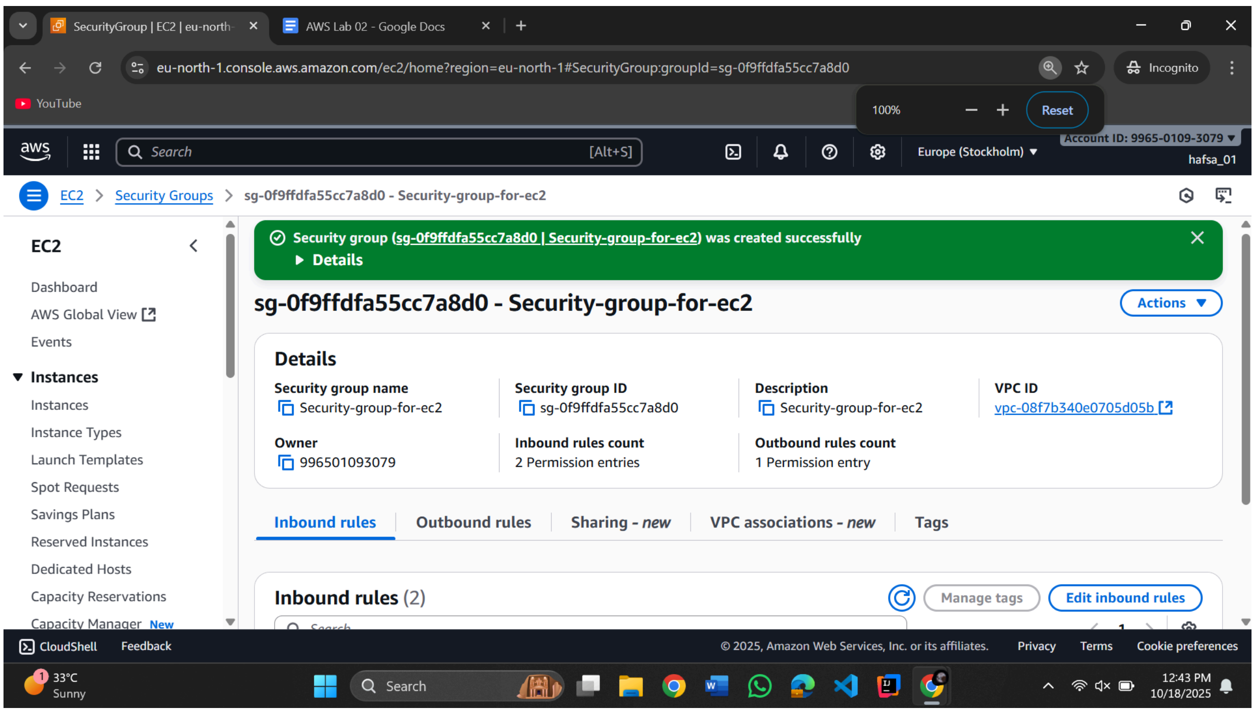


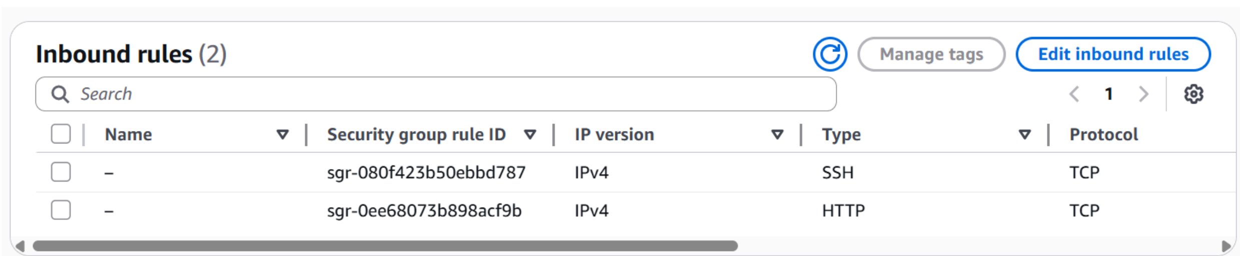


//Source: Anywhere Ipv4

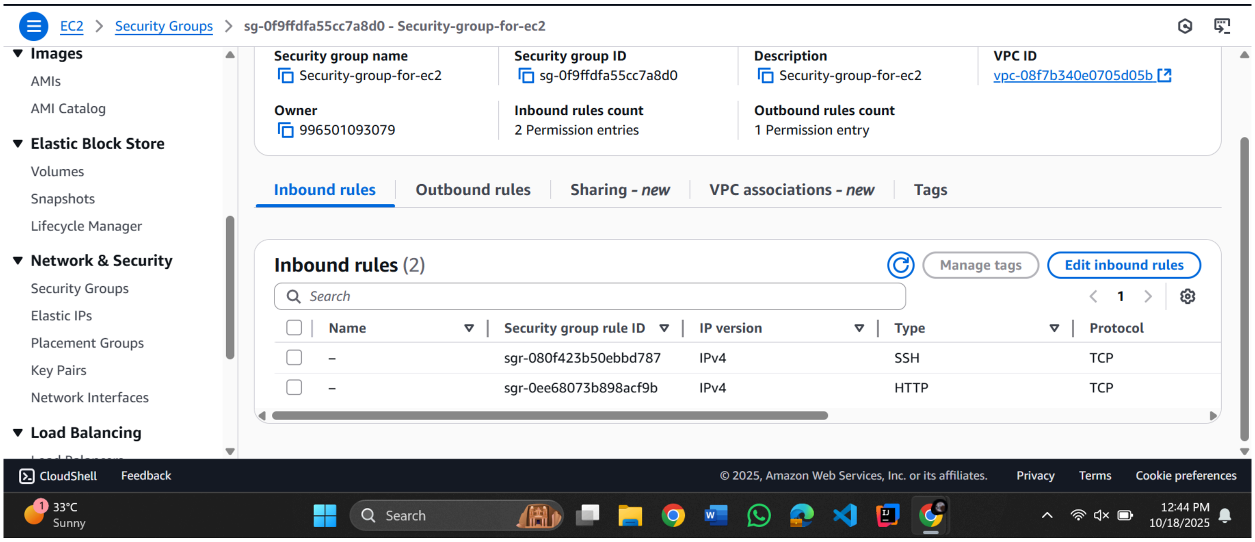


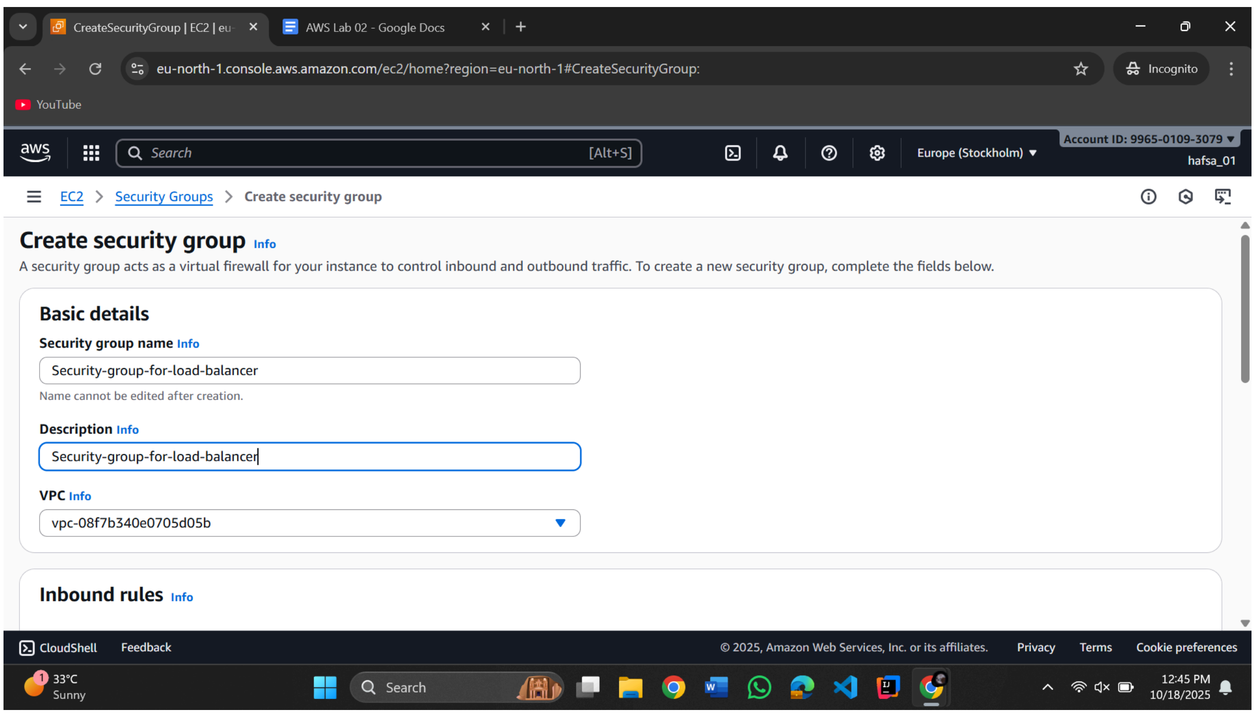
//Create Security Group

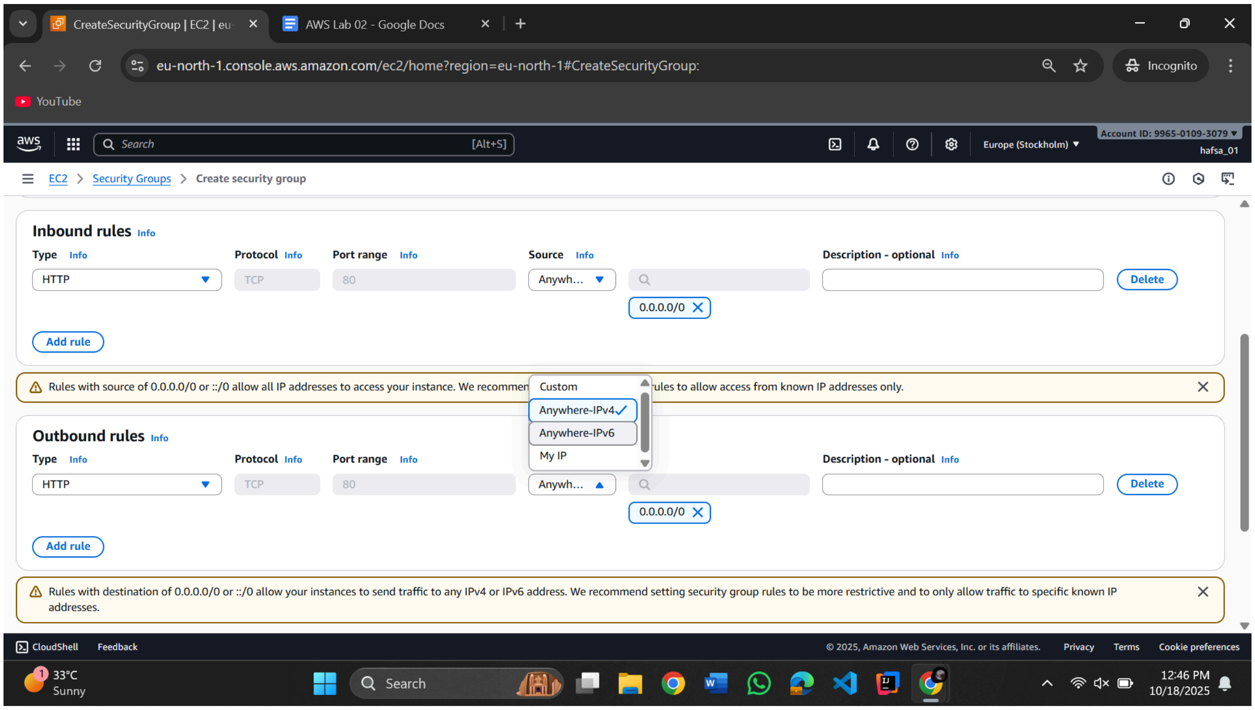




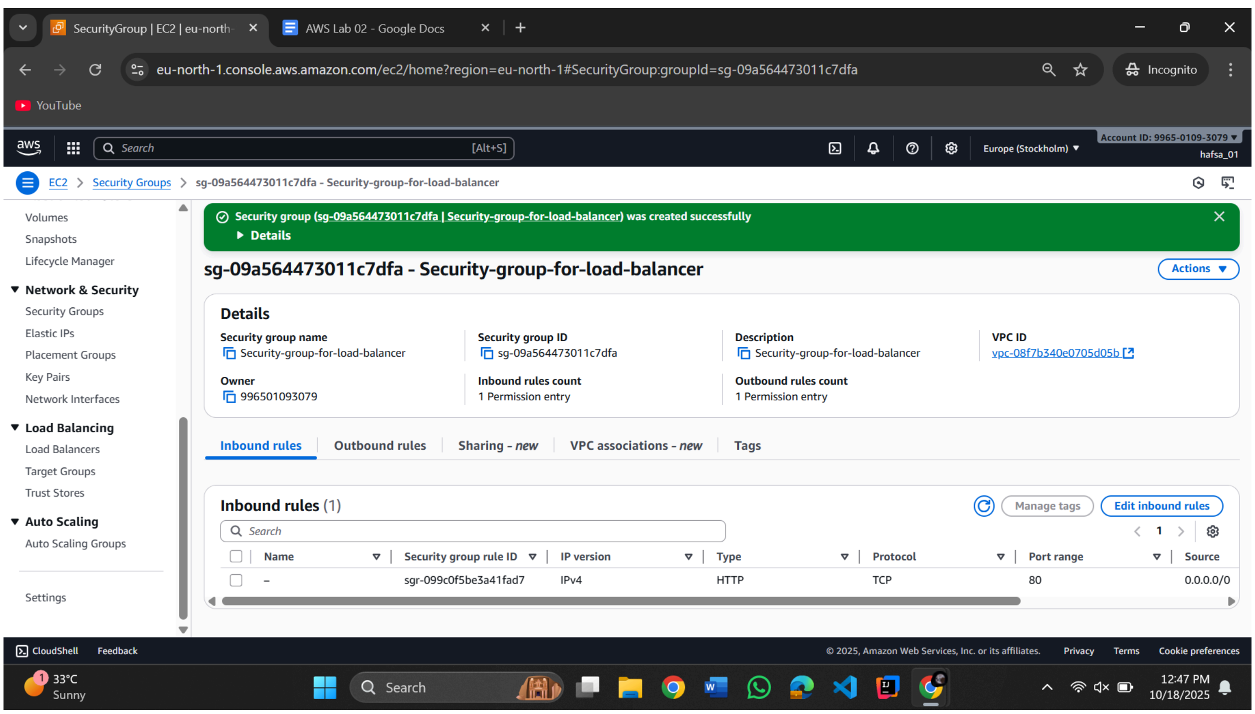
//Create another Security group


EC2 > Security Groups > sg-0f9ffdfa55cc7a8d0 - Security-group-for-ec2
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Security group name
Security-group-for-ec2
Security group ID
sg-0f9ffdfa55cc7a8d0
Description
Security-group-for-ec2
VPC ID
vpc-08f7b340e0705d05b
Owner
996501093079
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing - new
VPC associations - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
–
sgr-080f423b50ebbd787
IPv4
SSH
TCP
–
sgr-0ee68073b898acf9b
IPv4
HTTP
TCP
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
12:44 PM
10/18/2025


CreateSecurityGroup | EC2 | eu-
AWS Lab 02 - Google Docs
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateSecurityGroup:
Incognito
YouTube
aws
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Security Groups > Create security group
Create security group Info
A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.
Basic details
Security group name Info
Security-group-for-load-balancer
Name cannot be edited after creation.
Description Info
Security-group-for-load-balancer
VPC Info
vpc-08f7b340e0705d05b
Inbound rules Info
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
12:45 PM
10/18/2025

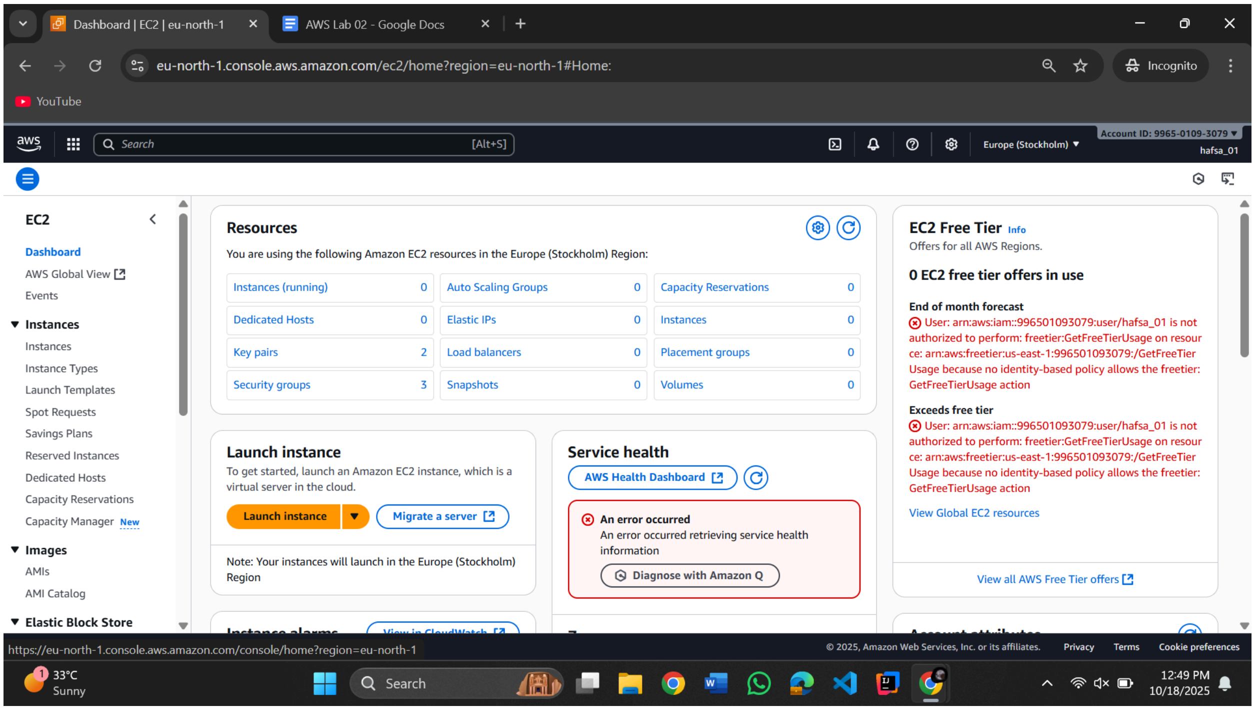


(Source: Anywhere-IPv4)

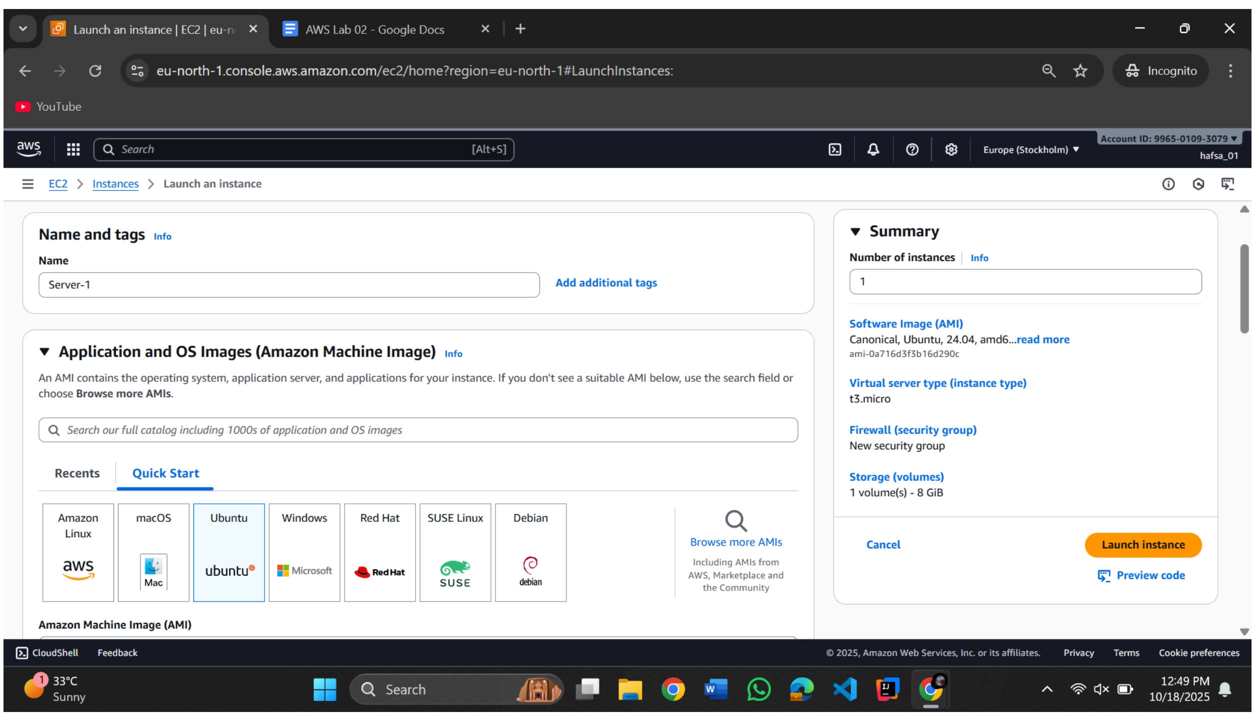


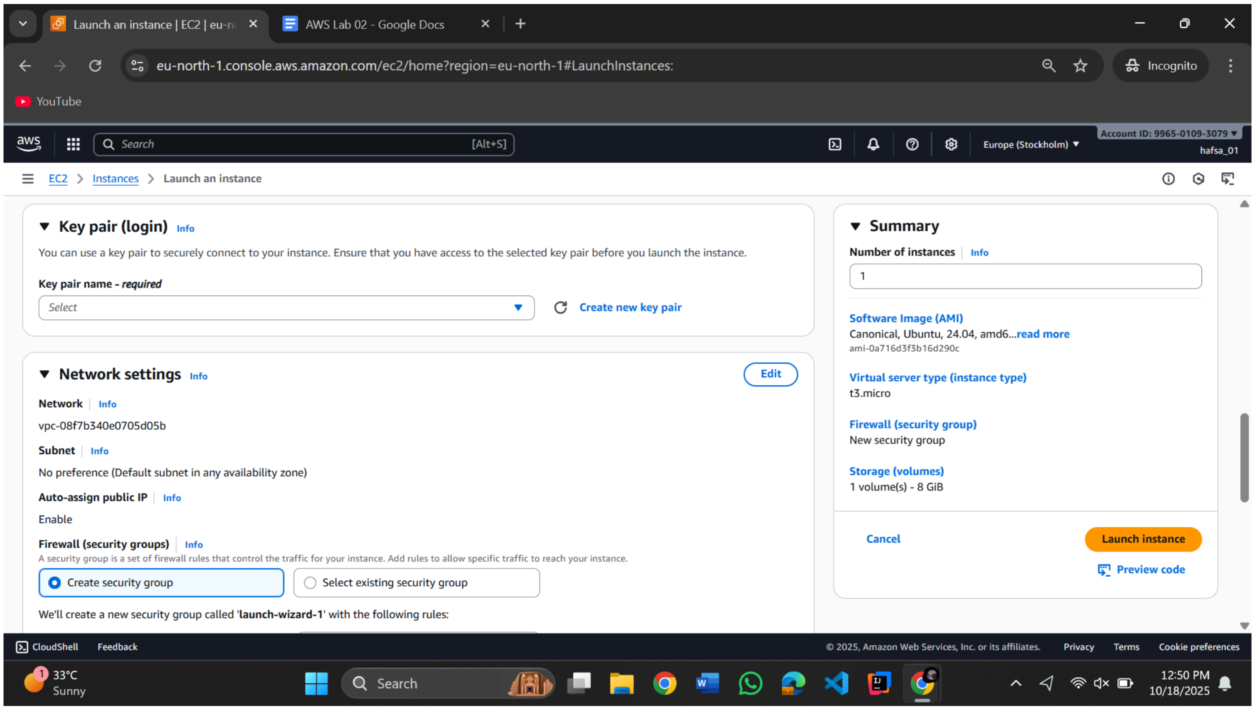
//Make a target group (that has 2 EC2)

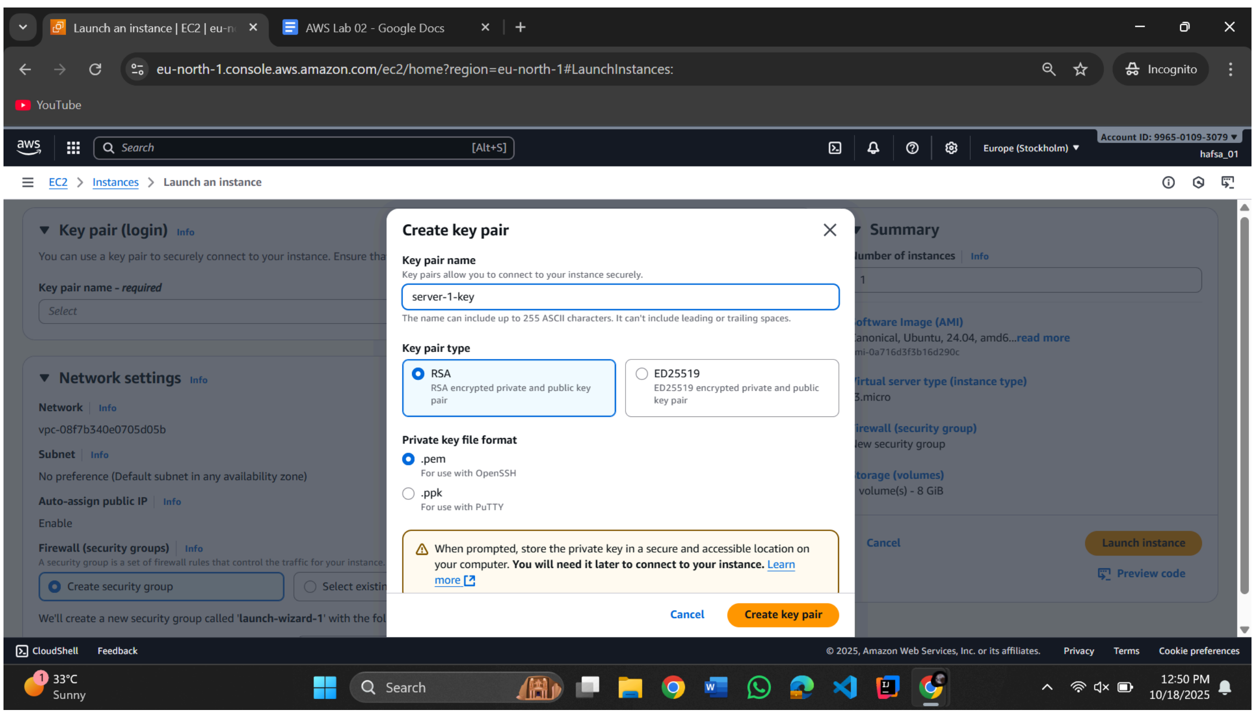
//Making EC2

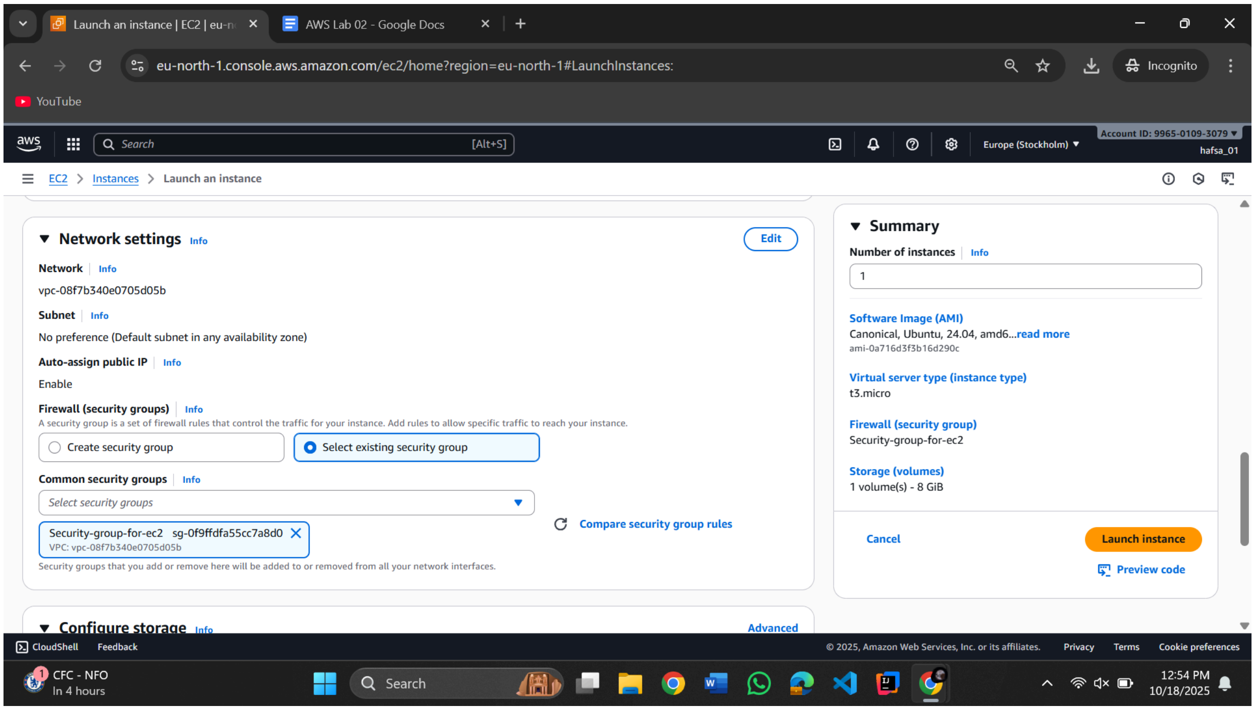


(Launch Instance)




Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Select
Create new key pair
Network settings Info
Edit
Network Info
vpc-08f7b340e0705d05b
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-1' with the following rules:
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0a716d3f3b16d290c
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
12:50 PM
10/18/2025

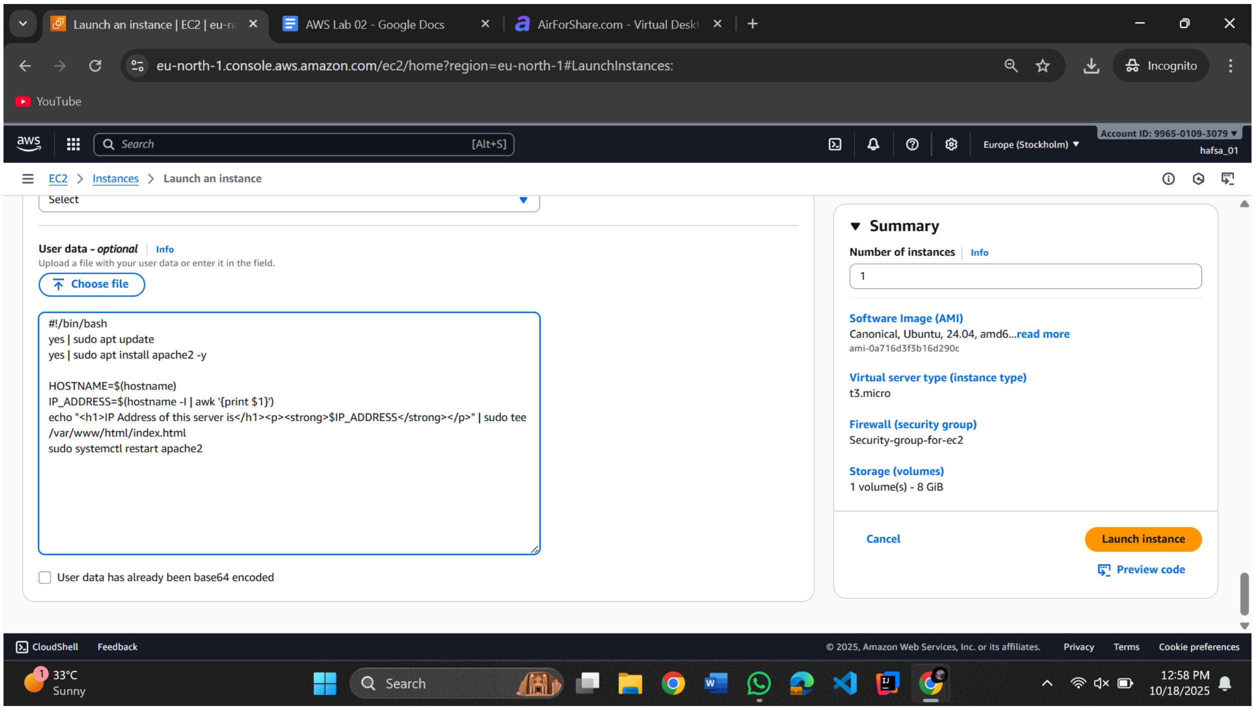

Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
server-1-key
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
12:50 PM
10/18/2025

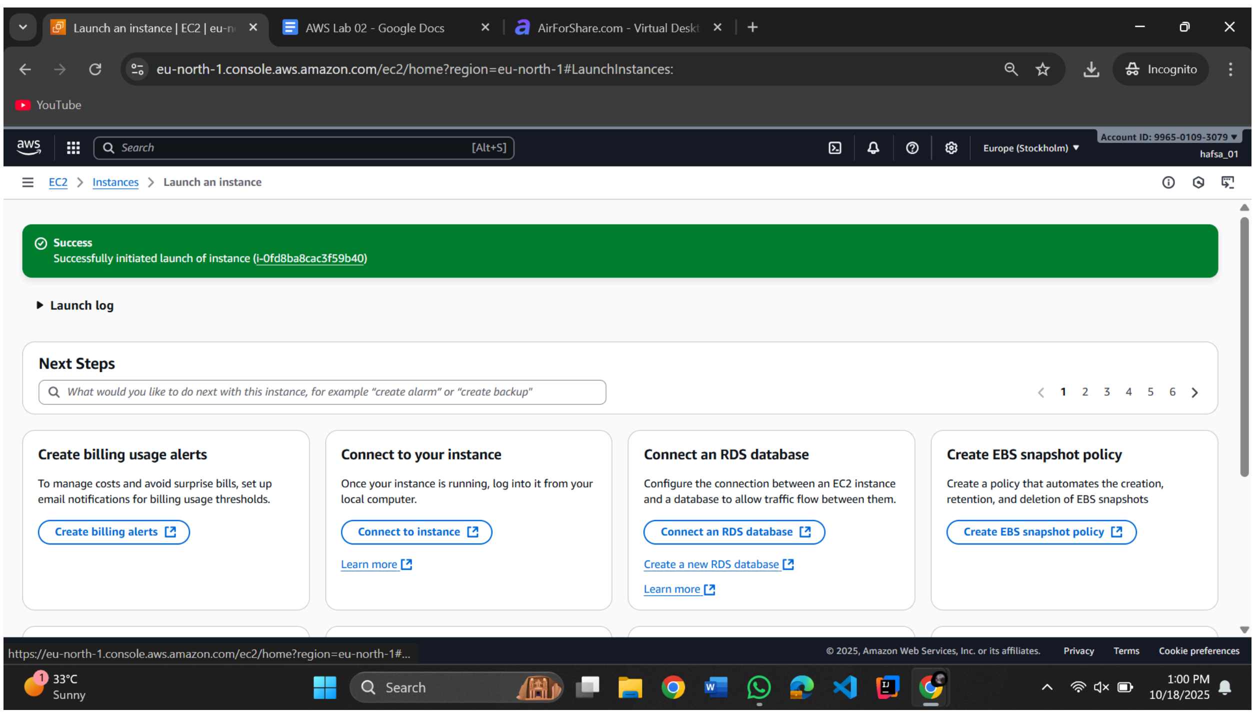


//Code

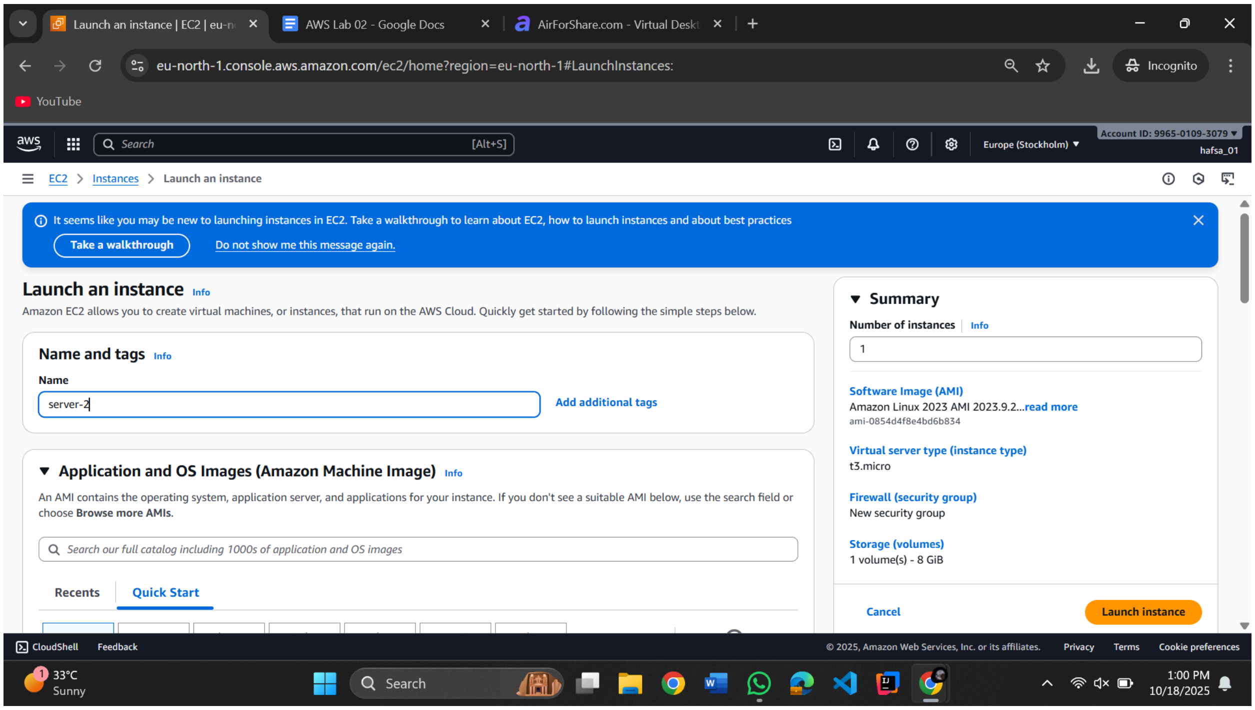
```
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2 -y

HOSTNAME=$(hostname)
IP_ADDRESS=$(hostname -I | awk '{print $1}')
echo "<h1>IP Address of this server is</h1><p><strong>$IP_ADDRESS</strong></p>" | sudo tee /var/www/html/index.html
sudo systemctl restart apache2
```

//In advance detail -> Copy paste this above code

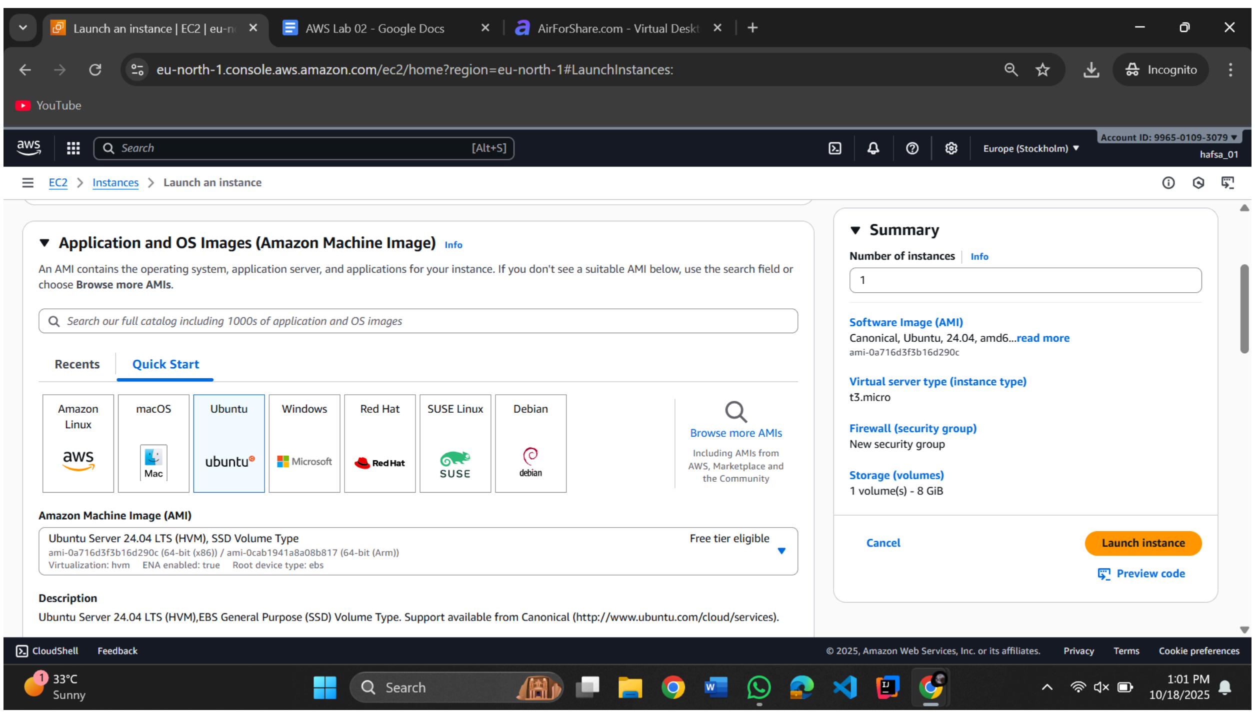




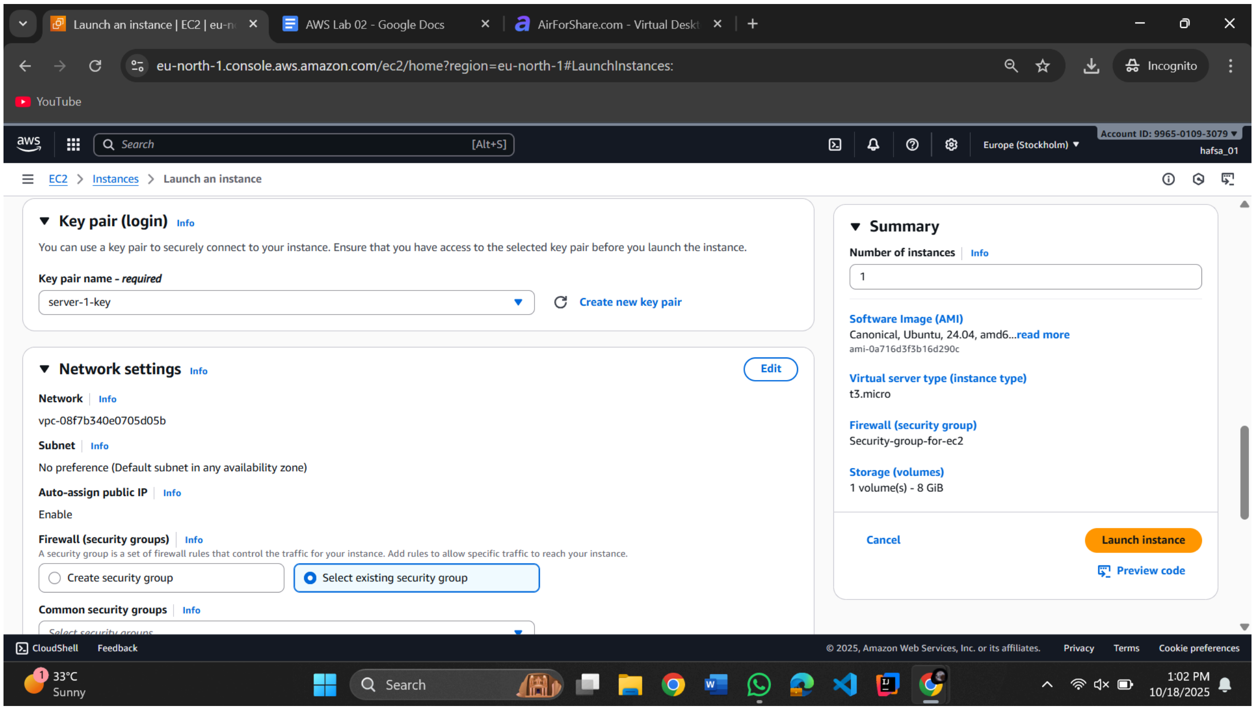
//Making Second EC2

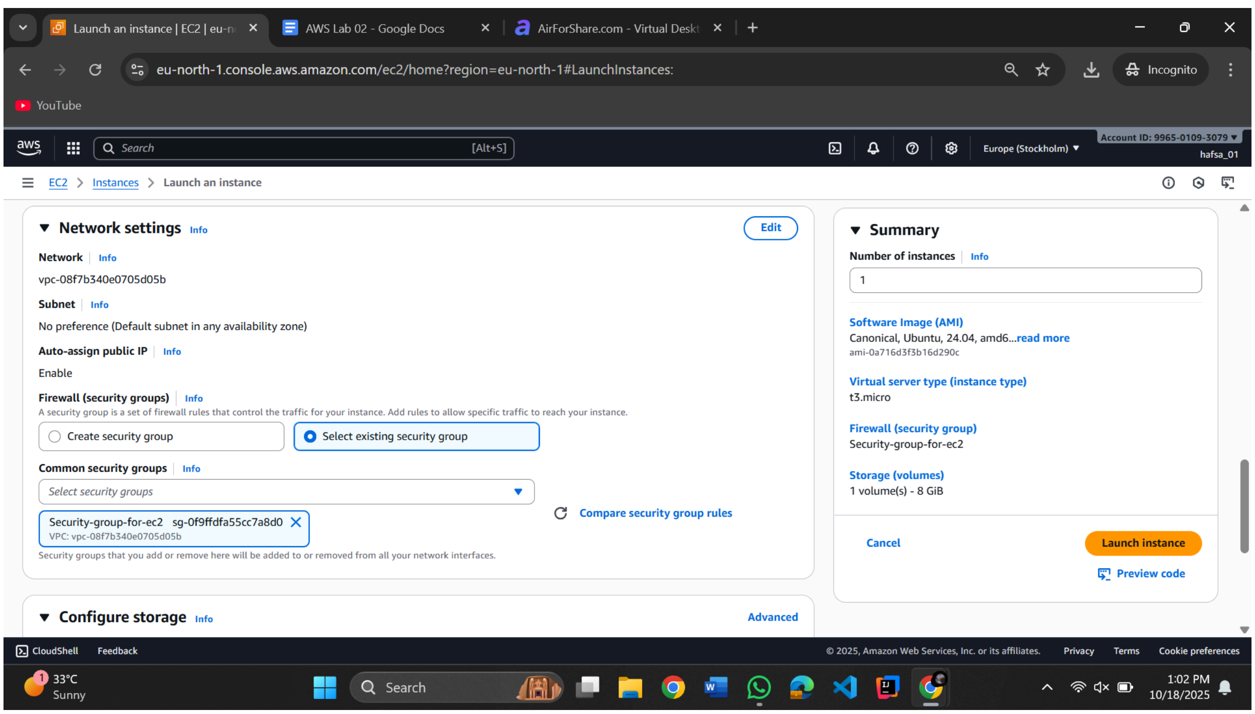


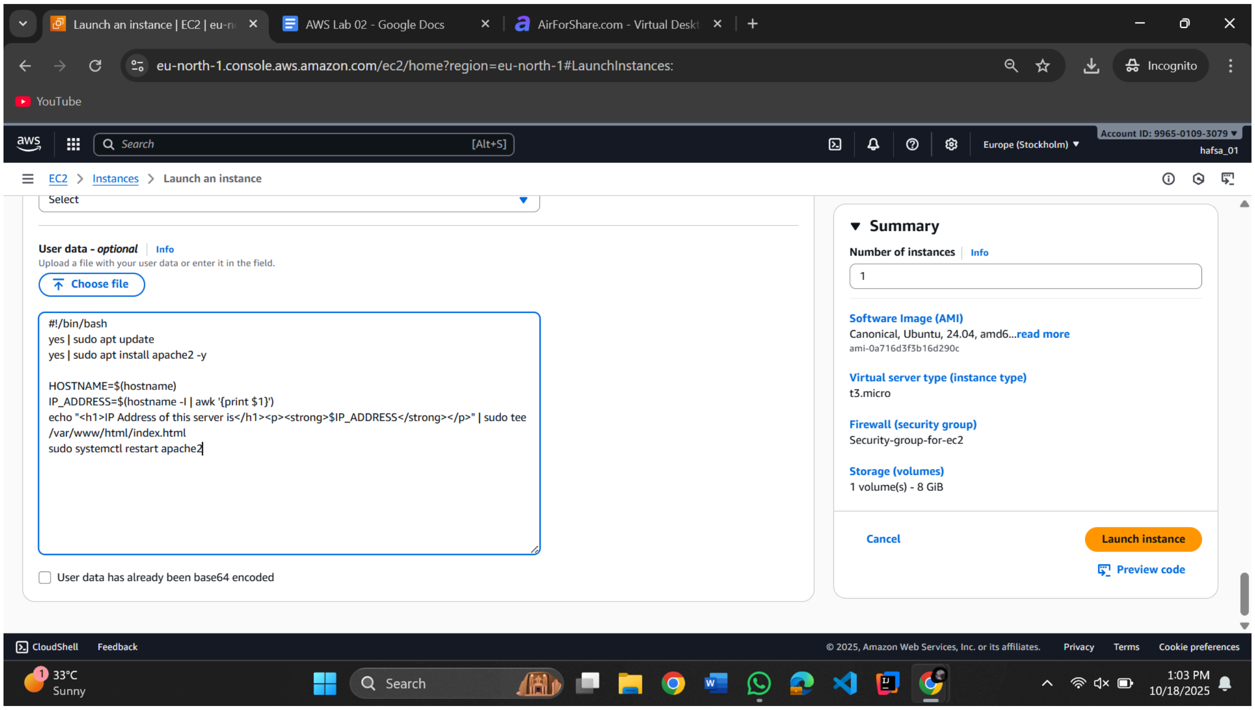
Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Desk
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
aws
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
server-2
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-0854d4f8e4bd6b834
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:00 PM
10/18/2025

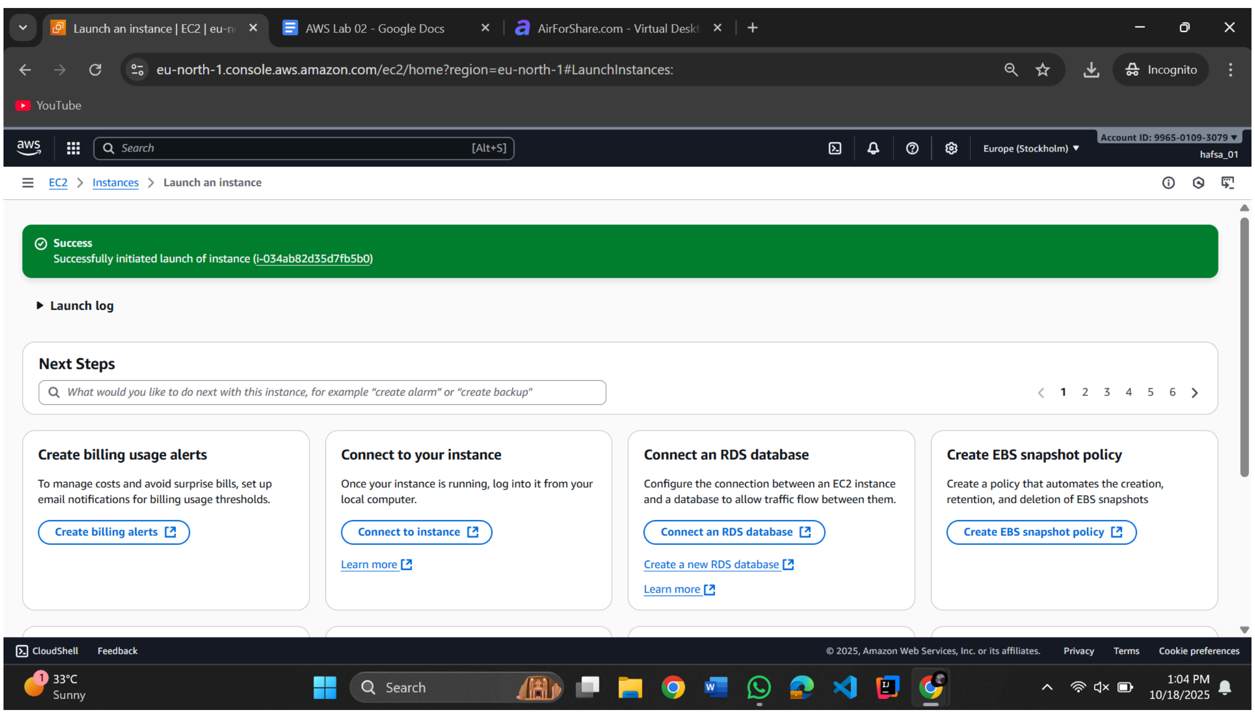


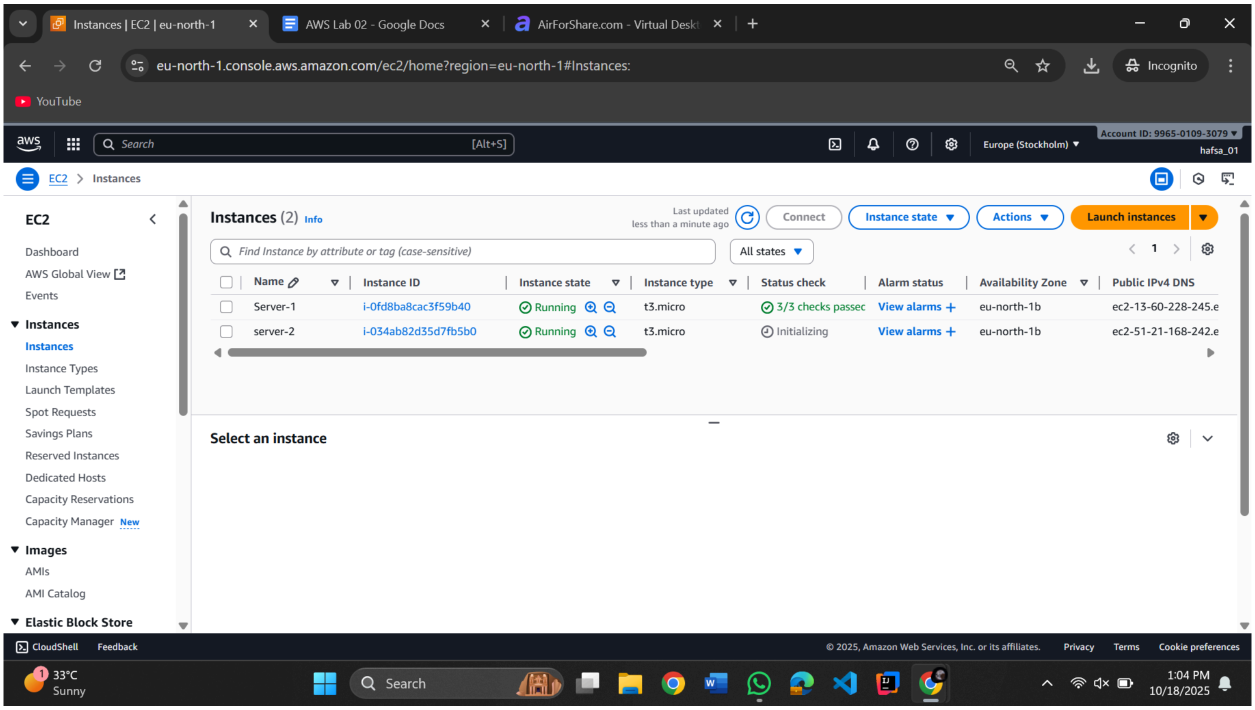
Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Desk
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
aws
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
Free tier eligible
ami-0a716d3f3b16d290c (64-bit (x86)) / ami-0cab1941a8a08b817 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0a716d3f3b16d290c
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:01 PM
10/18/2025


Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Deskt
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search [Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
server-1-key
Create new key pair
Network settings Info
Edit
Network Info
vpc-08f7b340e0705d05b
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0a716d3f3b16d290c
Virtual server type (instance type)
t3.micro
Firewall (security group)
Security-group-for-ec2
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:02 PM
10/18/2025

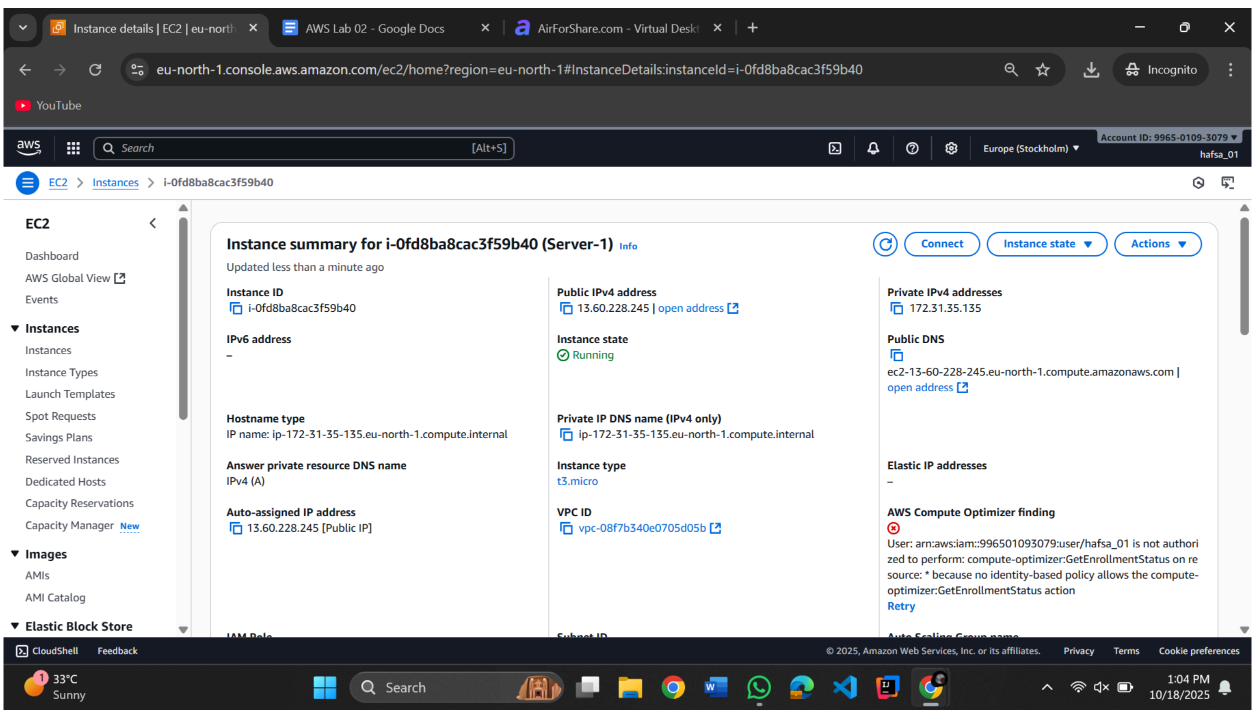

Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Deskt
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search [Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Network settings Info
Edit
Network Info
vpc-08f7b340e0705d05b
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
Security-group-for-ec2 sg-0f9ffdfa55cc7a8d0
VPC: vpc-08f7b340e0705d05b
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Configure storage Info
Advanced
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0a716d3f3b16d290c
Virtual server type (instance type)
t3.micro
Firewall (security group)
Security-group-for-ec2
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:02 PM
10/18/2025

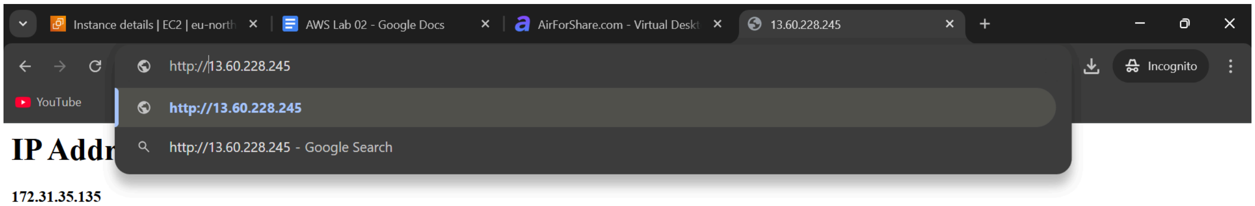

Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Deskt
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Select
User data - optional Info
Upload a file with your user data or enter it in the field.
Choose file
#!/bin/bash
yes | sudo apt update
yes | sudo apt install apache2 -y
HOSTNAME=$(hostname)
IP_ADDRESS=$(hostname -I | awk '{print $1}')
echo "<h1>IP Address of this server is</h1><p>$IP_ADDRESS</p>" | sudo tee
/var/www/html/index.html
sudo systemctl restart apache2
User data has already been base64 encoded
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-0a716d3f3b16d290c
Virtual server type (instance type)
t3.micro
Firewall (security group)
Security-group-for-ec2
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:03 PM
10/18/2025


Launch an instance | EC2 | eu-n
AWS Lab 02 - Google Docs
AirForShare.com - Virtual Deskt
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-034ab82d35d7fb5b0)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
1 2 3 4 5 6
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:04 PM
10/18/2025

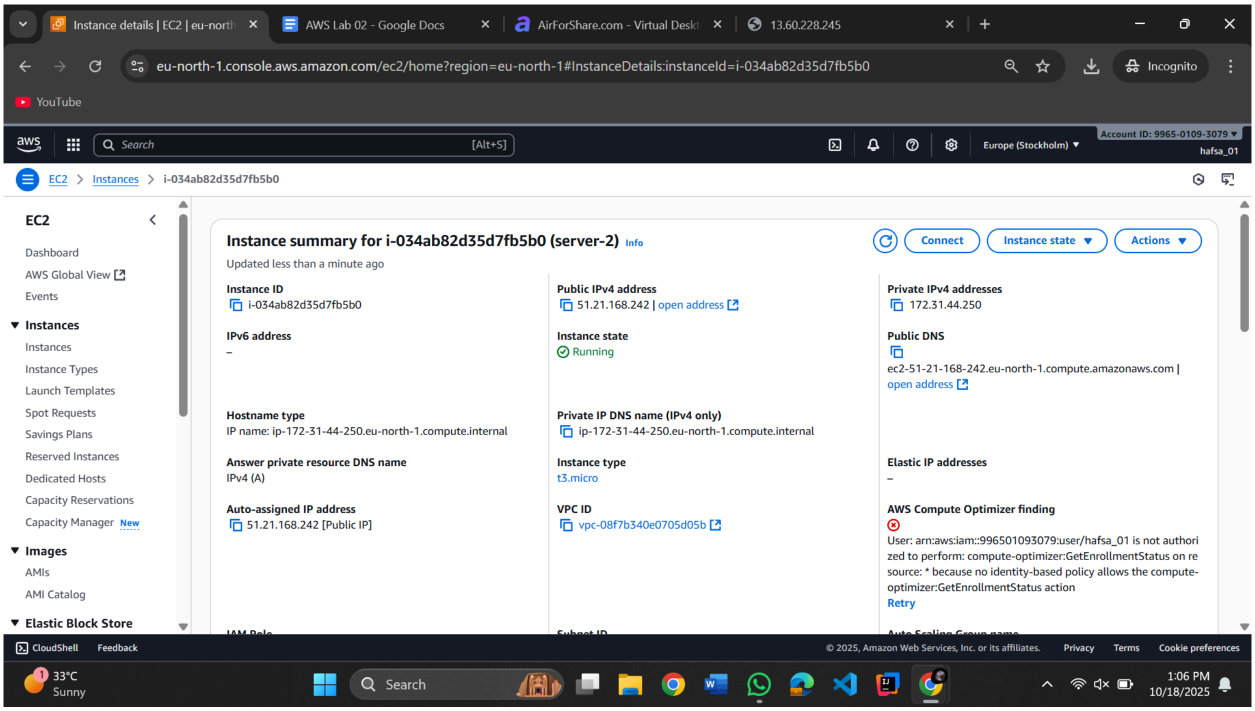


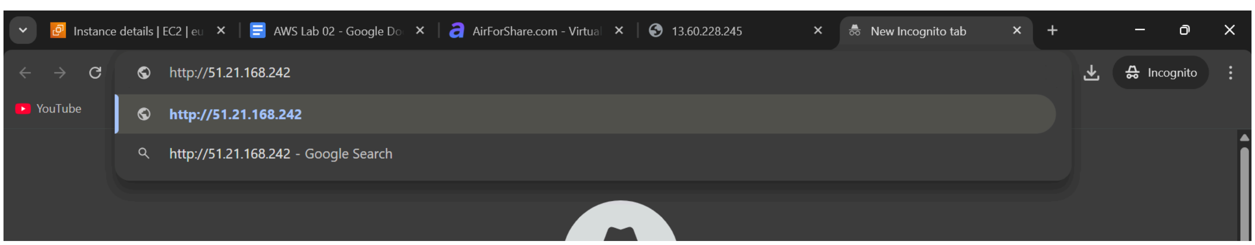
//Open server-01 and copy public IPv4

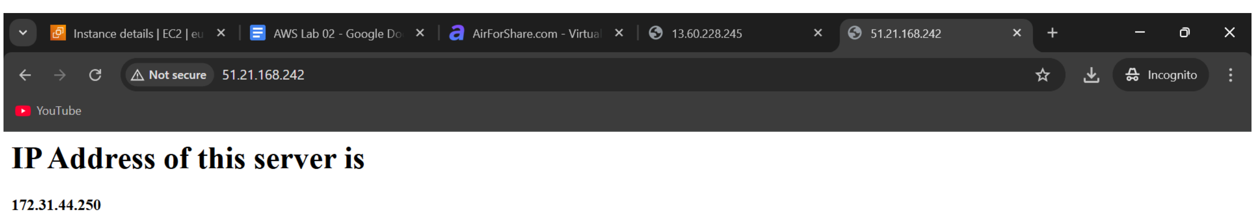




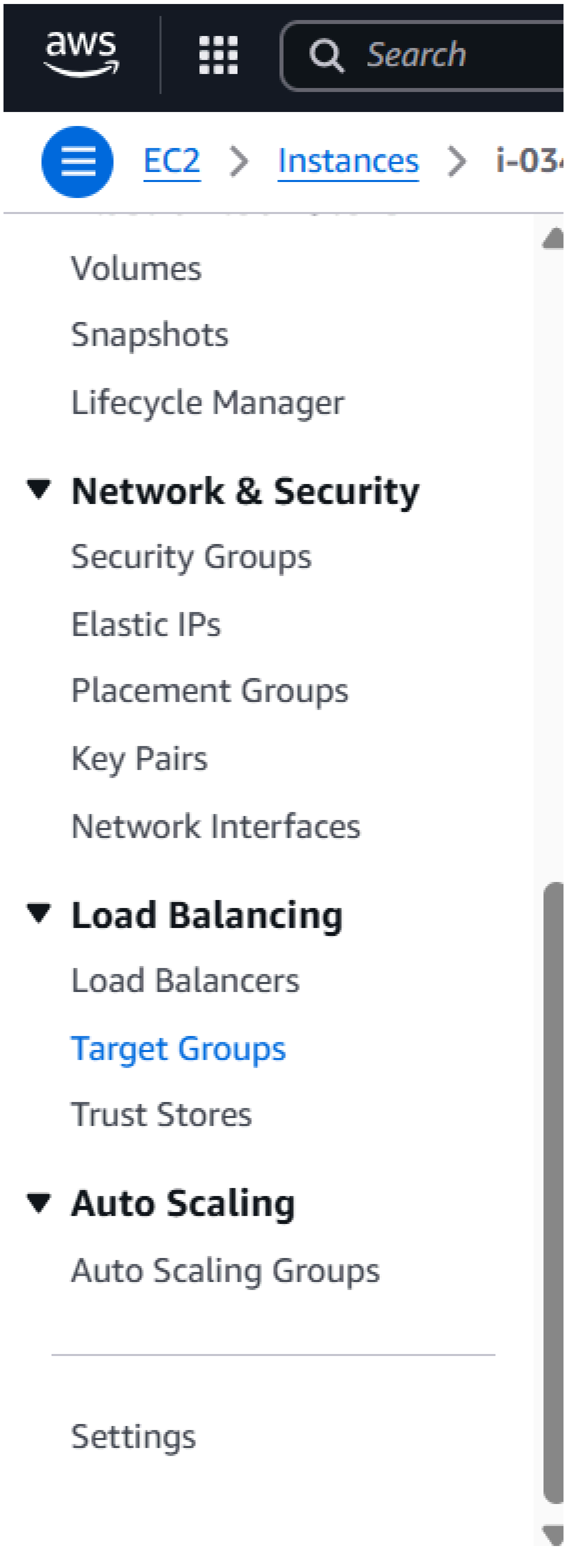
//Do the same with server-2 -> Copy public IPv4



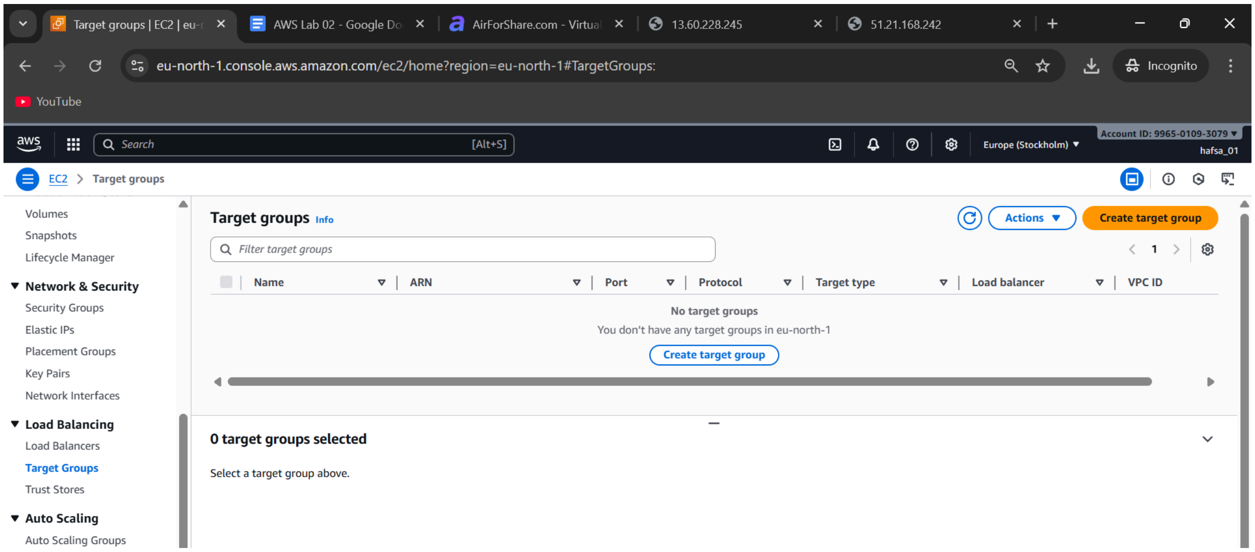
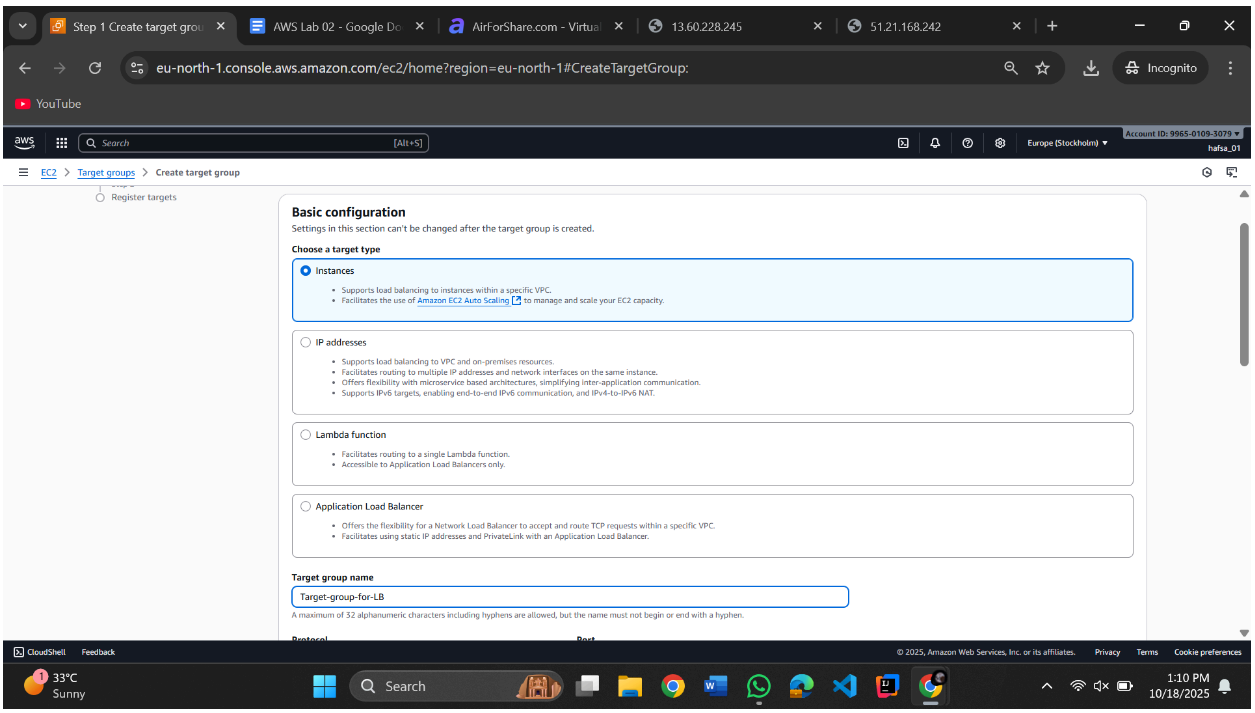


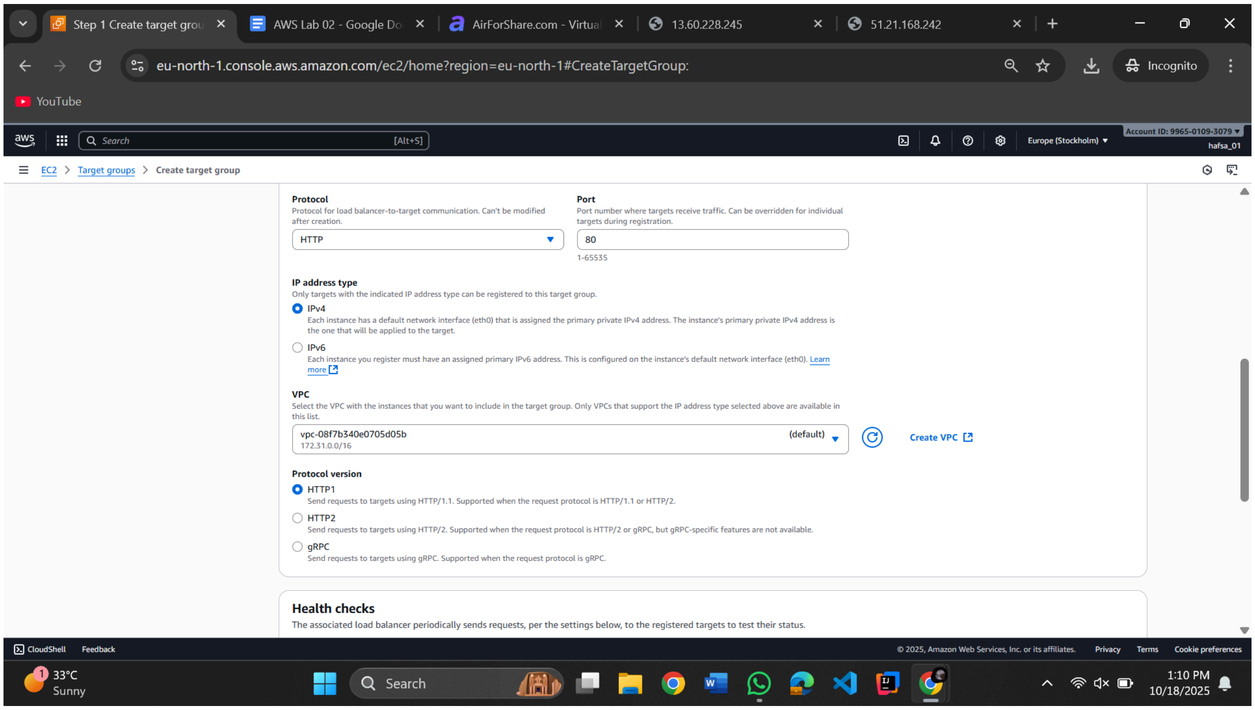


//Making target group now

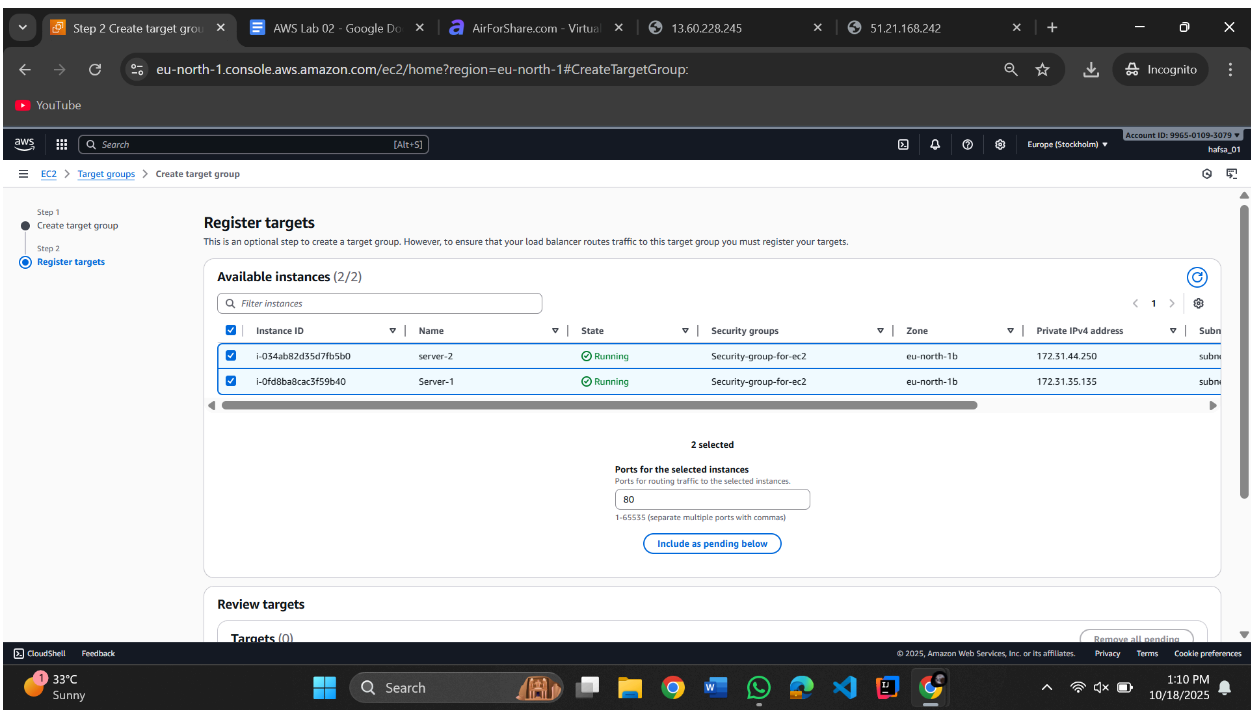


//Create target group

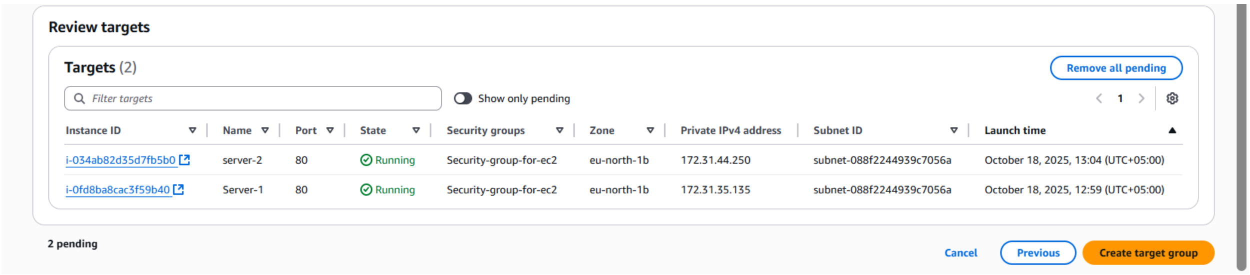

Target groups | EC2 | eu-
AWS Lab 02 - Google Do
AirForShare.com - Virtual
13.60.228.245
51.21.168.242
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:
Incognito
YouTube
Search
[Alt+S]
Account ID: 9965-0109-3079
Europe (Stockholm)
hafsa_01
EC2 > Target groups
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Target groups Info
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
Load balancer
VPC ID
No target groups
You don't have any target groups in eu-north-1
Create target group
0 target groups selected
Select a target group above.

Step 1 Create target grou
AWS Lab 02 - Google Do
AirForShare.com - Virtual
13.60.228.245
51.21.168.242
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateTargetGroup:
Incognito
YouTube
EC2 > Target groups > Create target group
Register targets
Basic configuration
Settings in this section can't be changed after the target group is created.
Choose a target type
Instances
Supports load balancing to instances within a specific VPC.
Facilitates the use of Amazon EC2 Auto Scaling to manage and scale your EC2 capacity.
IP addresses
Supports load balancing to VPC and on-premises resources.
Facilitates routing to multiple IP addresses and network interfaces on the same instance.
Offers flexibility with microservice based architectures, simplifying inter-application communication.
Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.
Lambda function
Facilitates routing to a single Lambda function.
Accessible to Application Load Balancers only.
Application Load Balancer
Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.
Target group name
Target-group-for-LB
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
33°C
Sunny
Search
1:10 PM
10/18/2025

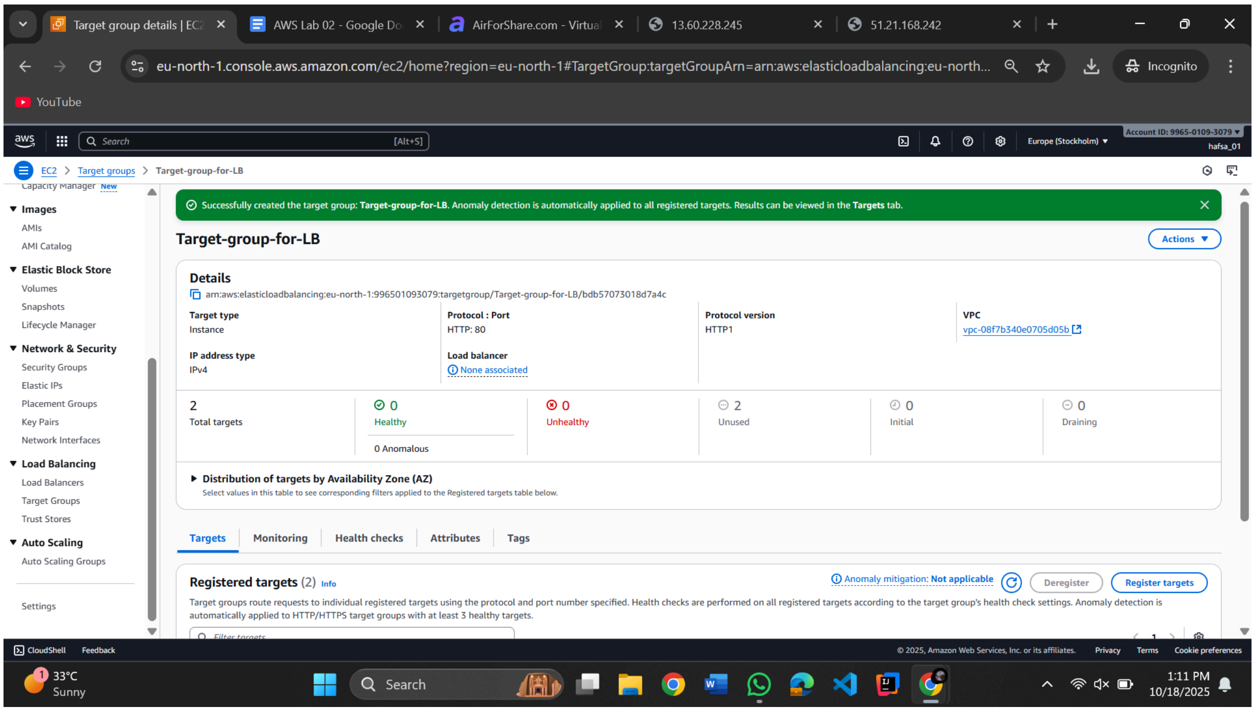


//Click next

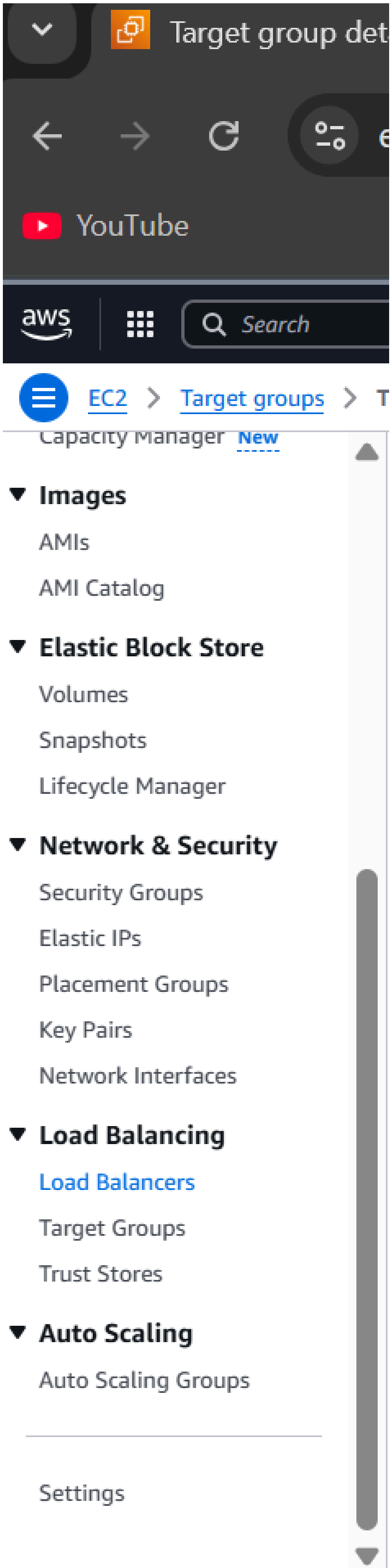


Click include as pending below (Port should be 80)

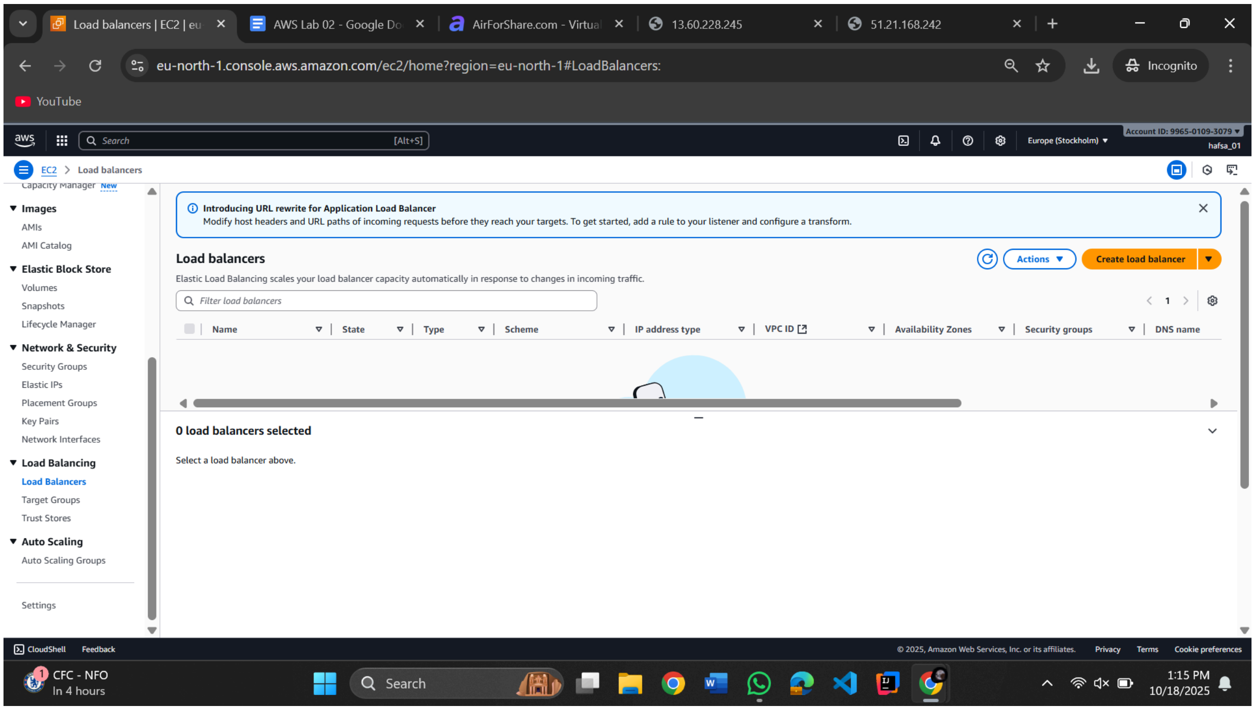




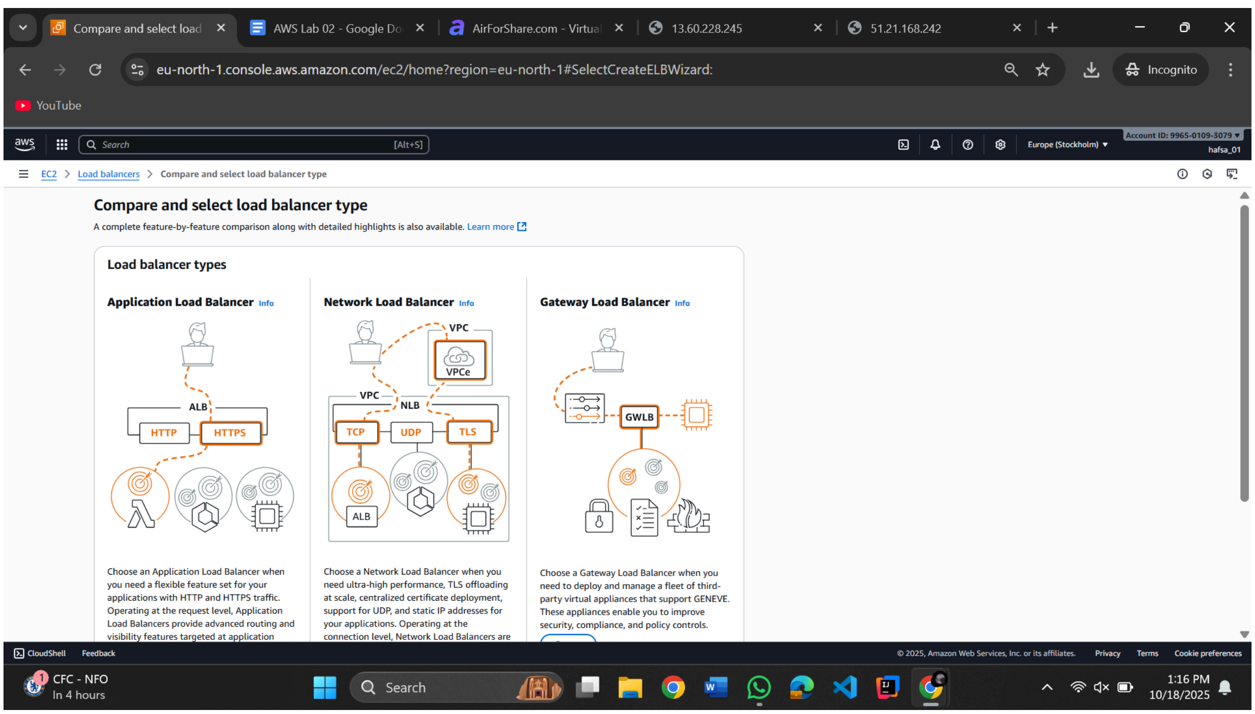
//Open Load balancer

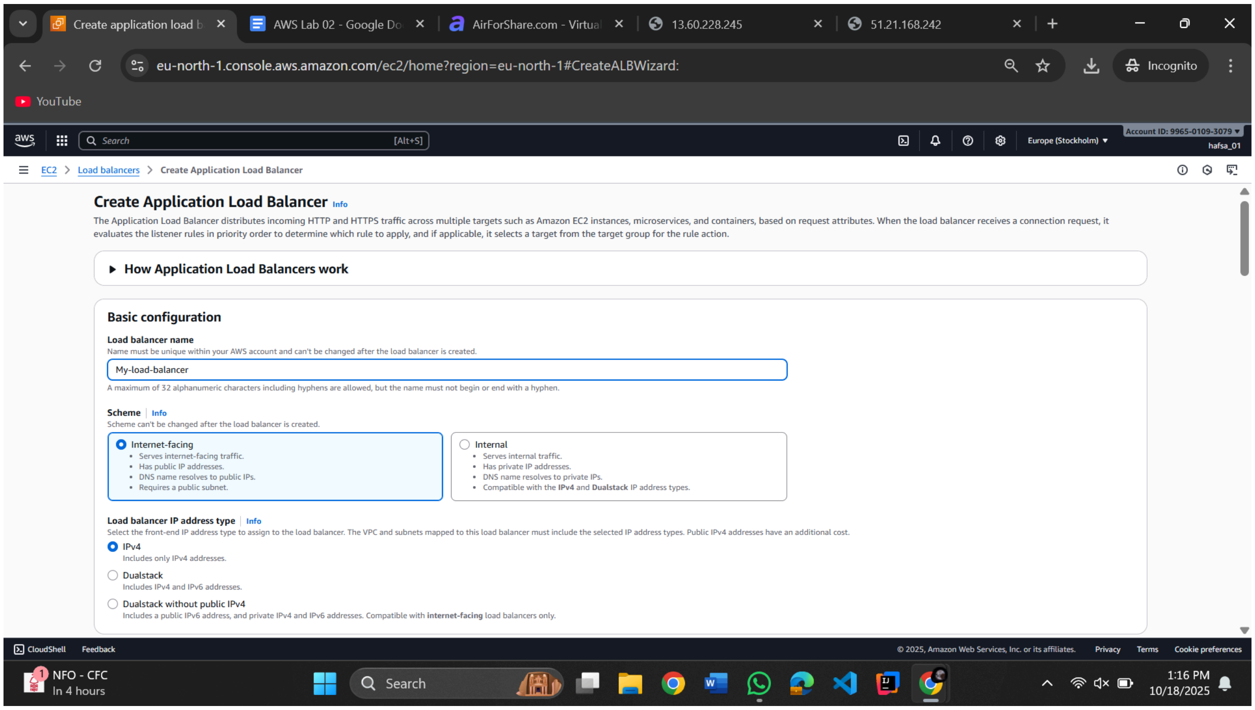

Target group det
YouTube
aws
Search
EC2
Target groups
T
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings

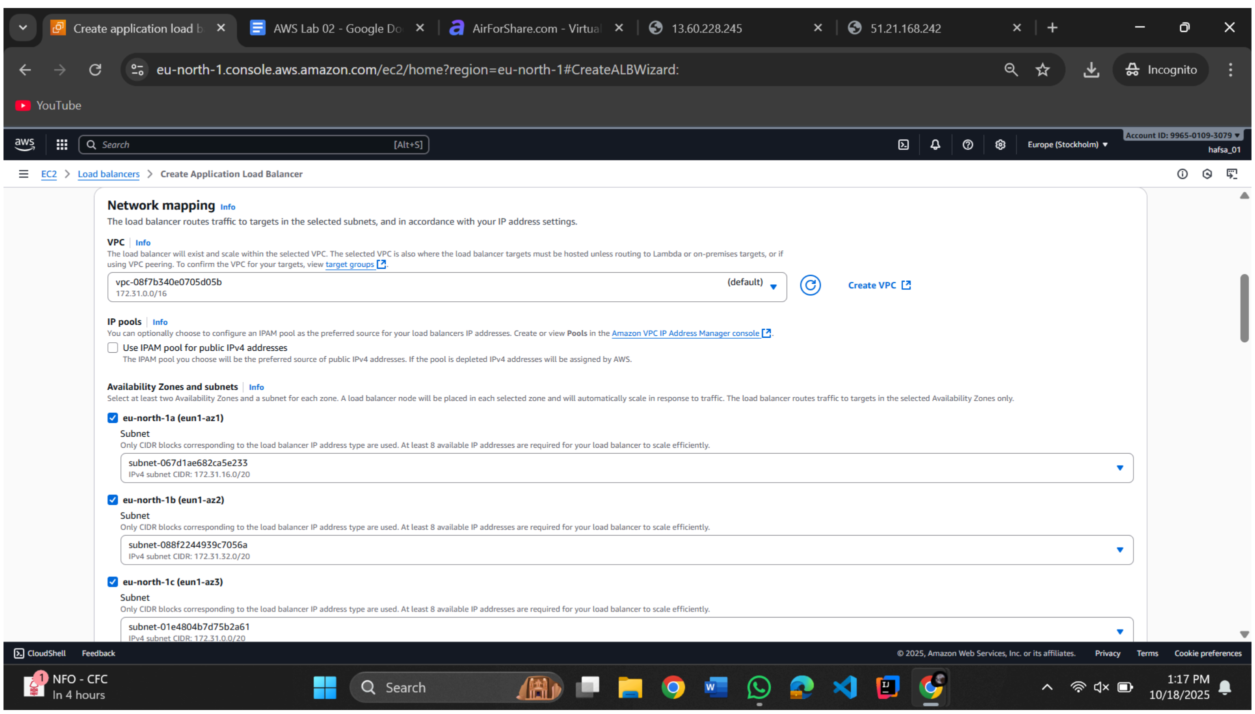
Create load balancer

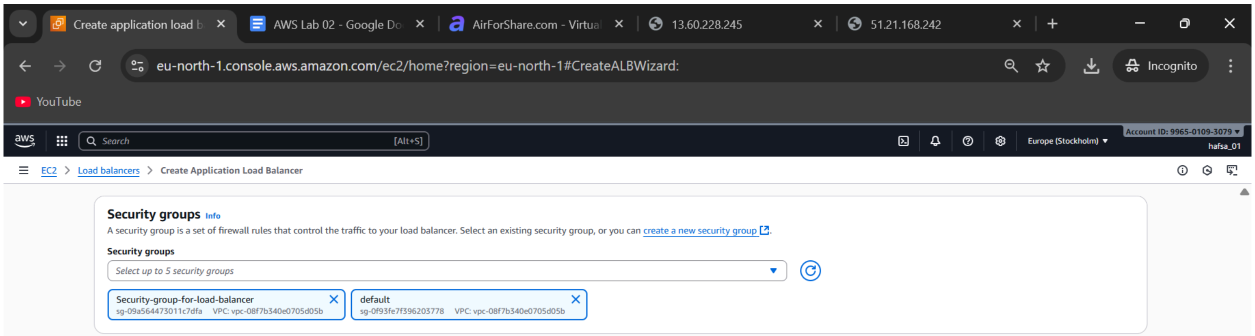


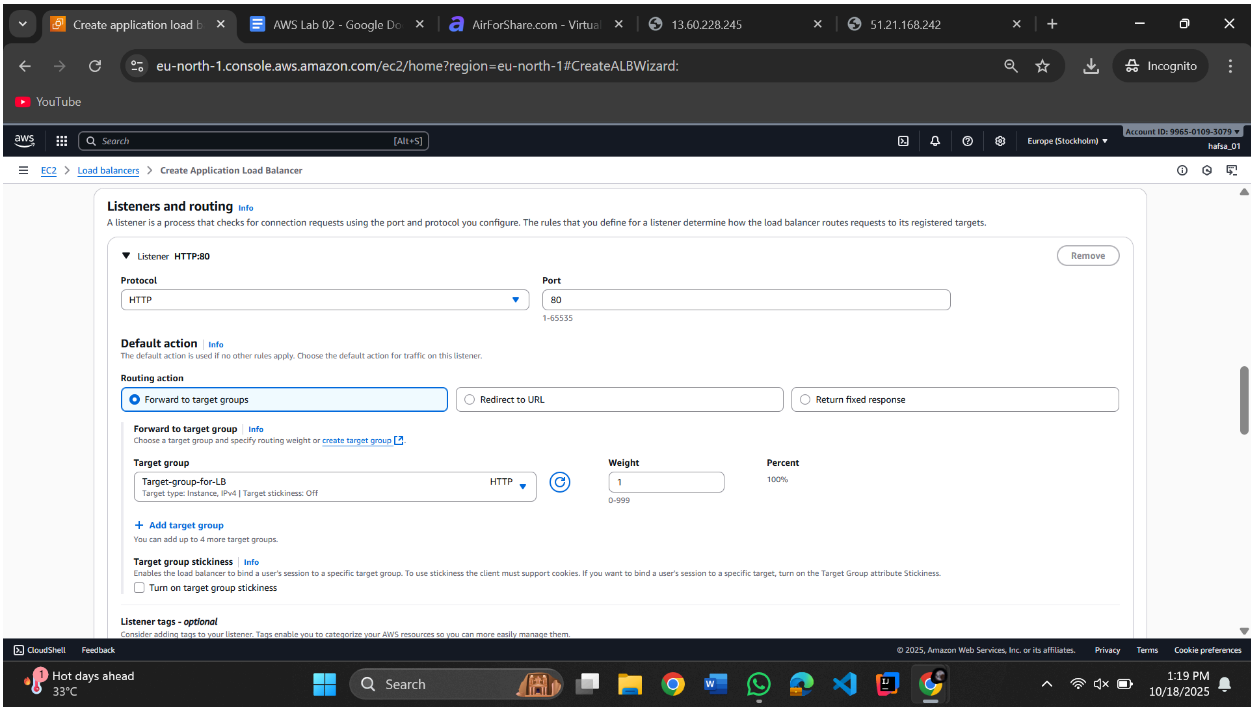
//Create application balancer -> Click



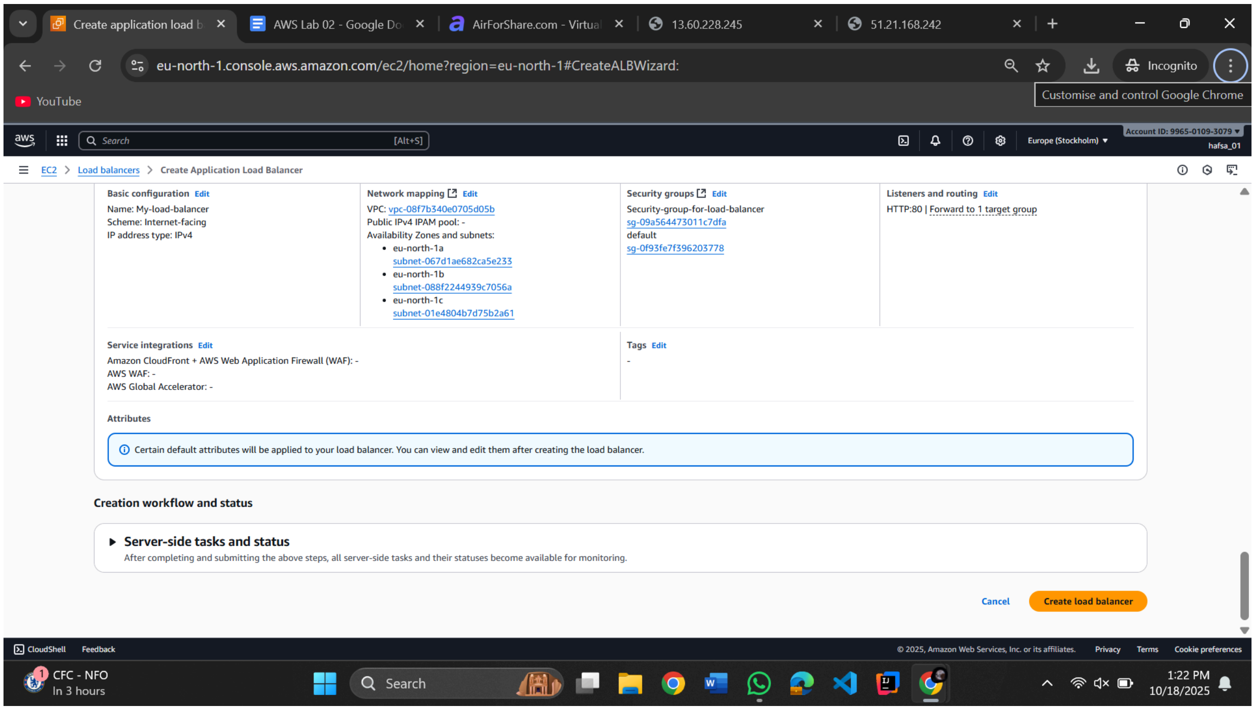

Create application load b
AWS Lab 02 - Google Do
AirForShare.com - Virtual
13.60.228.245
51.21.168.242
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateALBWizard:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Load balancers > Create Application Load Balancer
Create Application Load Balancer Info
The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.
How Application Load Balancers work
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
My-load-balancer
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
DNS name resolves to public IPs.
Requires a public subnet.
Internal
Serves internal traffic.
Has private IP addresses.
DNS name resolves to private IPs.
Compatible with the IPv4 and Dualstack IP address types.
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
NFO - CFC
In 4 hours
Search
1:16 PM
10/18/2025


Create application load b
AWS Lab 02 - Google Do
AirForShare.com - Virtual
13.60.228.245
51.21.168.242
eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#CreateALBWizard:
Incognito
YouTube
Search
[Alt+S]
Europe (Stockholm)
Account ID: 9965-0109-3079
hafsa_01
EC2 > Load balancers > Create Application Load Balancer
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups
vpc-08f7b340e0705d05b
172.31.0.0/16
(default)
Create VPC
IP pools Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the Amazon VPC IP Address Manager console.
Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.
Availability Zones and subnets Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.
eu-north-1a (eun1-az1)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-067d1ae682ca5e233
IPv4 subnet CIDR: 172.31.16.0/20
eu-north-1b (eun1-az2)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-088f2244939c7056a
IPv4 subnet CIDR: 172.31.32.0/20
eu-north-1c (eun1-az3)
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-01e4804b7d75b2a61
IPv4 subnet CIDR: 172.31.0.0/20
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
NFO - CFC
In 4 hours
Search
1:17 PM
10/18/2025

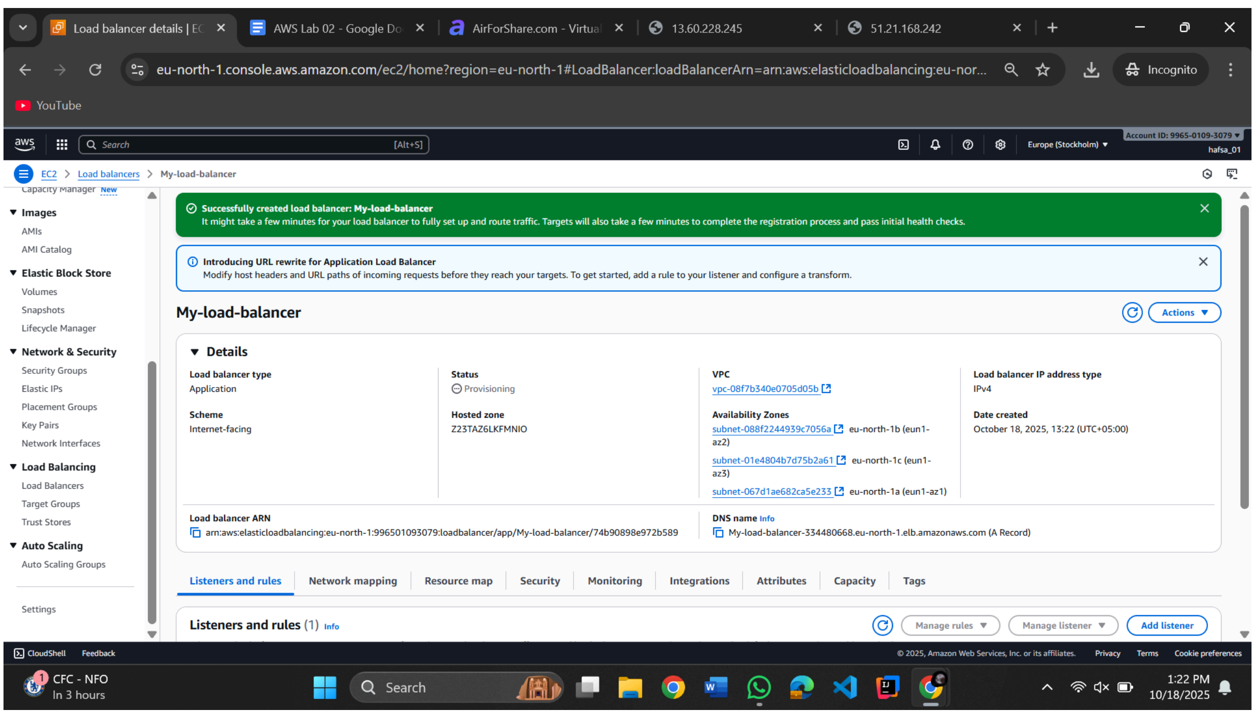




(Since we have one target group, we will choose the weight “1”)

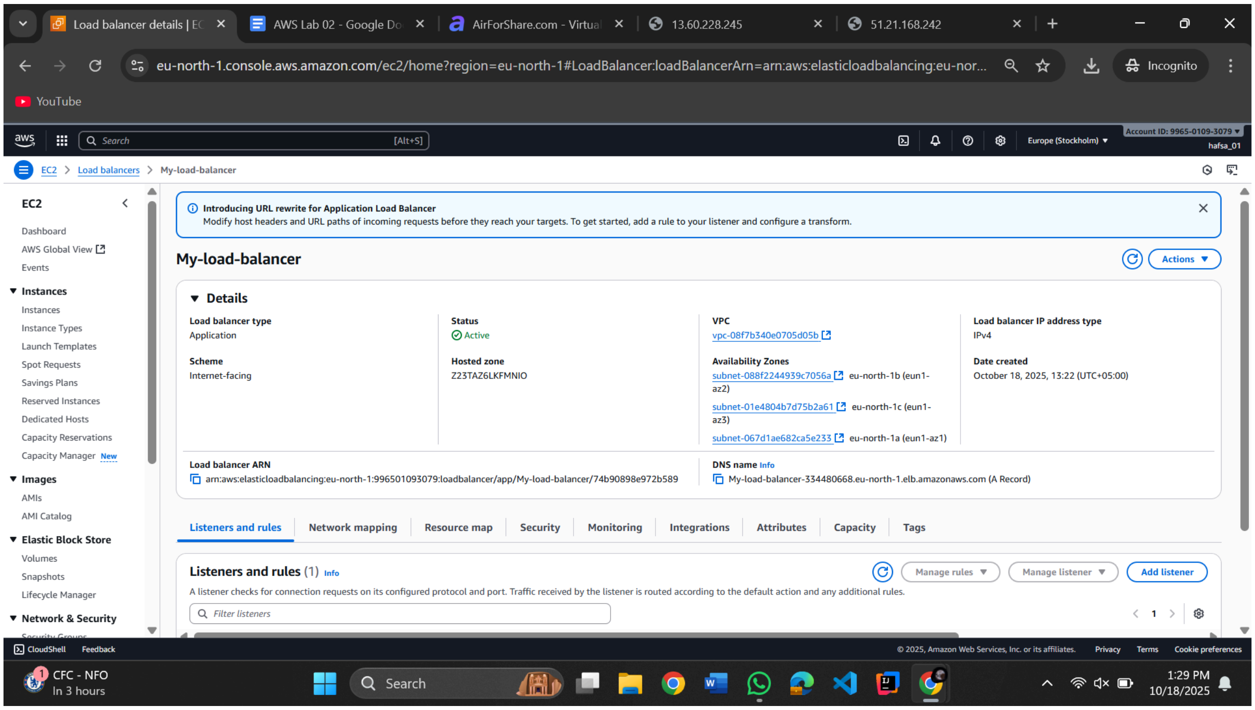


Create it!!!

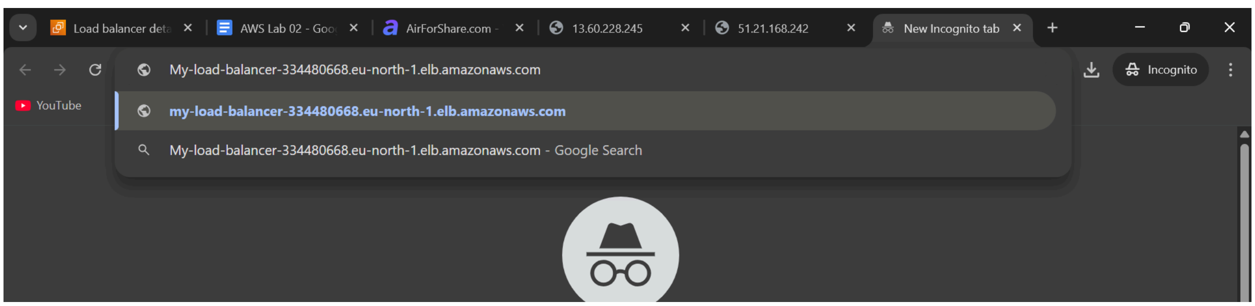


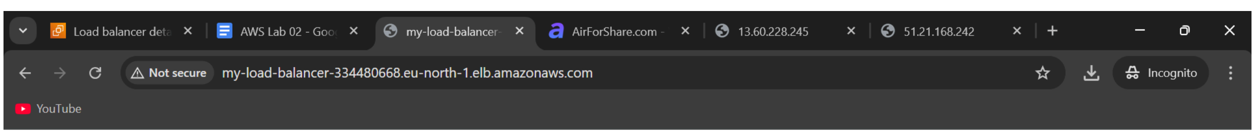
Whehehe

//Refresh (Status will change from provisioning to active)



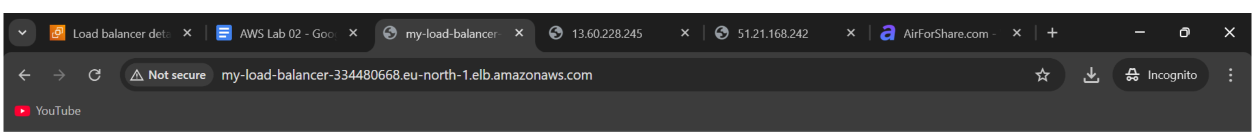
//Copy DNS Name





IP Address of this server is

172.31.44.250

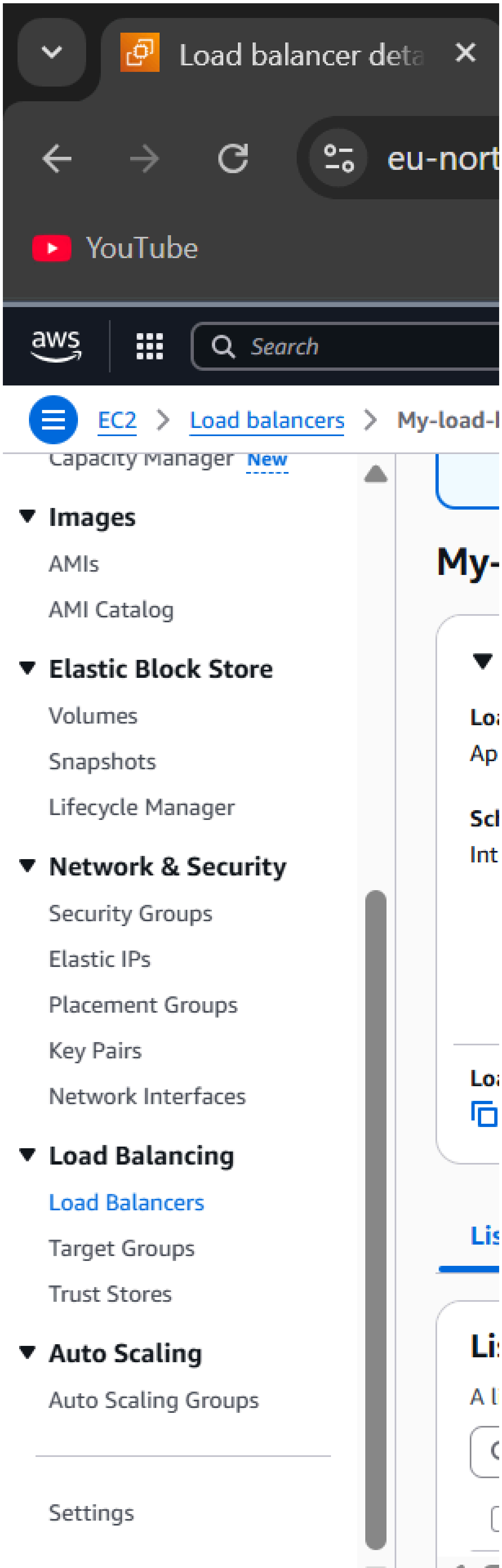


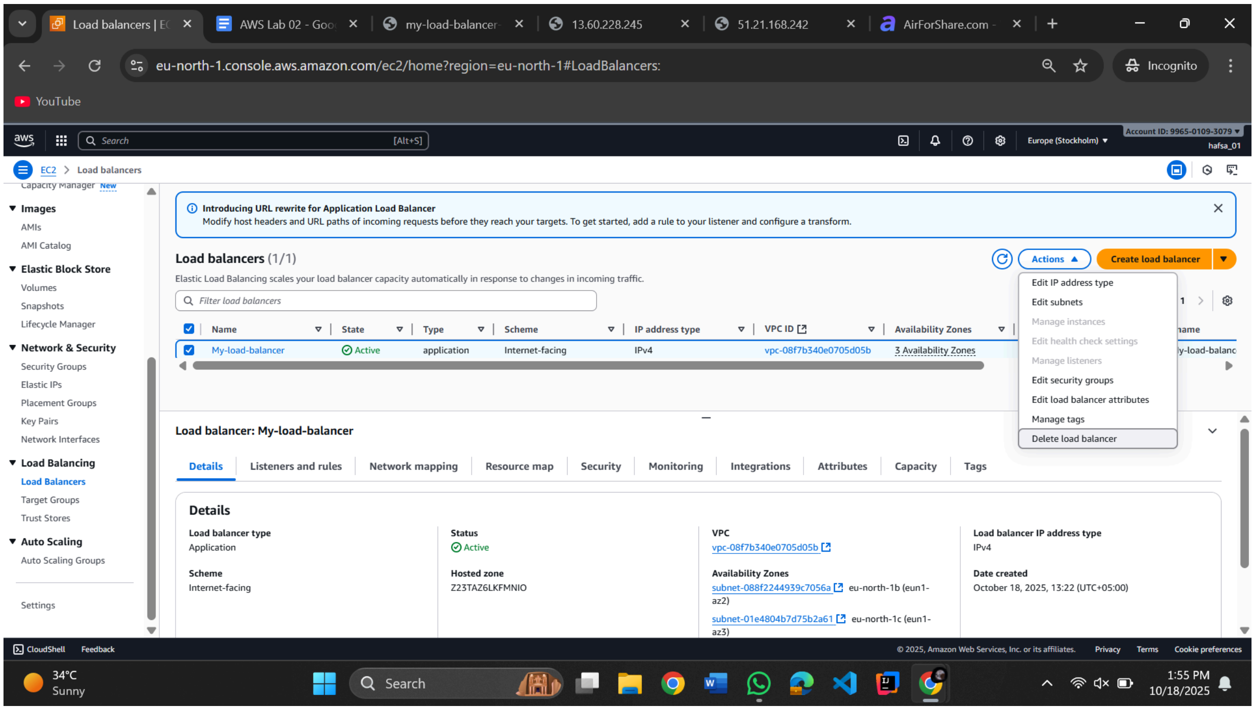
IP Address of this server is

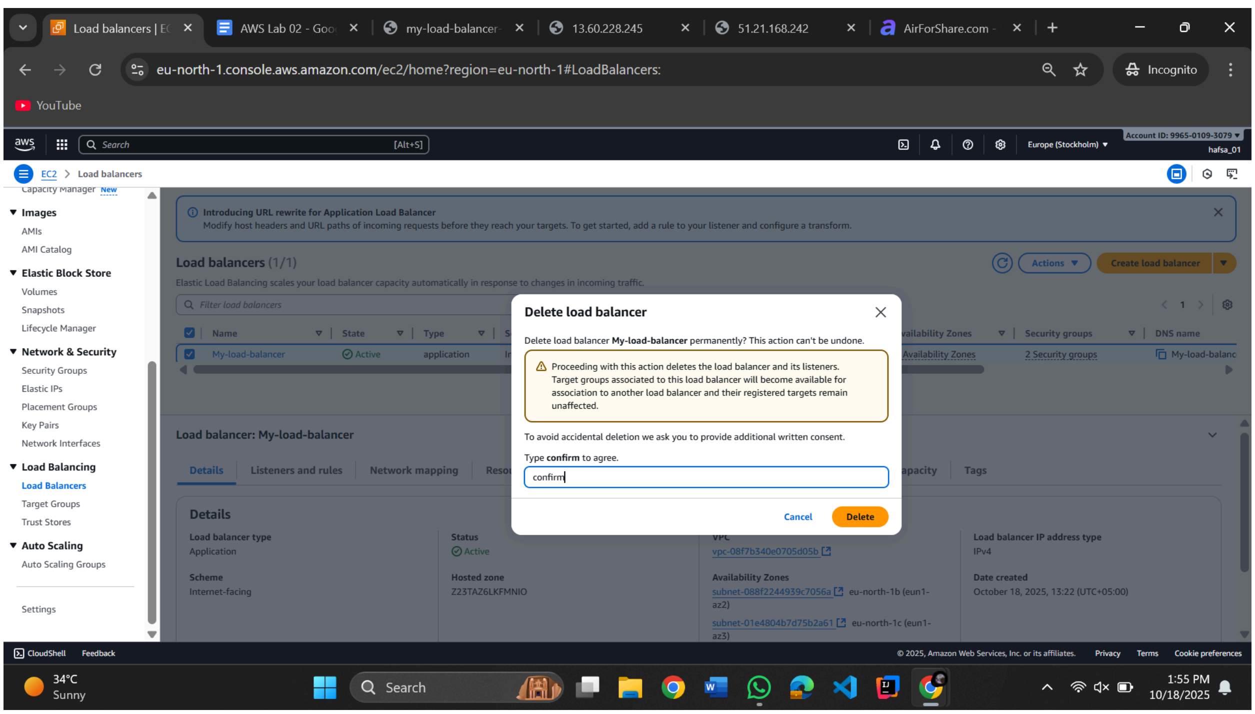
172.31.35.135

(Refresh the page. The load balancer DNS will connect to either one of the EC2. I will only work once the status is ACTIVE!)

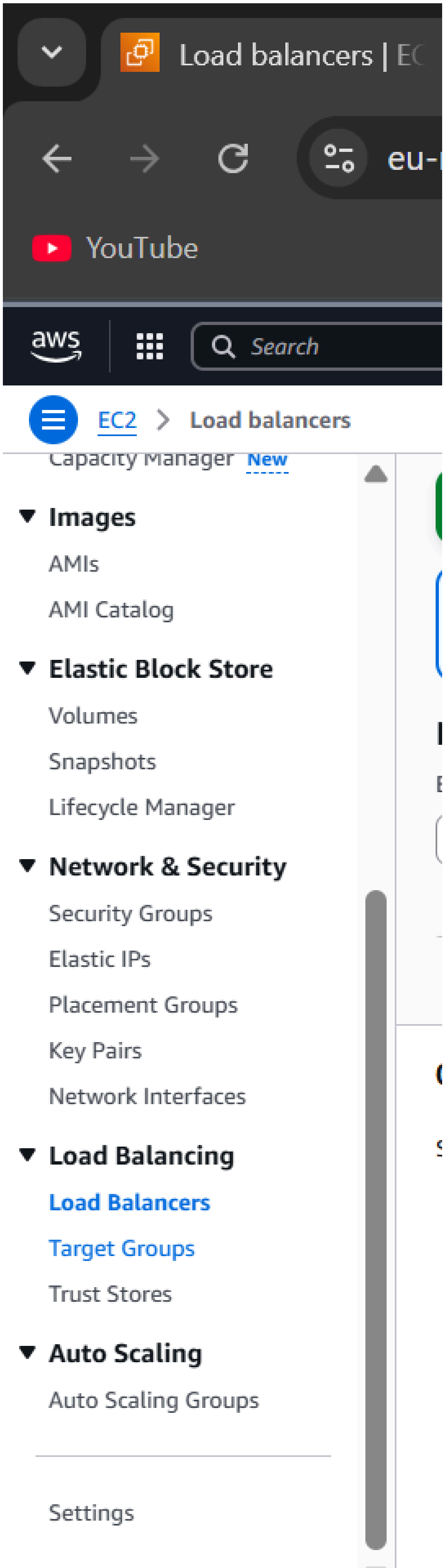
//Terminating the load balancer

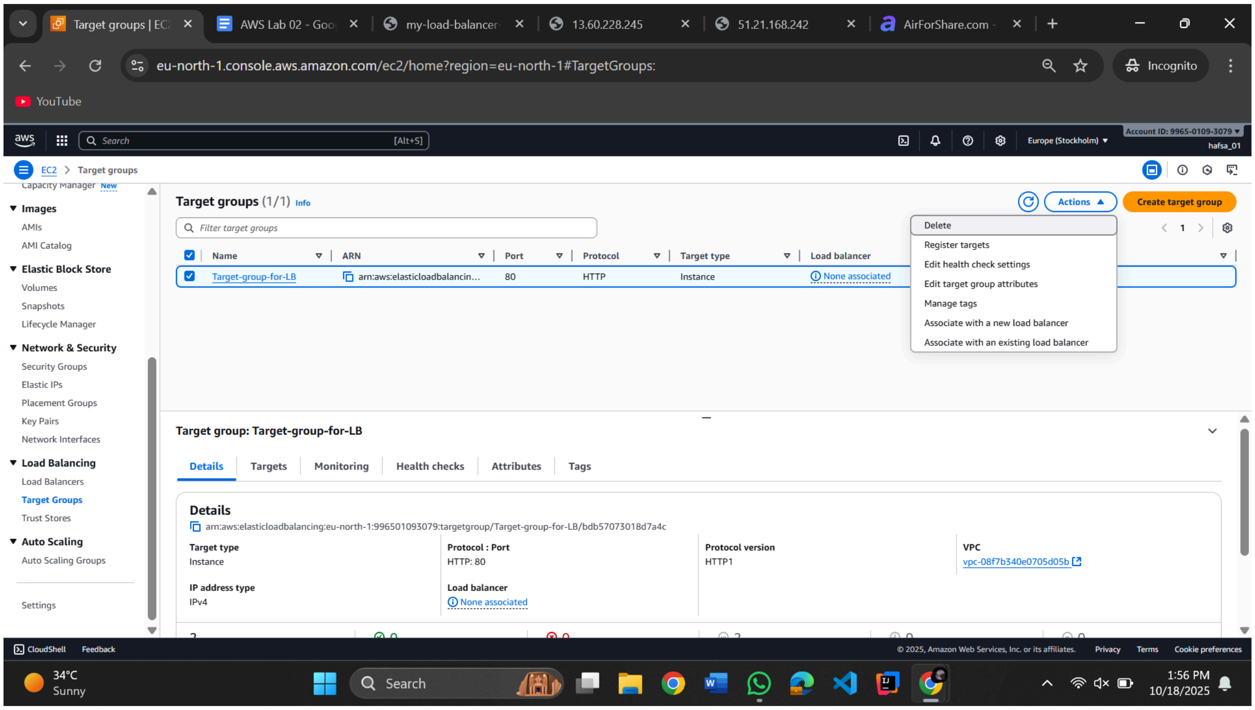

Load balancer deta
eu-nort
YouTube
aws
Search
EC2
Load balancers
My-load-l
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings

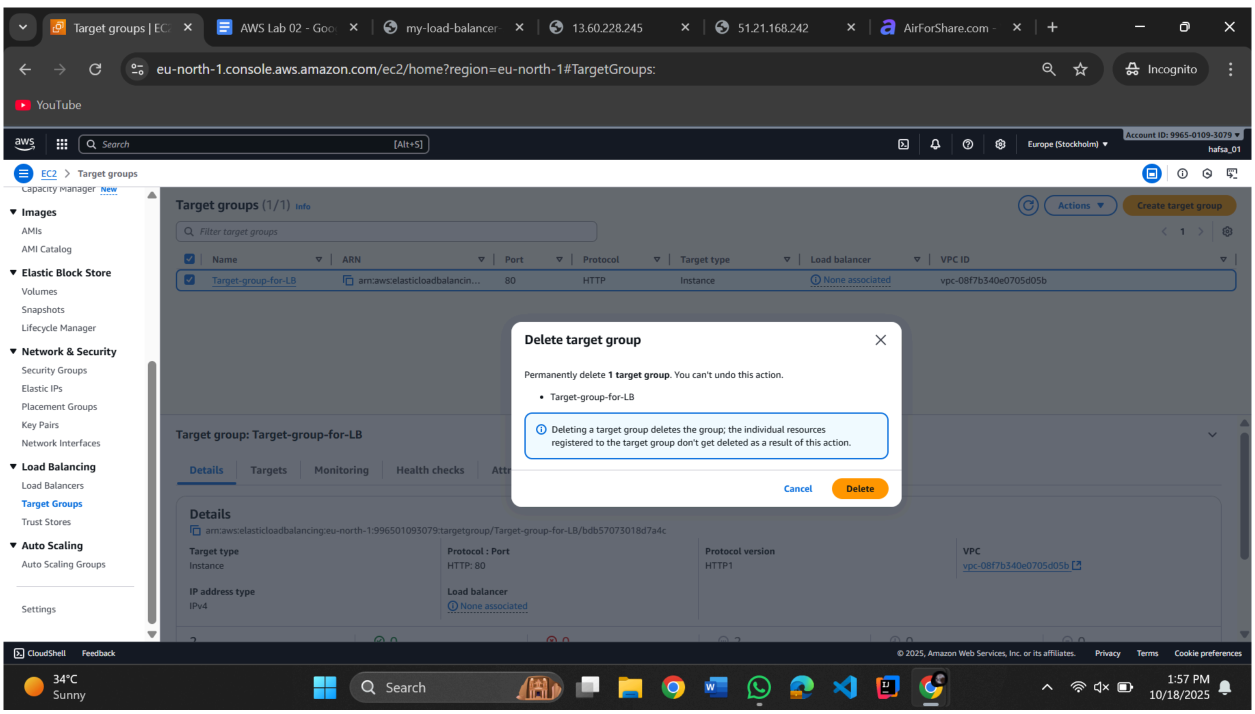


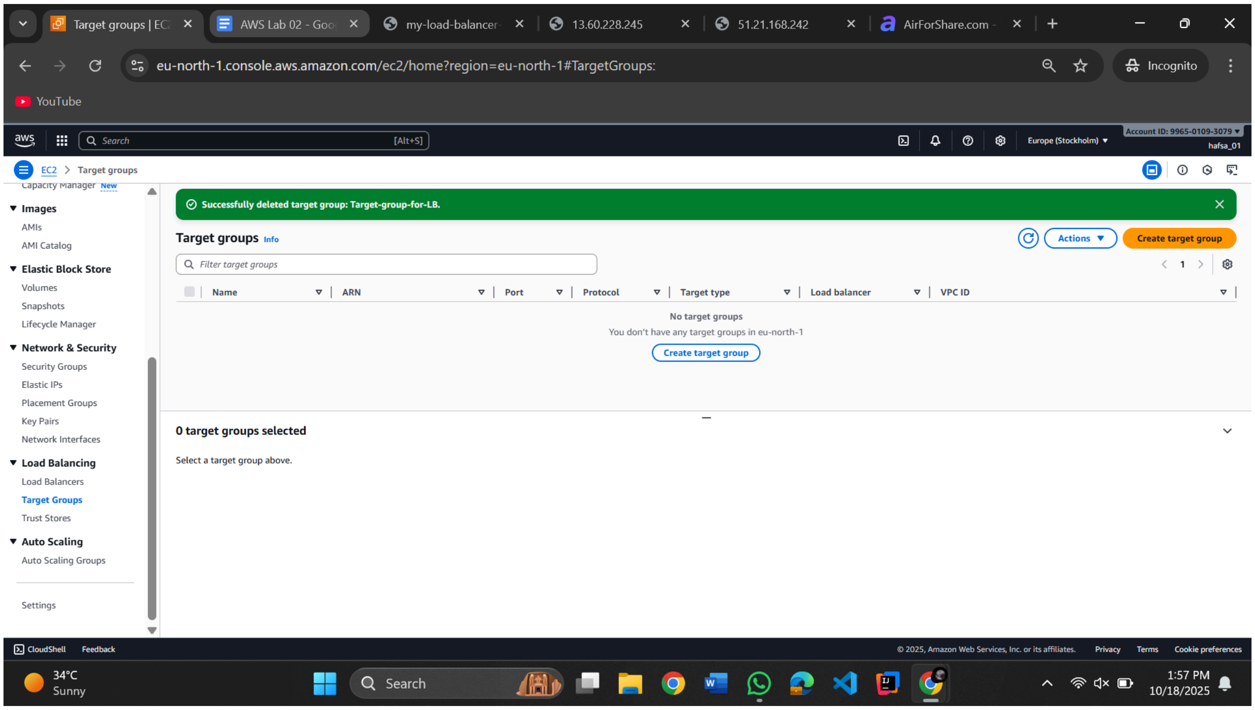


//Terminating Target groups

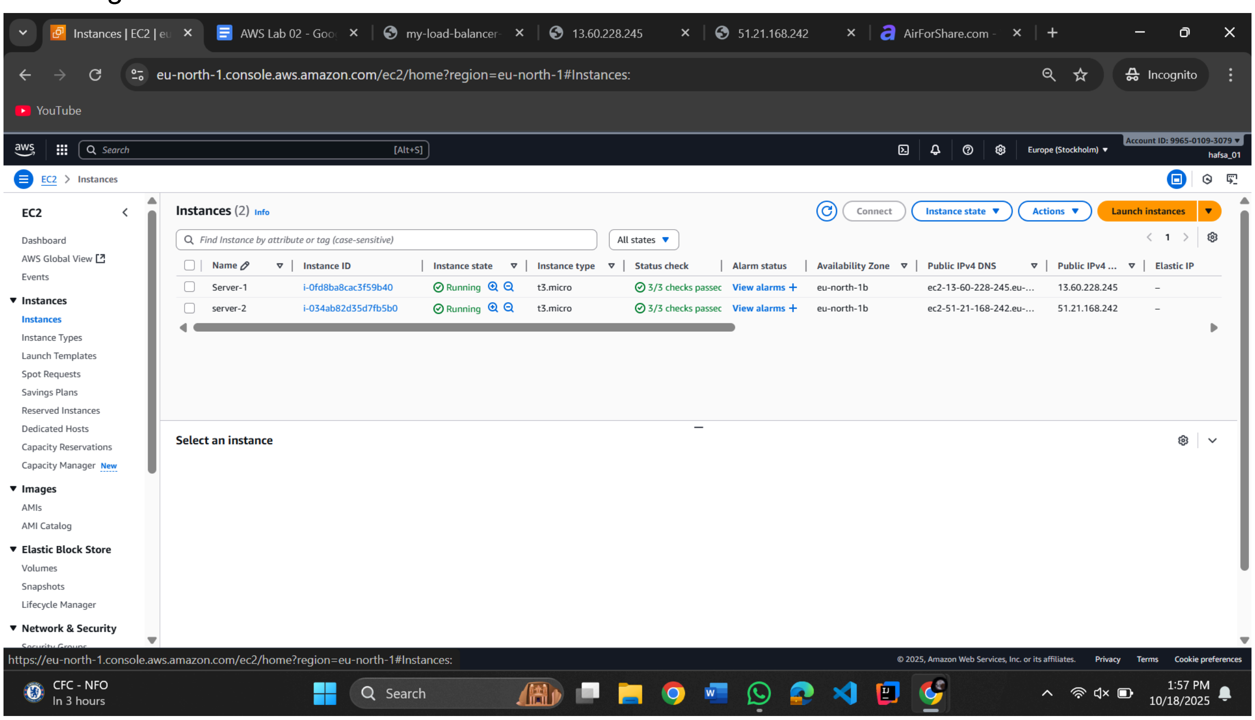

Load balancers | EC
eu-
YouTube
aws
Search
EC2
Load balancers
Capacity Manager
New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings


eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:
EC2 > Target groups
Target groups (1/1) Info
Actions
Create target group
Filter target groups
Name
ARN
Port
Protocol
Target type
Load balancer
Target-group-for-LB
arn:aws:elasticloadbalancin...
80
HTTP
Instance
None associated
Delete
Register targets
Edit health check settings
Edit target group attributes
Manage tags
Associate with a new load balancer
Associate with an existing load balancer
Target group: Target-group-for-LB
Details
Targets
Monitoring
Health checks
Attributes
Tags
arn:aws:elasticloadbalancing:eu-north-1:996501093079:targetgroup/Target-group-for-LB/bdb57073018d7a4c
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-08f7b340e0705d05b
IP address type
IPv4
Load balancer
None associated
1:56 PM
10/18/2025

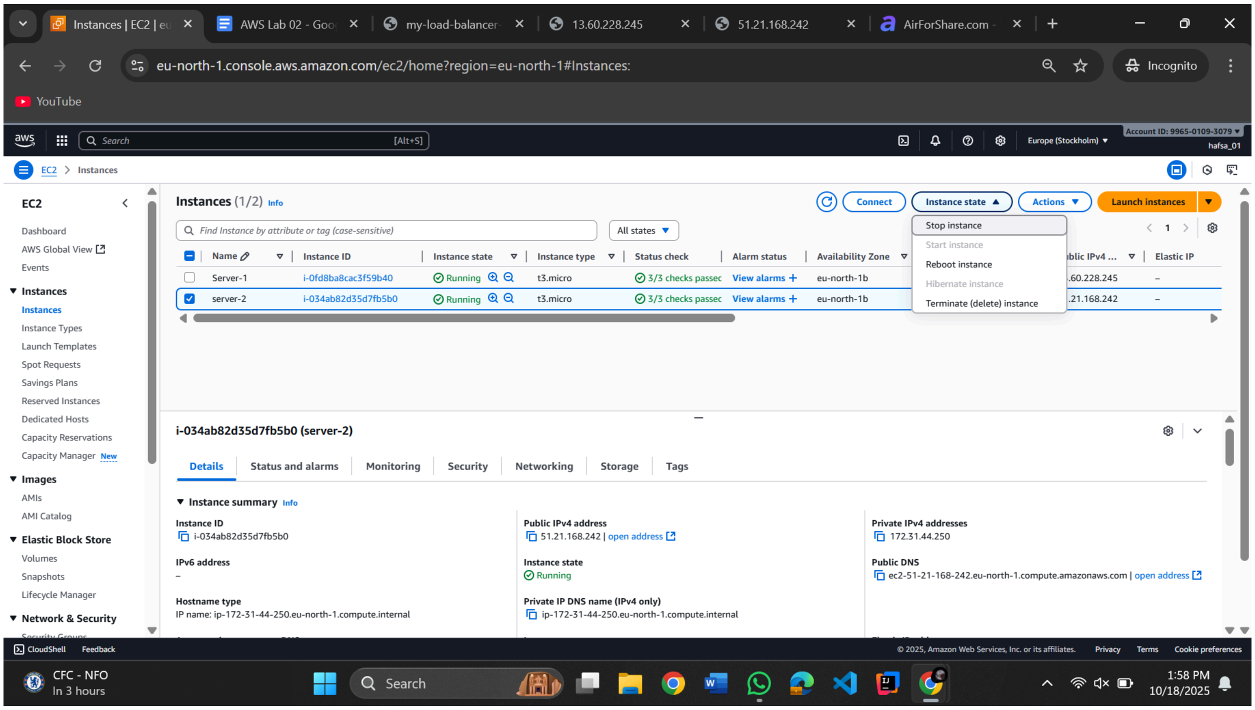

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#TargetGroups:
Delete target group
Permanently delete 1 target group. You can't undo this action.
Target-group-for-LB
Deleting a target group deletes the group; the individual resources registered to the target group don't get deleted as a result of this action.
Cancel
Delete
1:57 PM
10/18/2025

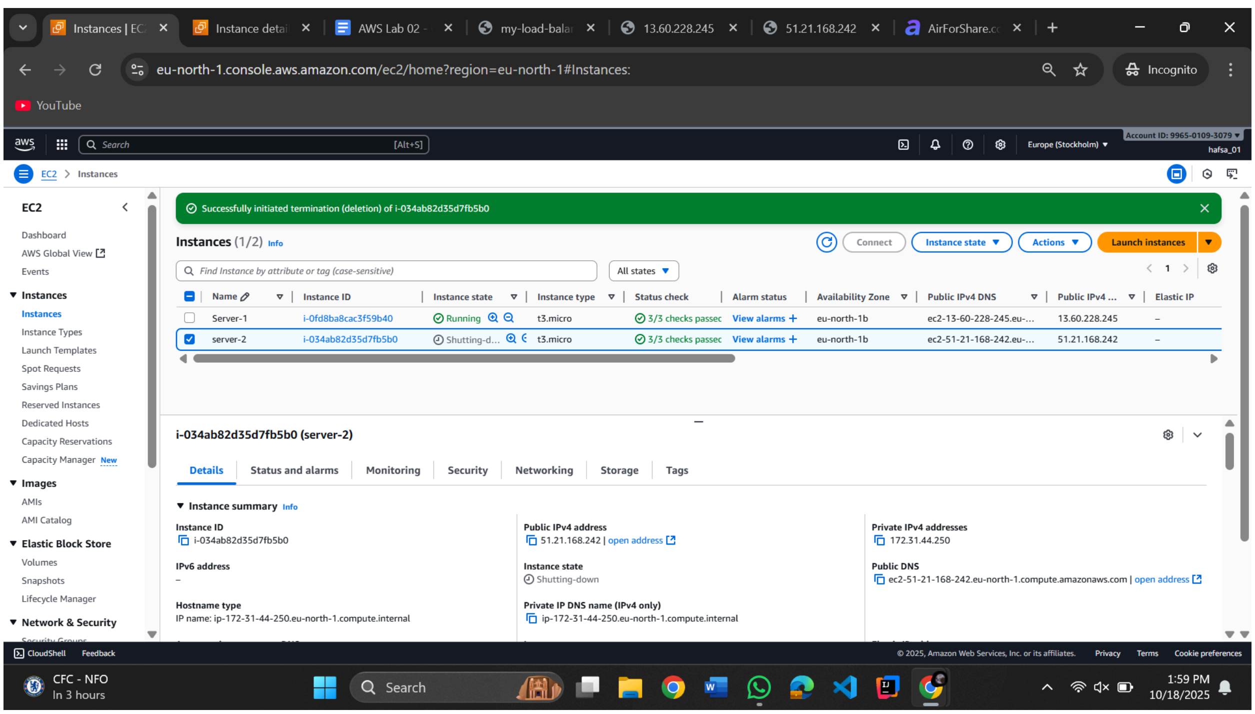


//Deleting



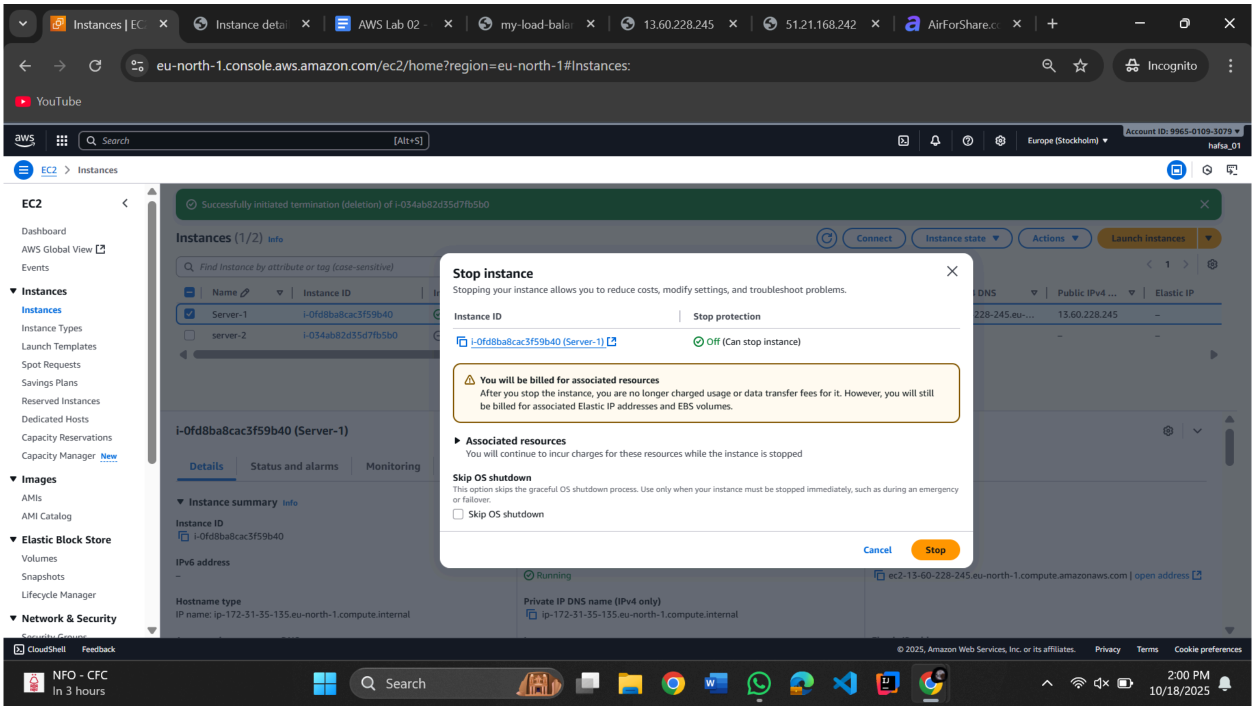
//Stopping instance



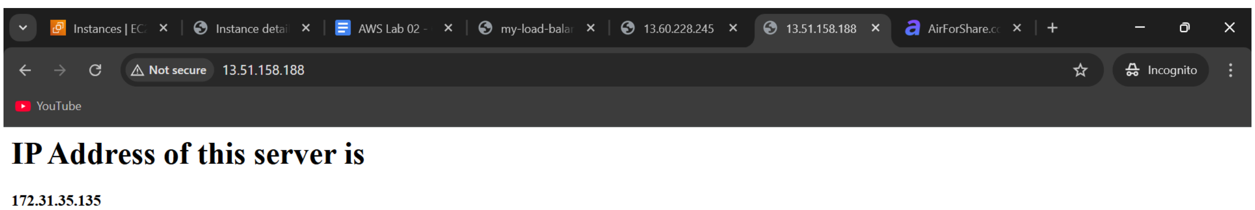


Terminate one of them

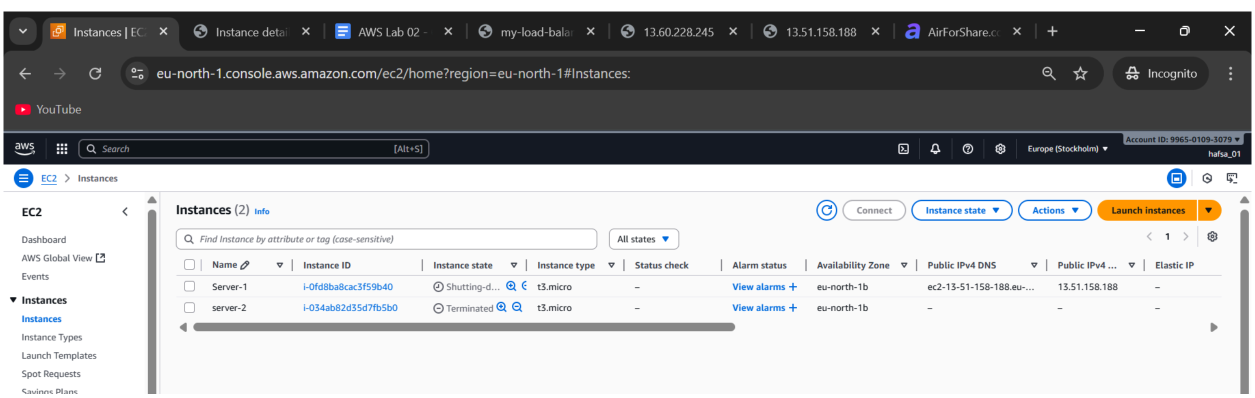
And stop one of them (Before stopping copy their instance ID)



Restart the stopped instance, It will change it

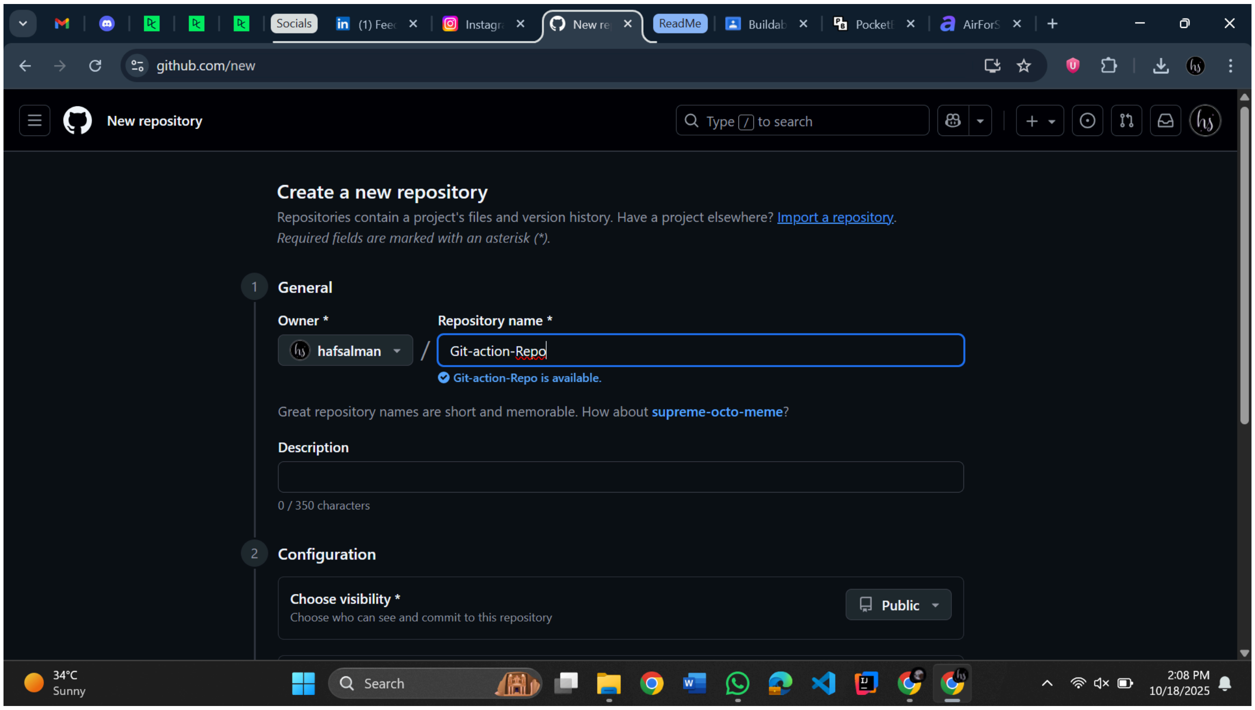


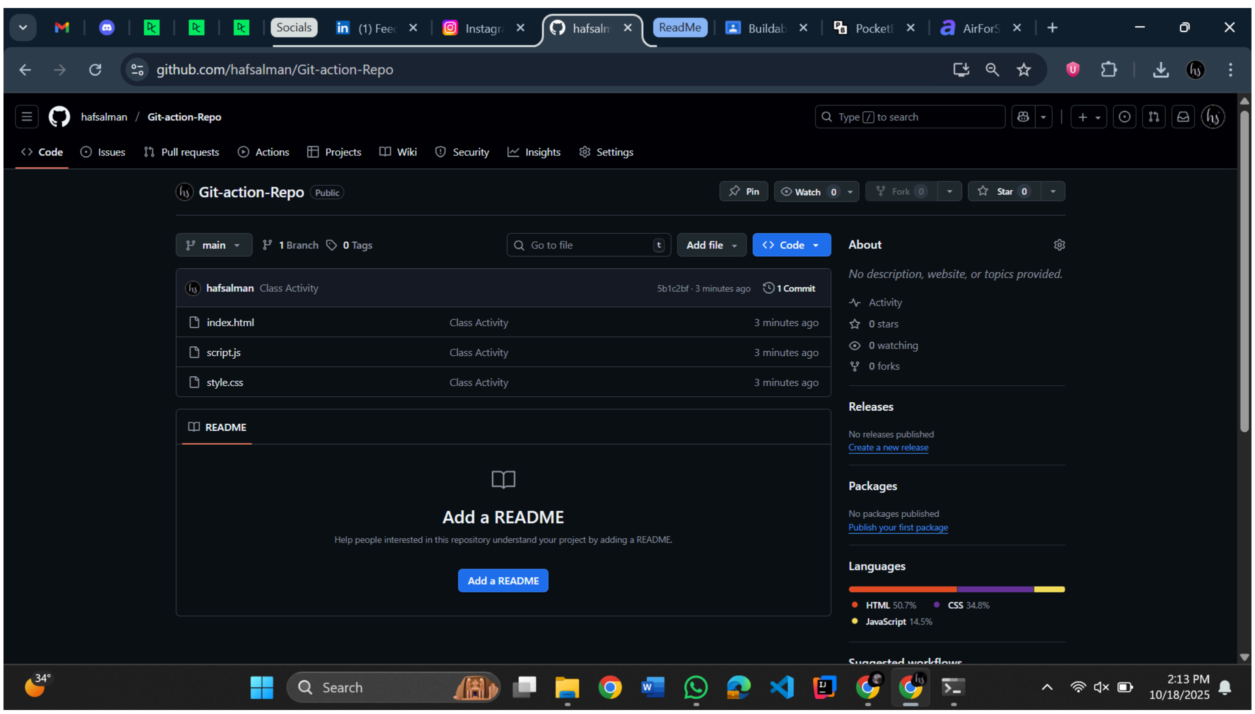
Terminate the instance

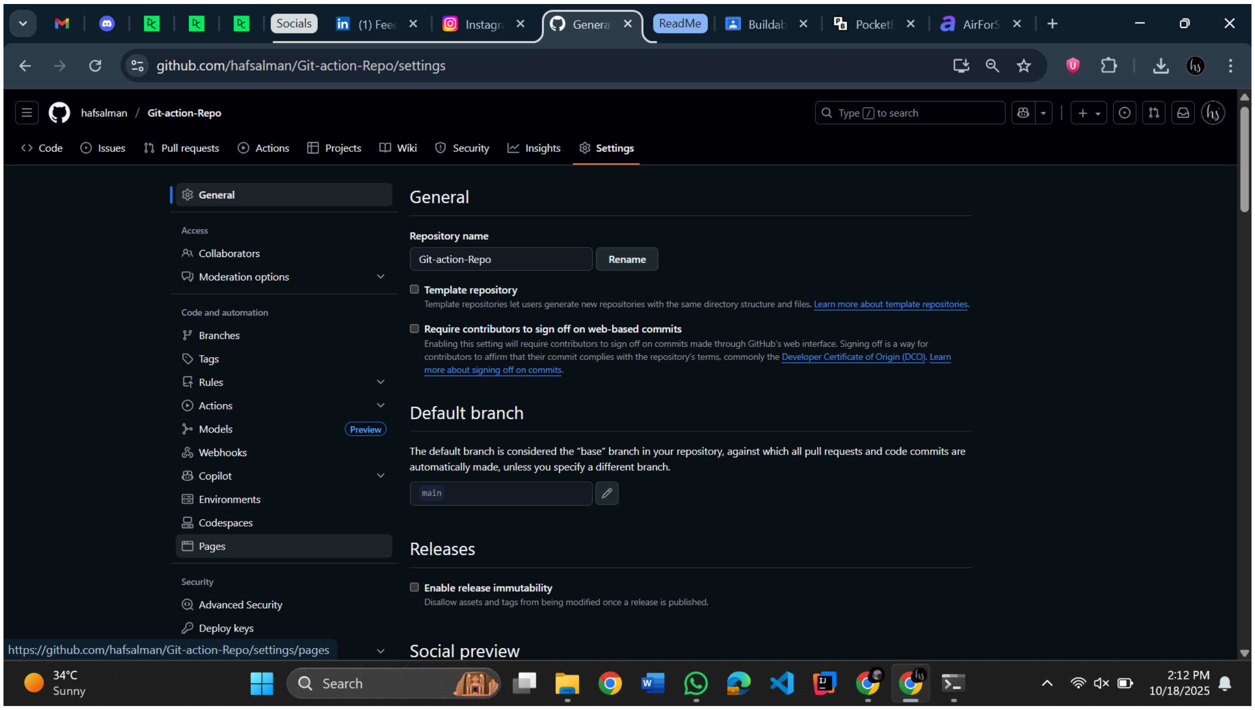


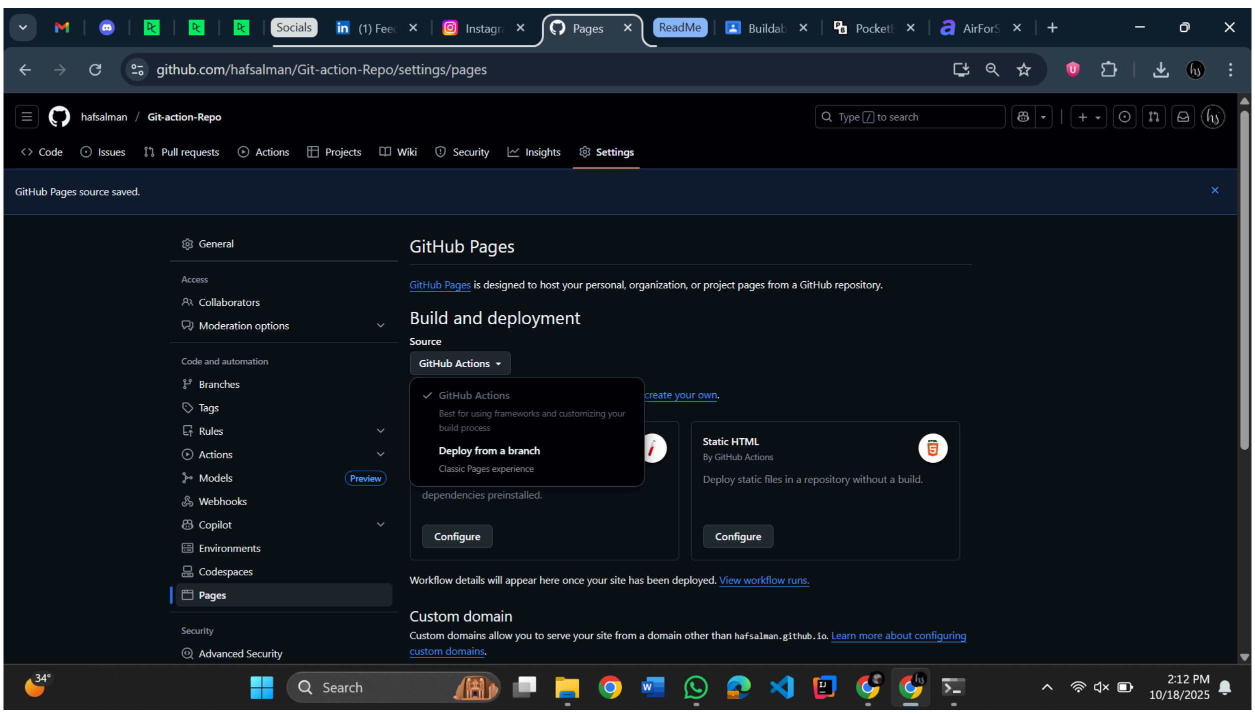
//GITHUB TIMEEEE

- Open GitHub (We will deploy application on GitHub. Next lab main we will deploy it on EC2)
- Files link (I will upload them on GitHub so you can clone them)
- Create repo


github.com/new
New repository
Type / to search
Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1
General
Owner *
hafsalman
Repository name *
Git-action-Repo
Git-action-Repo is available.
Great repository names are short and memorable. How about supreme-octo-meme?
Description
0 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
34°C
Sunny
Search
2:08 PM
10/18/2025

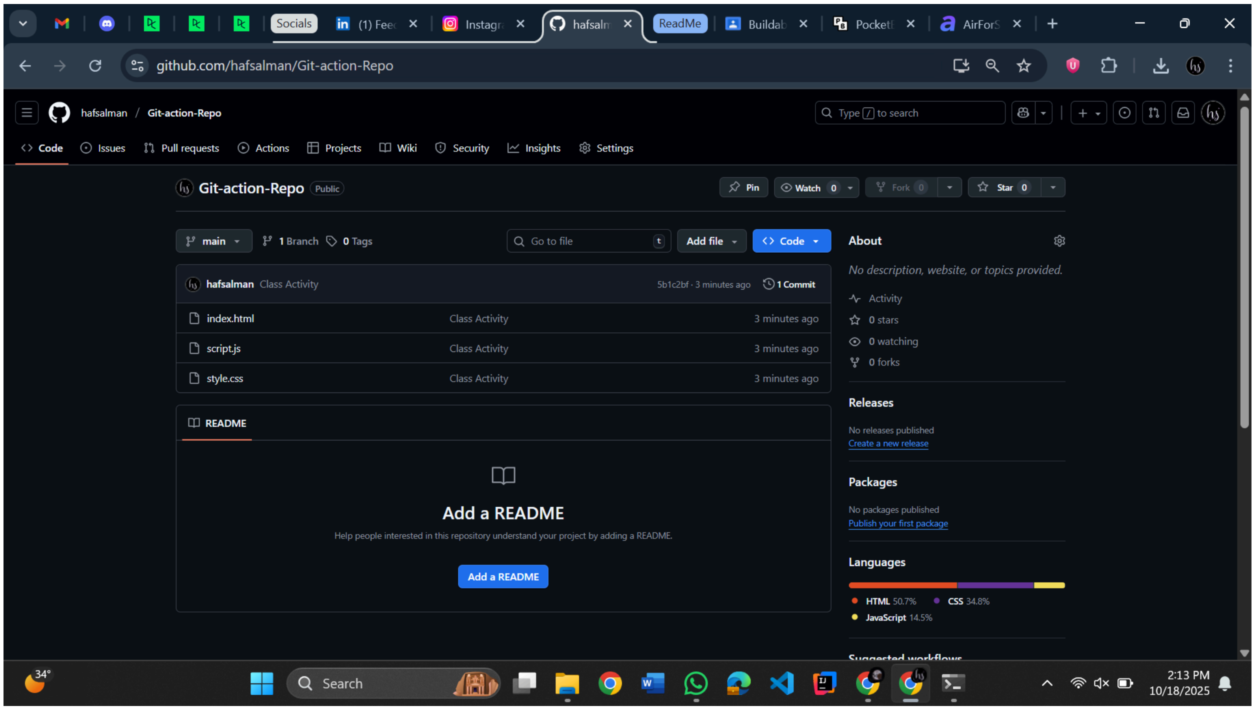

github.com/hafsalman/Git-action-Repo
hafsalman / Git-action-Repo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Git-action-Repo
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
hafsalman Class Activity
5b1c2bf · 3 minutes ago
1 Commit
index.html
Class Activity
3 minutes ago
script.js
Class Activity
3 minutes ago
style.css
Class Activity
3 minutes ago
README
Add a README
Help people interested in this repository understand your project by adding a README.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Languages
HTML 50.7%
CSS 34.8%
JavaScript 14.5%
34°
Search
2:13 PM
10/18/2025



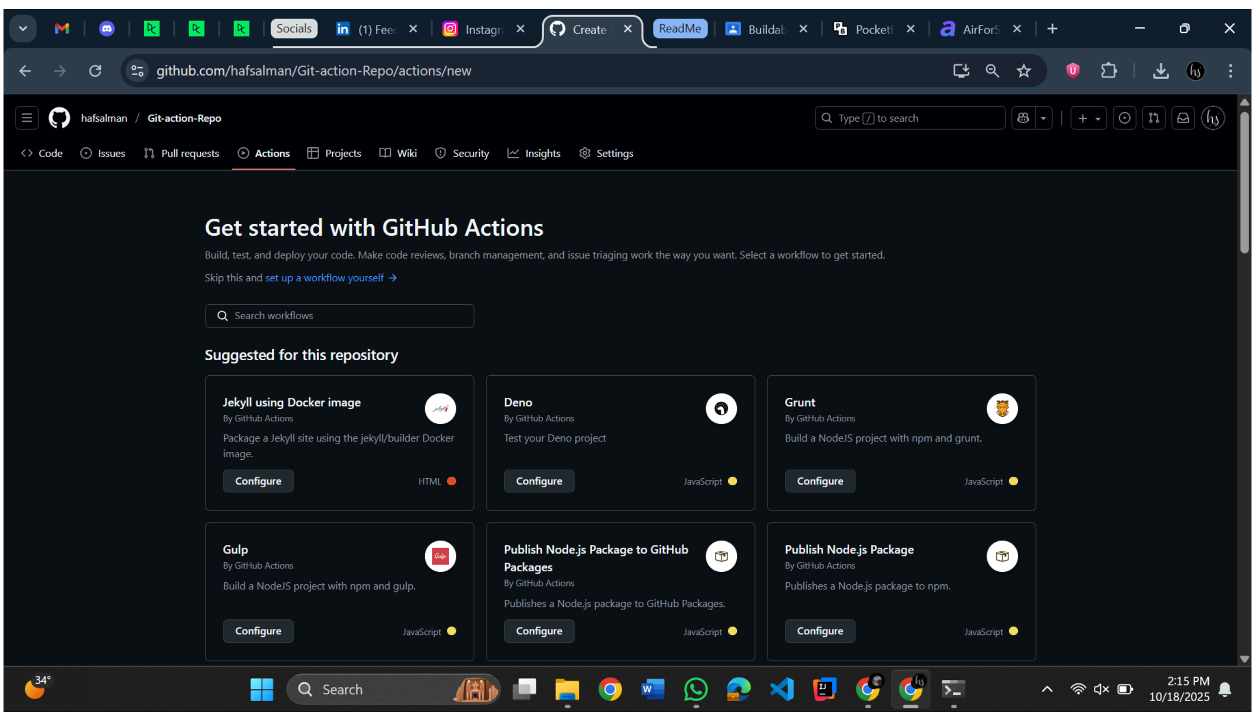


(Select GitHub Actions)

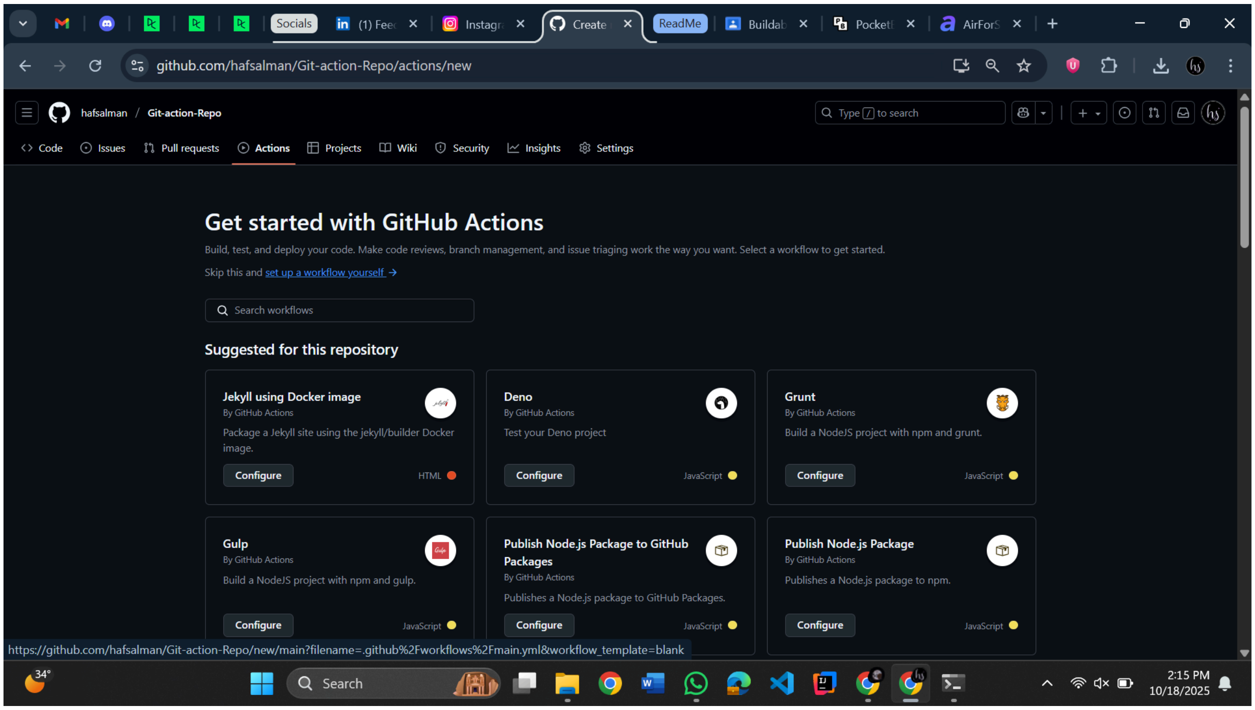
Go back to the code



Go to actions



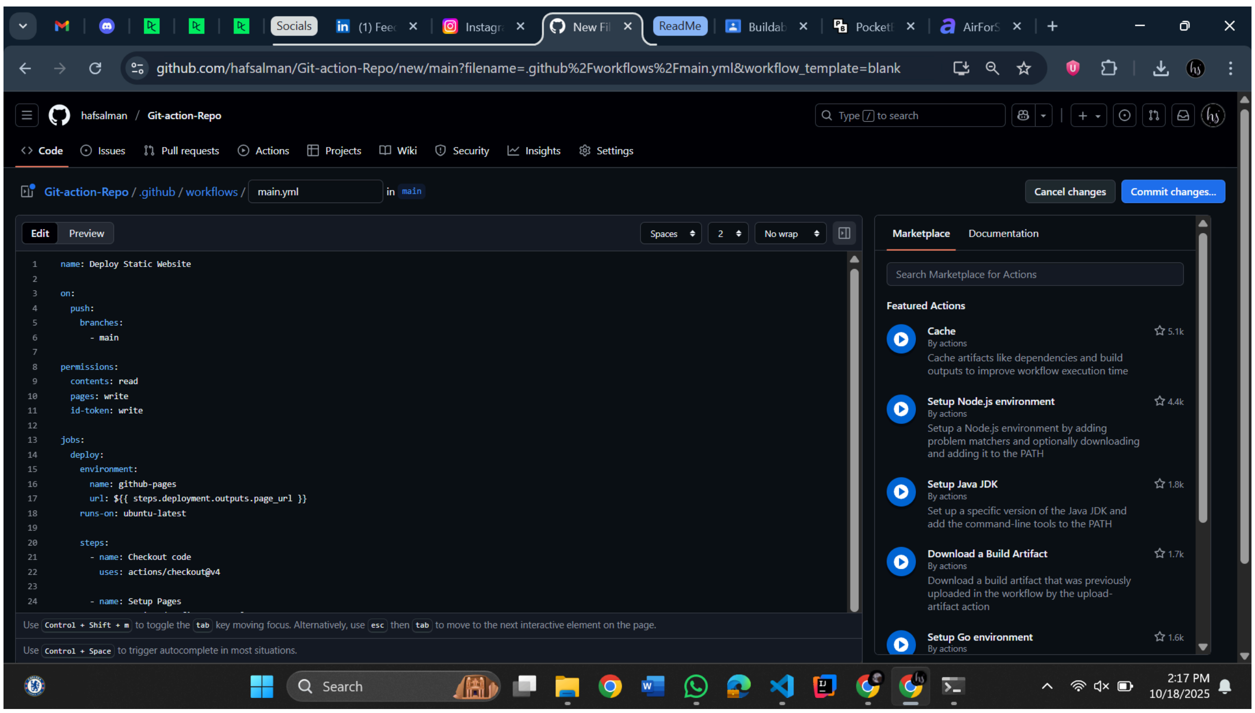
Click “Set up a workflow yourself”

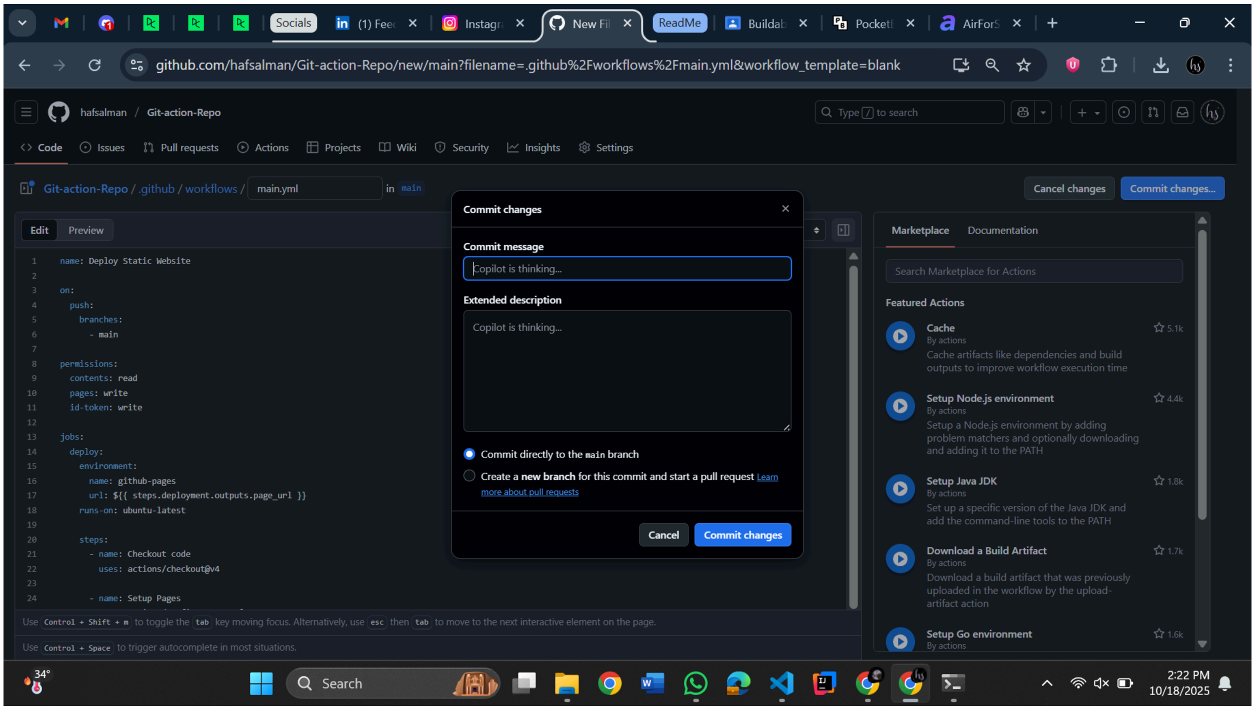


//Copy paste deploy.yml

```yaml
name: Deploy Static Website

on:
  push:
    branches:
      - main

permissions:
  contents: read
  pages: write
  id-token: write

jobs:
  deploy:
    environment:
      name: github-pages
      url: ${{ steps.deployment.outputs.page_url }}
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4
```

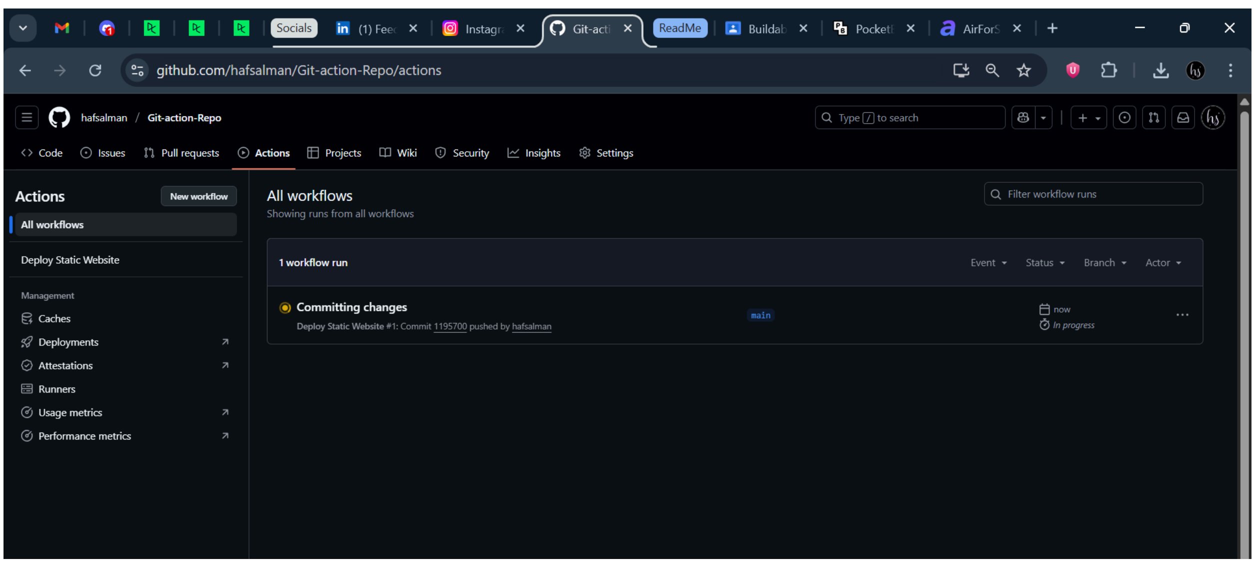
```yaml
- name: Setup Pages
  uses: actions/configure-pages@v5

- name: Upload website files
  uses: actions/upload-pages-artifact@v3
  with:
    path: ./  # Path to your static files

- name: Deploy to GitHub Pages
  id: deployment
  uses: actions/deploy-pages@v4
```



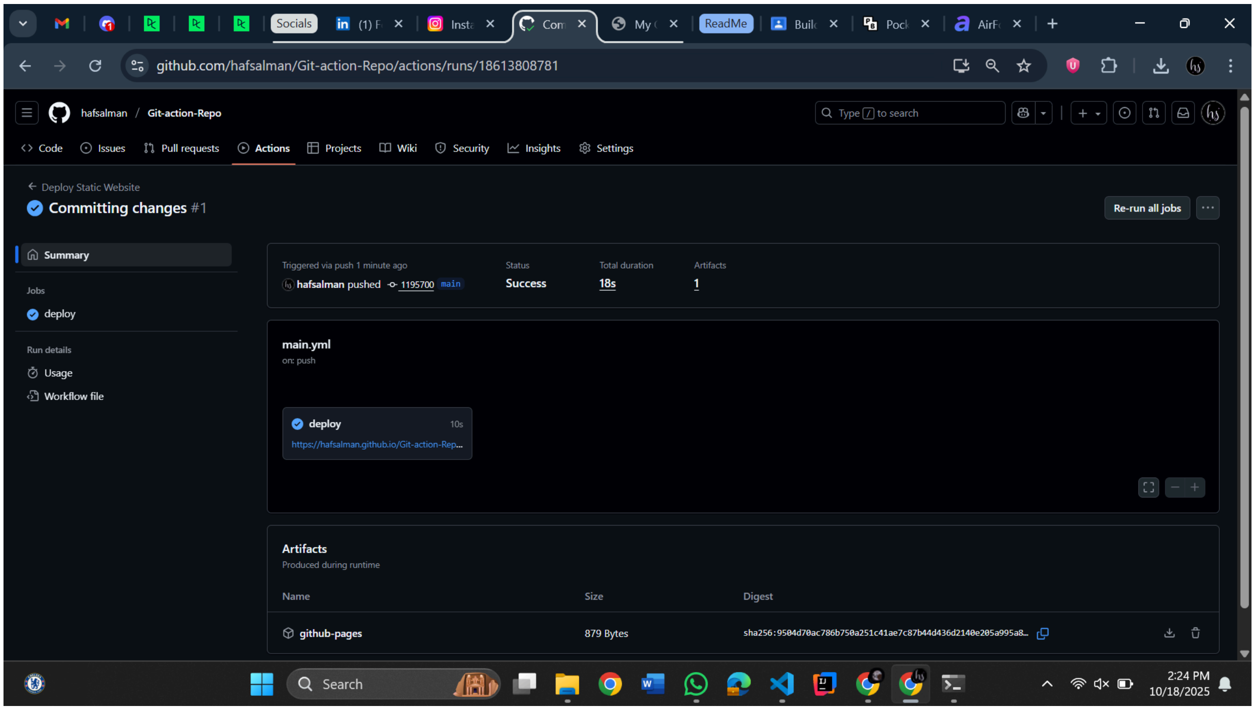


Commit changes (I wrote the "Committing Changes")

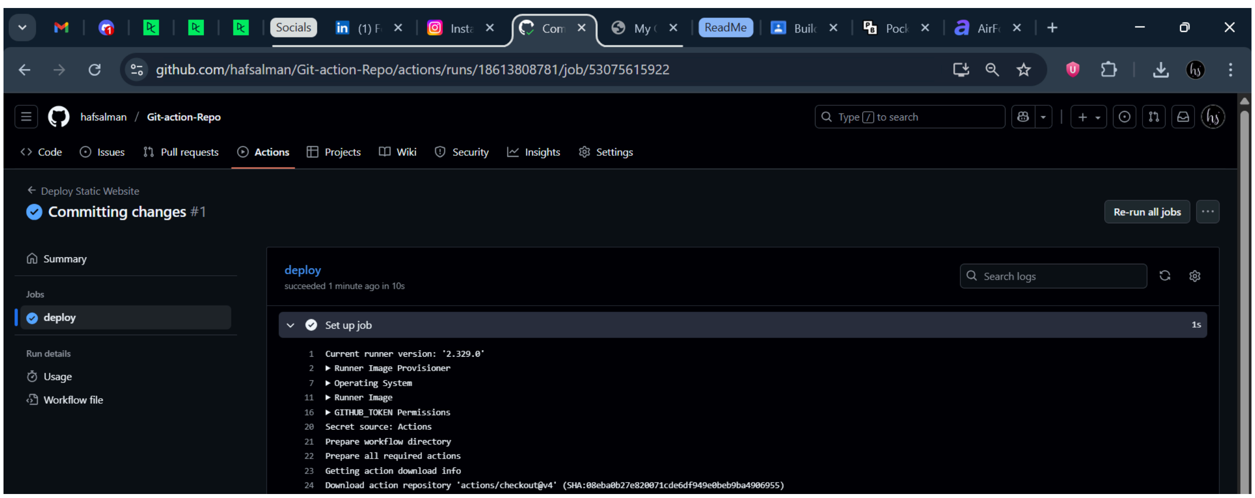
Go to actions -> All workflows

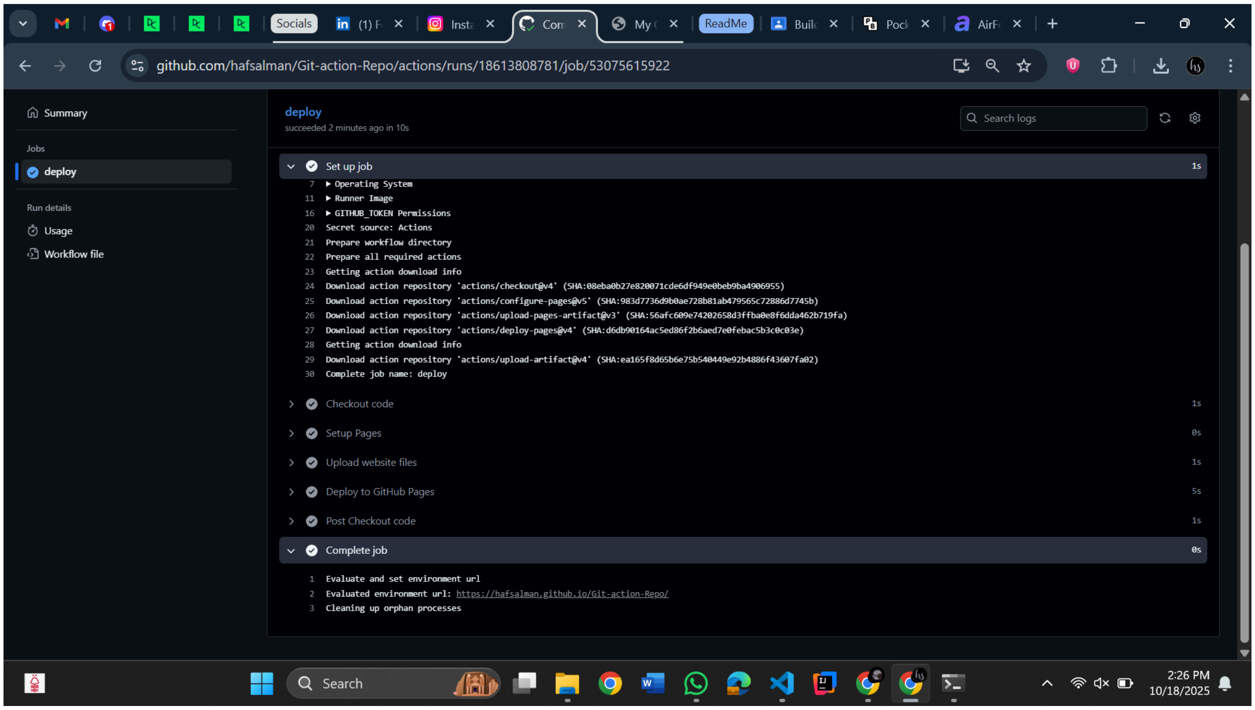


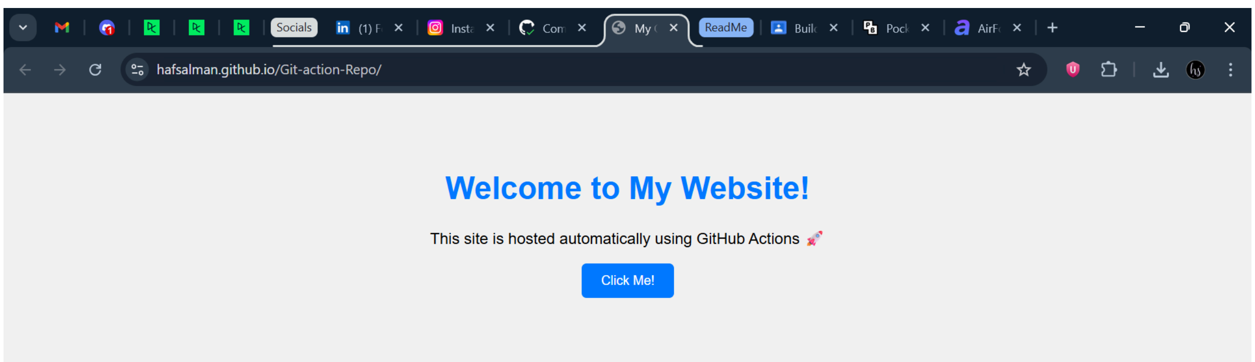
Click on "Committing Changes”



Click Deploy





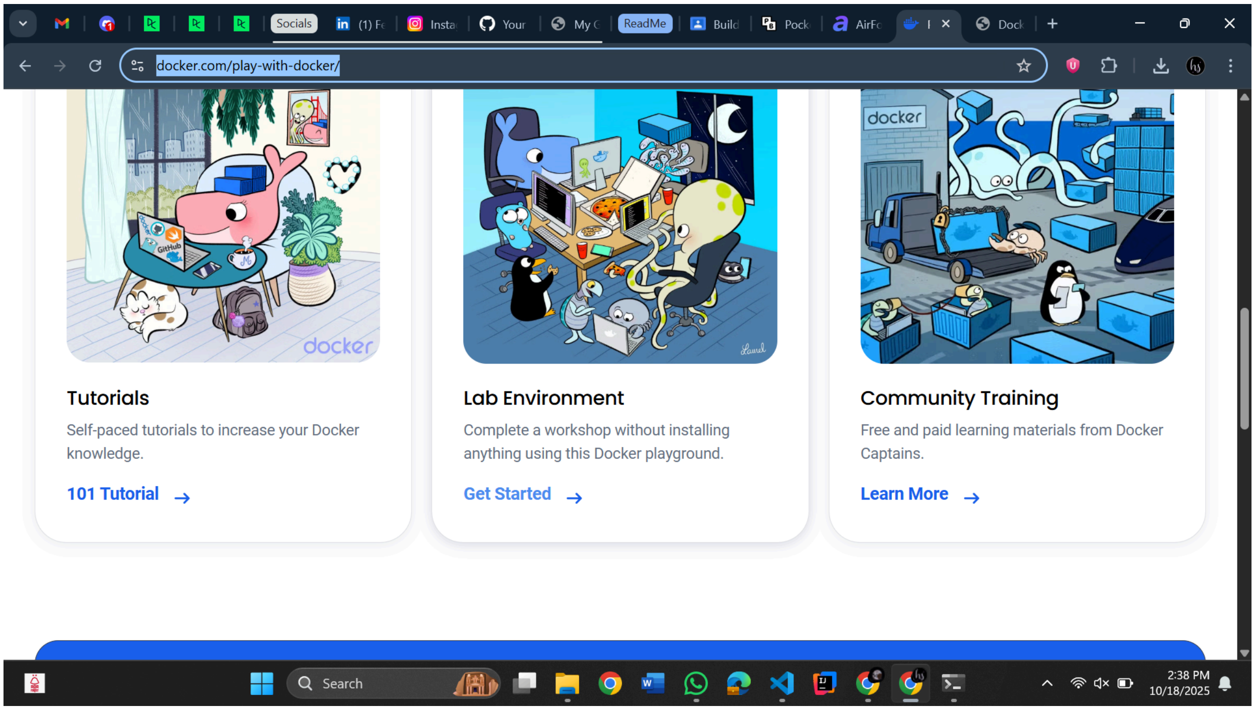
Click the link given in the last


TADAAAAAAAAAAAAAAAAAAAAAa

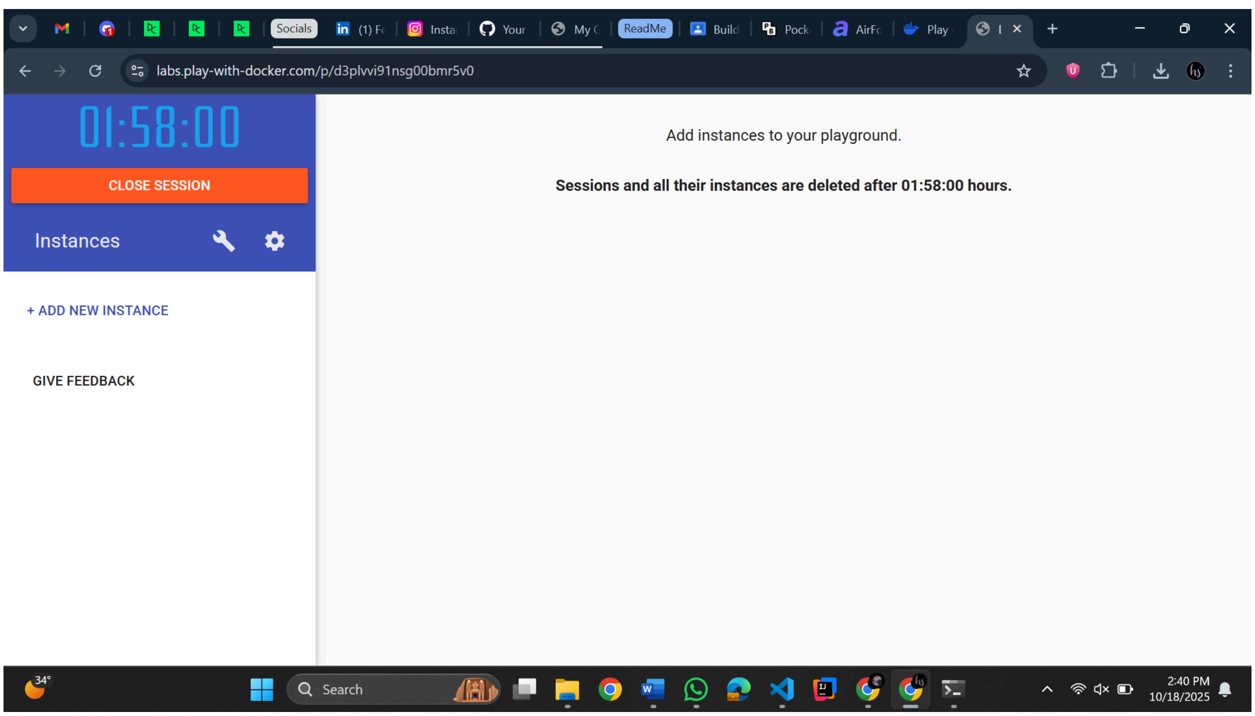
DOCKER TIME

I forgot to screenshot, sorrrryyyyyyyyyyyyy :(

1. https://www.docker.com/play-with-docker/
2. Go to lab Environment

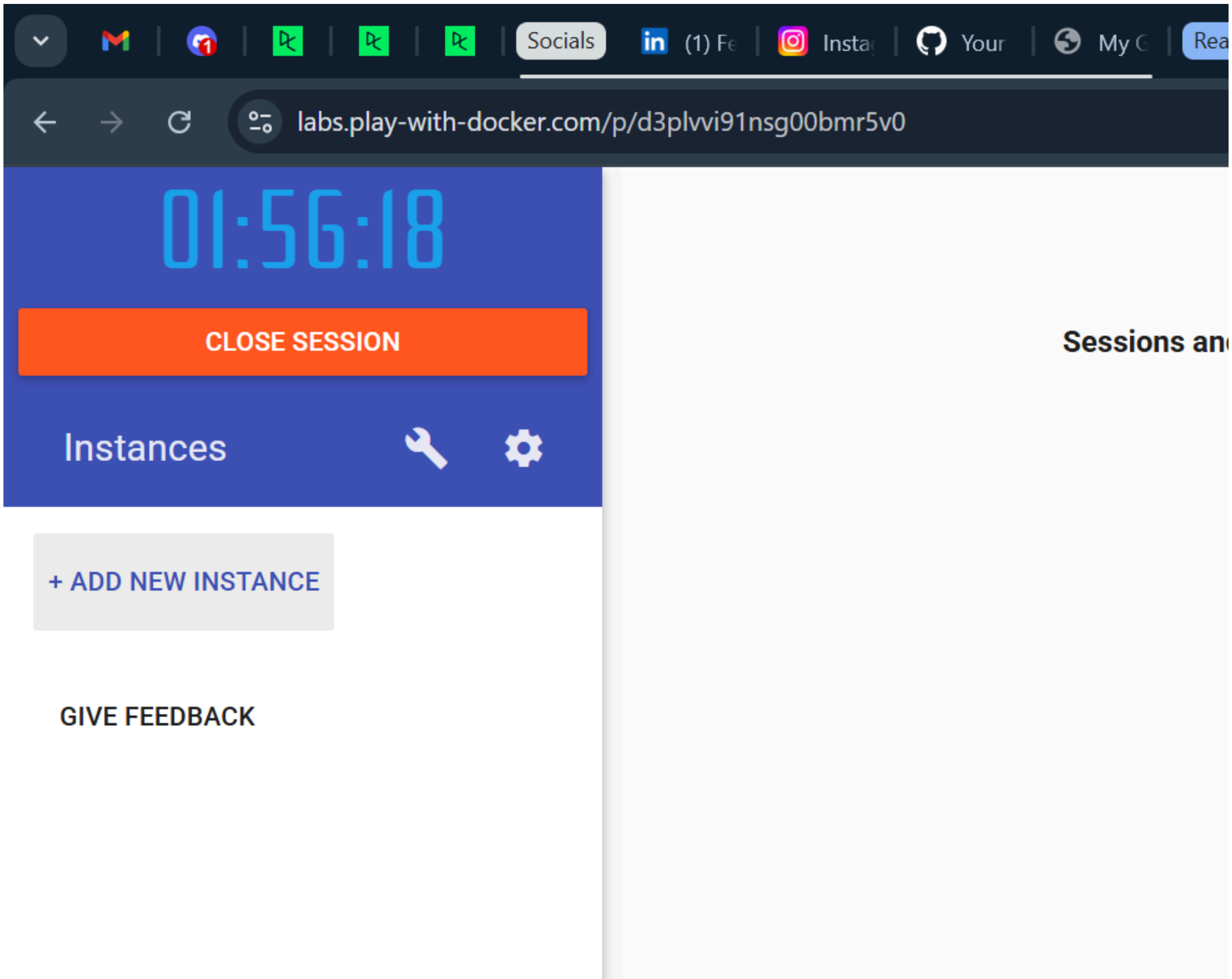


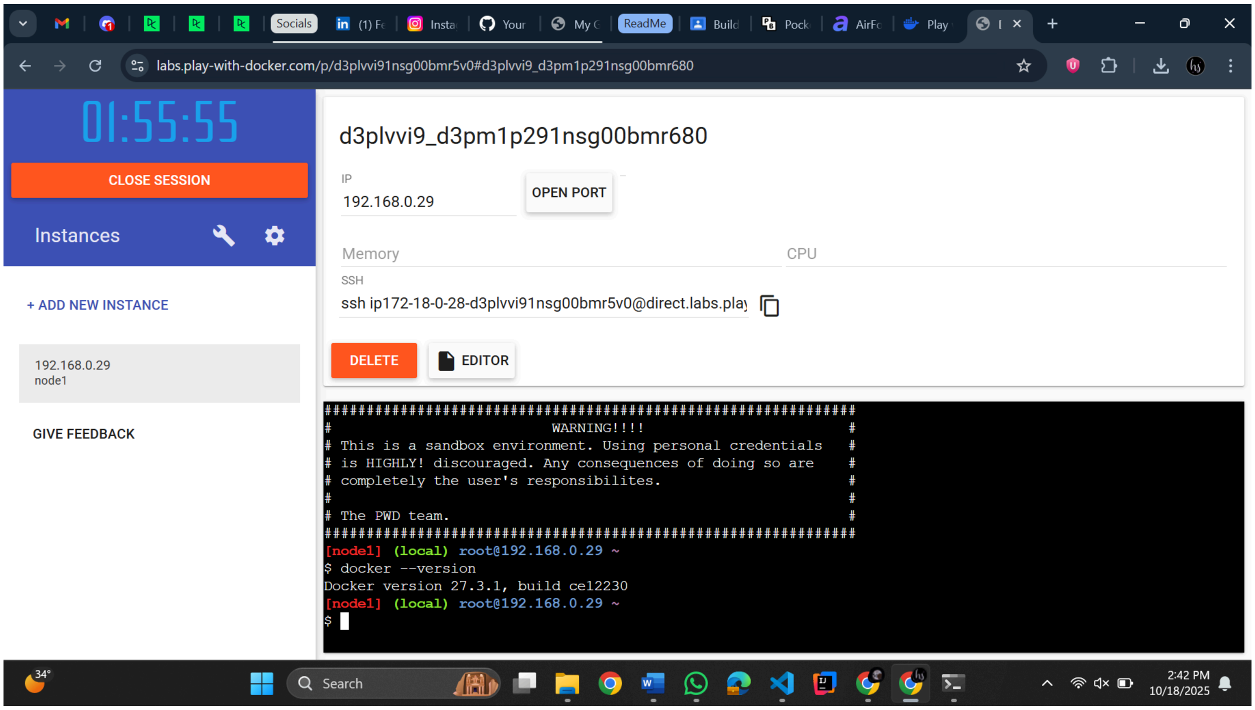
3. Get started
4. Sign up
5. Give username
6. Start



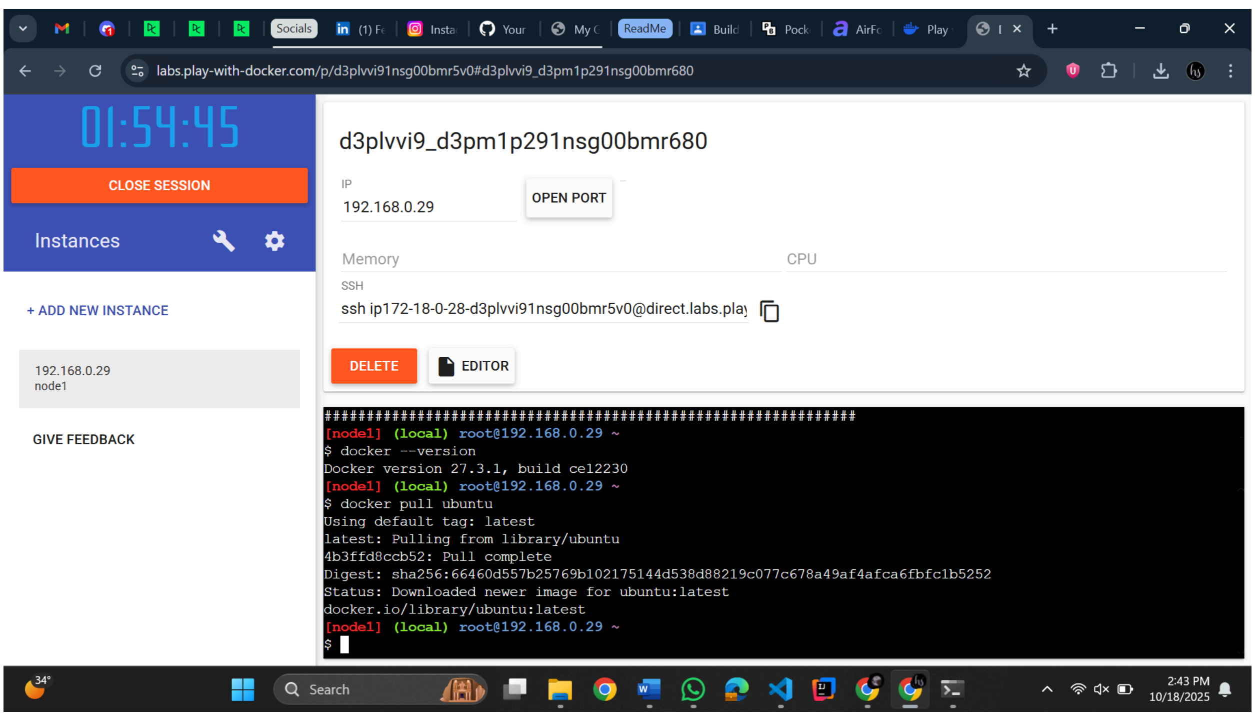
You will end up here

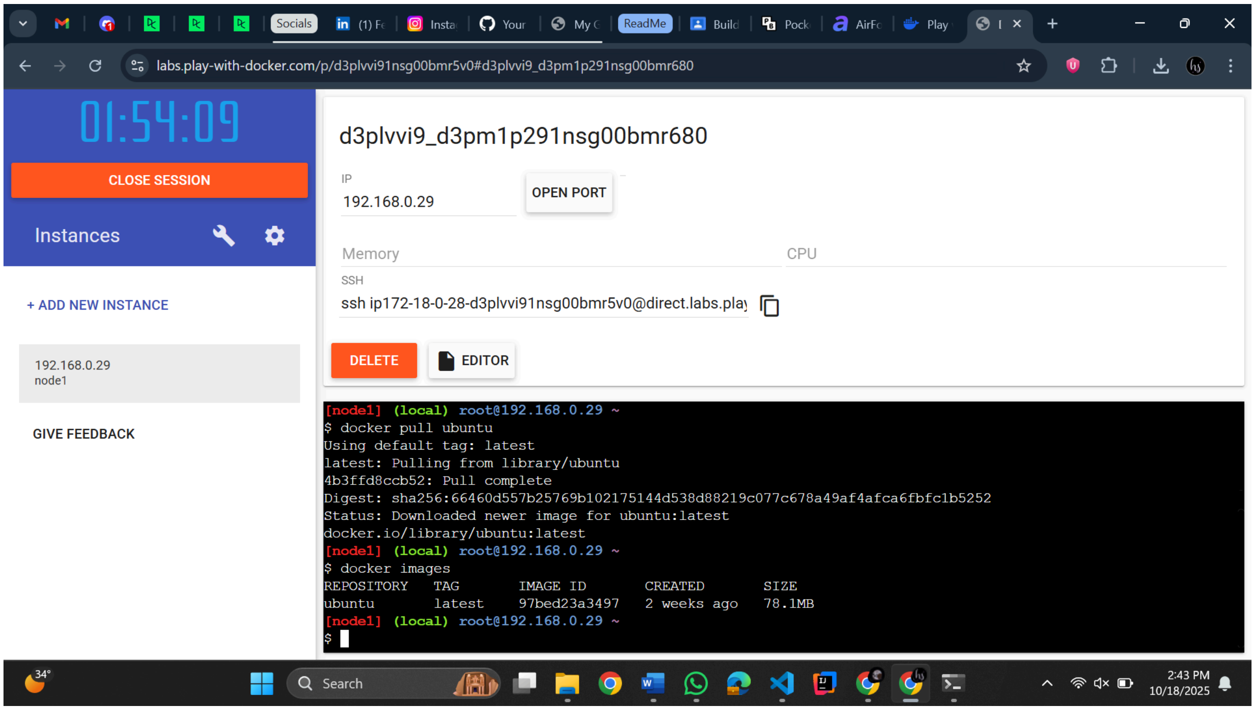
7. Open the docker word file (I will upload that on the repo as well)
8. Create instance

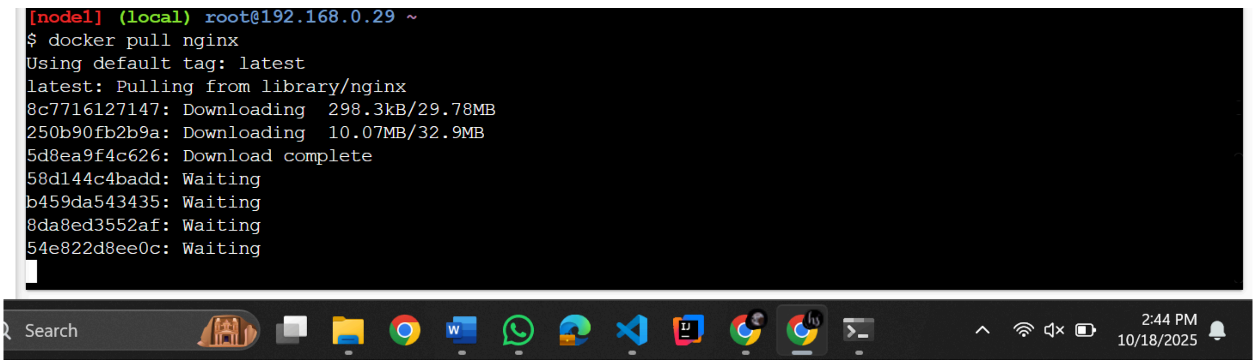

Socials
(1) Fe
Insta
Your
My
Rea
labs.play-with-docker.com/p/d3plvvi91nsg00bmr5v0
01:56:18
CLOSE SESSION
Instances
+ ADD NEW INSTANCE
GIVE FEEDBACK
Sessions an

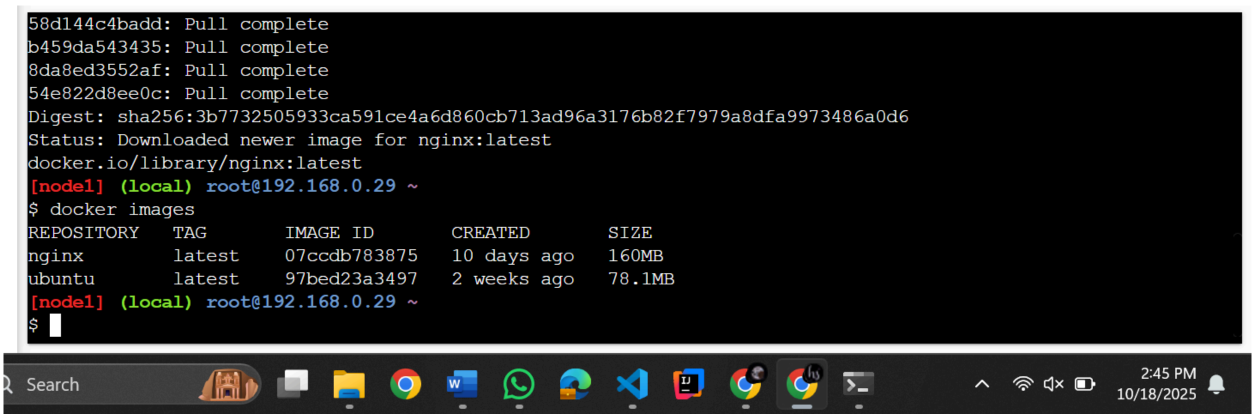


Run docker –version




labs.play-with-docker.com/p/d3plvvi91nsg00bmr5v0#d3plvvi9_d3pm1p291nsg00bmr680
01:54:09
CLOSE SESSION
Instances
+ ADD NEW INSTANCE
192.168.0.29
node1
GIVE FEEDBACK
d3plvvi9_d3pm1p291nsg00bmr680
IP
192.168.0.29
OPEN PORT
Memory
CPU
SSH
ssh ip172-18-0-28-d3plvvi91nsg00bmr5v0@direct.labs.play
DELETE
EDITOR
[node1] (local) root@192.168.0.29 ~
$ docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
4b3ffd8ccb52: Pull complete
Digest: sha256:66460d557b25769b102175144d538d88219c077c678a49af4afca6fbfc1b5252
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
ubuntu latest 97bed23a3497 2 weeks ago 78.1MB
[node1] (local) root@192.168.0.29 ~
$
Search
2:43 PM
10/18/2025


[node1] (local) root@192.168.0.29 ~
$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
8c7716127147: Downloading 298.3kB/29.78MB
250b90fb2b9a: Downloading 10.07MB/32.9MB
5d8ea9f4c626: Download complete
58d144c4badd: Waiting
b459da543435: Waiting
8da8ed3552af: Waiting
54e822d8ee0c: Waiting
Search
2:44 PM
10/18/2025


58d144c4badd: Pull complete
b459da543435: Pull complete
8da8ed3552af: Pull complete
54e822d8ee0c: Pull complete
Digest: sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 07ccdb783875 10 days ago 160MB
ubuntu latest 97bed23a3497 2 weeks ago 78.1MB
[node1] (local) root@192.168.0.29 ~
$
Search
2:45 PM
10/18/2025

```
54e822d8ee0c: Pull complete
Digest: sha256:3b7732505933ca591ce4a6d860cb713ad96a3176b82f7979a8dfa9973486a0d6
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
[node1] (local) root@192.168.0.29 ~
$ docker images
REPOSITORY   TAG       IMAGE ID       CREATED       SIZE
nginx        latest    07ccdb783875   10 days ago   160MB
ubuntu       latest    97bed23a3497   2 weeks ago   78.1MB
[node1] (local) root@192.168.0.29 ~
$ docker create --name myubuntu ubuntu
eabb8709cddec6057fd9e095085612d02e891a5ac7b7518145a4684d7f572d7d
[node1] (local) root@192.168.0.29 ~
$
```

Search

2:46 PM
10/18/2025

```
[node1] (local) root@192.168.0.29 ~
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[node1] (local) root@192.168.0.29 ~
$
```

```
[node1] (local) root@192.168.0.29 ~
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[node1] (local) root@192.168.0.29 ~
$ docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED         STATUS    PORTS     NAMES
eabb8709cdde   ubuntu    "/bin/bash"   2 minutes ago   Created             myubuntu
[node1] (local) root@192.168.0.29 ~
$
```

```
[node1] (local) root@192.168.0.29 ~
$ docker run myubuntu
Unable to find image 'myubuntu:latest' locally
docker: Error response from daemon: pull access denied for myubuntu, repository does not exist or may requi
re 'docker login': denied: requested access to the resource is denied.
See 'docker run --help'.
[node1] (local) root@192.168.0.29 ~
$
```

Search

2:52 PM
10/18/2025